No Space Like Home?
Selection and Efficiency in Hospital-at-Home*

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ABSTRACT: Hospital capacity is costly to expand, and the value of an available inpatient bed varies with congestion. Hospital-at-Home (HaH) provides inpatient-level care in patients' residences and can release hospital beds without requiring permanent capacity expansion. However, this only occurs when the hospital offers HaH and the patient accepts. I study how these decisions jointly determine the use and value of flexible capacity. Using clinical, accounting, census, and staffing data from four hospitals along with a novel patient survey, I model and estimate hospital offers and patient acceptance, accounting for time-invariant preferences and endogenous offer selection. Hospitals balance financial returns, clinical needs, and capacity relief while valuing access for patients who prefer home care. At the two largest hospitals, the estimated private value of released capacity is equivalent to $3,410–$4,660 per discounted bed-day at median occupancy. A counterfactual 10% reimbursement increase for HaH care expands potential inpatient bed-days released by 18.5% and raises expected value to patients from access to HaH by 19.6%. Although induced transfers increase treatment costs more than reimbursement, higher payments and filling 50–100% of released bed-days increase hospital profits by $3.3–$6.2 million. These findings show how consumer participation and provider incentives jointly shape the deployment and economic value of flexible hospital capacity.

KEYWORDS: Hospital-at-home, capacity management, consumer participation, healthcare innovation
JEL CLASSIFICATION: D12, D24, I11, I18, L21, L23

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1 Introduction

The United States healthcare system faces an overcrowding problem. This challenge is particularly acute in inpatient hospital care, with many facilities operating at or near capacity. Available bed capacity has declined since the Covid-19 public health emergency, even as patient demand has remained strong. Population aging is expected to increase annual hospitalizations by approximately 11 percent between 2025 and 2035, pushing national average hospital occupancy to 85 percent as early as 2032 if bed supply does not expand.[1] This presents both a clinical and economic problem. When beds and staff are scarce, accommodating one patient can limit access or compromise care for others, forcing hospitals to balance competing patient needs with the resources available.[2] Hospitals cannot costlessly eliminate these losses by simply building more beds. Physical expansion requires substantial upfront investment that is slow to deliver and difficult to reverse. Capacity built to accommodate high demand may remain underused when demand recedes, making permanent expansion risky even when hospitals are currently crowded. Financing constraints further affect hospital investment, while certificate-of-need requirements and related regulation can further constrain expansion. Against this backdrop, the recent expansion of Hospital-at-Home (HaH), an innovative care delivery model that provides inpatient-level care in patients' homes, offers a flexible way to relieve capacity pressure without requiring permanent capacity expansion.

Realizing this flexibility depends on decisions by both hospitals and patients. Hospitals assess patients' suitability for HaH and decide whether to offer home treatment. Patients who receive an offer choose whether to accept it or to continue treatment in the brick-and-mortar (B&M) hospital, meaning that capacity is released only when an offer is accepted. HaH can improve allocative efficiency by matching clinically suitable patients to their preferred care setting, in turn freeing hospital beds for patients who require scarce facility-based care. These interacting decisions motivate the paper's central question: how do firm incentives and consumer preferences shape the use and value of flexible capacity when conventional production is capacity constrained? I estimate a joint model of hospital offers and patient acceptance to characterize these decisions and recover the value hospitals place on released capacity. The framework then evaluates how reimbursement, information about patient preferences, and HaH program scale affect access to home treatment, potential capacity relief, and hospital financial returns.

[1] These projections are from Leuchter et al. (2025). An occupancy rate of 85 percent is a commonly used healthcare benchmark for elevated capacity pressure because hospitals require slack to accommodate stochastic demand and patient flows. Occupancy at this level is associated with regular bed shortages and periodic capacity crises (Bagust et al., 1999; National Guideline Centre, 2018). However, recent evidence suggests that 85 percent is a conservative upper bound for the onset of capacity strain, which may occur at lower occupancy rates depending on the ward type and care setting (Pratt and Wood, 2021).

[2] When inpatient beds are unavailable, admitted patients may remain in the emergency department, a practice known as boarding. Longer boarding is associated with higher inpatient mortality and longer stays (Singer et al., 2011).

The empirical analysis links four complementary sources of information from four hospitals in a large southeastern U.S. health system between September 2021 and February 2026. Detailed electronic health records (EHR) identify eligibility, clinical characteristics, health outcomes, and each stage of the HaH screening, offer, and acceptance process. Granular census, capacity, and staffing records measure the hospital state in which these decisions are made and trace how admissions, HaH transfers, and discharges affect subsequent congestion. Contracted payor rates and patient-level accounting data reveal the pecuniary incentives associated with treating a patient at home rather than in the hospital. Finally, I link the EHR to a novel survey that I designed and administered to patients and caregivers. The survey provides supplementary evidence on patients' expectations about care quality, comfort, safety, and convenience, together with household non-medical spending and caregiver circumstances not captured in administrative records.

I jointly estimate a hospital offer model and a panel model of patient acceptance using 31,944 inpatient stays from 21,354 unique patients. Repeated encounters help distinguish time-invariant patient preferences from history dependence in choices. I allow unobserved factors influencing hospital offers and patient choices to be correlated, accounting for endogenous offer selection. Hospitals decide whether to offer HaH based on expected acceptance, financial returns, patient surplus, clinical needs, HaH program strain, and the value of released B&M capacity. The expected remaining B&M stay determines how many bed-days a transfer releases, while higher hospital occupancy increases the value of each released day.

The model estimates show that hospitals value expected patient surplus alongside the financial and capacity consequences of accepted transfers. The private value of released B&M capacity rises with occupancy and, at hospital-specific median levels, is approximately \$4,660 and \$3,410 per discounted bed-day at the two largest hospitals. This reflects the value hospitals place on having an additional bed available to accommodate another patient or preserve operating slack. Debate over retaining payment parity with B&M care makes reimbursement central to HaH policy (Medicare Payment Advisory Commission, 2024). Rate changes affect both payor spending and hospitals' incentives to provide home treatment, shaping patient access and the resulting capacity relief. A 10% reimbursement increase generates 2,640 additional potential B&M bed-days and raises expected patient option value by 19.6%. Although induced transfers increase treatment costs more than additional payments, higher payments on baseline placements yield a net increase in hospital profits of \$1.6 million. Filling 50–100% of the additional released bed-days resulting from HaH transfers raises the hospital gain to \$3.3–\$6.2 million.[3] Conversely, a 10% reimbursement reduction decreases potential capacity relief by 1,778 bed-days and patient option value by 13.1%.

[3]Induced transfers increase treatment costs by \$4.4 million and payments by \$1.9 million, while higher payments on baseline placements contribute \$4.1 million. Hospital financial returns need not reflect social resource savings, because reimbursements simply transfer funds between payors and providers, while hospital accounting may omit household costs borne by patients and caregivers.

Additional simulations examine how occupancy-targeted payments, improved information about patient preferences, and expanded HaH program scale affect access, capacity relief, and hospital returns.

Clinical studies establish the feasibility of HaH for treating selected patients and report improvements in satisfaction or reductions in resource use in some settings (Leff et al., 2005; Levine et al., 2020; Shepperd et al., 2009; Caplan et al., 2012; Arsenault-Lapierre et al., 2021). Recent evidence under the federal Acute Hospital Care at Home initiative likewise reports generally favorable short-run outcomes among selected patients, while highlighting patient selection and variation across programs (Adams et al., 2023; Centers for Medicare & Medicaid Services, 2024; Vakkalanka et al., 2026). I examine the economic forces behind patient selection into HaH, focusing on how hospital incentives and patient preferences jointly determine offers, accepted transfers, and the resulting capacity relief.

Firms across the economy face the challenge of satisfying uncertain demand without investing in capacity that may later sit idle. Pindyck (1988) analyzes capacity choice when investment is irreversible, while studies of cement (Ryan, 2012) and bulk shipping (Kalouptsidi, 2014) show how sunk costs and construction delays limit capacity adjustment. HaH creates scope for capital avoidance by allowing hospitals to accommodate inpatient demand without committing to permanent B&M expansion. This connects my paper to research on hospital reserve capacity, which examines the benefits of keeping beds available to accommodate fluctuations in demand alongside the costs of unused capacity (Joskow, 1980; Gaynor and Anderson, 1995; Carey, 1998). Reimbursement and financing conditions also shape hospital investment (Wedig et al., 1989; Adelino et al., 2015; Gao et al., 2022), while certificate-of-need requirements can constrain bed expansion (Chiu, 2021; Courtemanche and Garuccio, 2025).

The paper's first central contribution is to recover the model-implied shadow value of released B&M capacity, which I refer to as the private capacity value. Freedman (2016) shows that bed availability affects discretionary admissions, while Kleiner (2019) shows that it influences hospital assignment. Congestion also affects care delivery and patient health (Woodworth, 2020; Hoe, 2022; Francetic et al., 2024), with longer emergency department wait times raising treatment costs for seriously ill patients (Woodworth and Holmes, 2020). Most closely related, Gowrisankaran et al. (2026) examine how patient selection and congestion interact to determine hospital outcomes during the Covid-19 pandemic. I study a within-hospital margin, in which transferring care to the home makes B&M capacity available for another admission or operating slack. This connects the paper to research on the opportunity cost of committing scarce capacity and the value of flexible production arrangements (Jofre-Bonet and Pesendorfer, 2003; Jeziorski and Krasnokutskaya, 2016). Allowing hospitals to value patient benefits alongside financial returns also connects the offer model to research on provider objectives (Ellis and McGuire, 1986; Jack, 2005; Brekke et al.,

2011). Patient-level offer decisions reveal how hospitals value capacity relief relative to accounting returns, conditional on expected patient acceptance.

The paper's second central contribution is to model consumer participation as a determinant of effective supply. In HaH, this participation involves co-production, with hospitals providing clinical services and patients making their homes available for treatment. Outside healthcare settings, Ely et al. (2017) study how airlines reallocate scarce seats by repurchasing tickets from passengers willing to relinquish them, while Wolak (2011) shows how hourly electricity prices induce consumers to reduce peak-period use, easing pressure on available generation capacity. Gandhi (2023) provides a close healthcare comparison by studying how capacity constraints, financial incentives, and heterogeneous patient preferences shape skilled nursing facility admissions. Firms weigh the returns from an admission against the capacity it commits, making reimbursement and anticipated length of stay central to access. I study a complementary margin in which hospitals can release B&M capacity by offering home treatment, with the release contingent on patient acceptance. The benefit of an additional offer therefore depends on both the patient's willingness to accept and the hospital's gains from the resulting transfer under prevailing operating conditions.

More broadly, patient demand shapes both the allocation of healthcare production and the development and diffusion of medical technologies (Chandra et al., 2016; Acemoglu and Linn, 2004; Hamilton et al., 2021). The value patients place on treatment options relates to research on hospital-network access (Capps et al., 2003), while hospital targeting connects to work on how provider incentives shape hospital choice (Ho and Pakes, 2014). To characterize heterogeneity in willingness to receive home treatment, I use a survey to elicit beliefs about care and household circumstances that administrative records do not capture (Stantcheva, 2023; Ferrario and Stantcheva, 2025). I combine these measures with observed choices in the acceptance model (Ben-Akiva et al., 1994) and examine how expected future household expenses are associated with treatment choices (Allcott and Wozny, 2014). Together, these data help characterize the beliefs and household tradeoffs underlying acceptance of home treatment.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 develops a framework which distinguishes the private and social value of HaH. Section 4 describes the novel data and reduced-form evidence that supports the model. Section 5 develops a joint model of patient acceptance and hospital offers, and Section 6 discusses estimation and identification. Section 7 interprets the results and Section 8 presents the counterfactual experiments. Section 9 concludes.

2 Institutional Background: Hospital-at-Home

Hospital-at-Home (HaH) provides acute, hospital-level care in a patient's residence as a substitute for treatment that would otherwise require a brick-and-mortar inpatient admission. It is distinct from conventional home health, which primarily provides intermittent post-acute or chronic care, and from telemedicine alone. HaH programs combine clinician oversight, in-person visits, and remote support. Programs are commonly classified as either admission-avoidance models, in which patients receive care at home without first being admitted to a B&M ward, or early-transfer models, in which an inpatient episode begun in the B&M hospital is completed at home (Leong et al., 2021). The program studied in this paper contains elements of both: patients generally enter through a hospital emergency department and may transfer to HaH immediately or after receiving inpatient care in the B&M hospital.

Modern clinical experimentation with home-based acute care dates to randomized trials in the United Kingdom in the 1970s (Hill et al., 1978). In the United States, Johns Hopkins developed an early physician-led HaH program in the 1990s. Its pilot and subsequent multi-site evaluation demonstrated the feasibility of treating selected older patients outside a conventional hospital (Leff et al., 1999, 2005). Nevertheless, U.S. adoption remained limited for the next two decades. Operating a program required hospitals to develop dedicated clinical and logistical infrastructure, while reimbursement for home-based inpatient care remained uncertain (Brody et al., 2019; Appelbaum and Batt, 2023). HaH consequently remained concentrated in demonstration programs, integrated delivery systems, and alternative-payment arrangements.

The Covid-19 pandemic created a capacity shock that prompted regulatory changes supporting HaH expansion in the United States. In March 2020, the Centers for Medicare & Medicaid Services (CMS) launched Hospitals Without Walls, allowing hospitals to treat inpatients at alternative care sites while continuing to receive Medicare hospital payments. CMS subsequently introduced the Acute Hospital Care at Home (AHCAH) initiative in November 2020, permitting approved Medicare-certified hospitals to provide inpatient care in patients' homes (Centers for Medicare & Medicaid Services, 2020). Patients must enter from an emergency department or inpatient unit and receive an in-person clinical evaluation before home care begins. Home care requires daily physician or advanced-practice provider evaluation, daily nursing assessment, and at least two in-person visits per day by registered nurses or paramedics. Programs must also ensure immediate remote access to the care team, an in-home emergency response within 30 minutes, and hospital transfer when needed. The initiative has been extended beyond the Covid-19 public-health emergency and is currently authorized through September 30, 2030 (Centers for Medicare & Medicaid Services, 2026). Regulatory flexibility, inpatient reimbursement, and pandemic-era capacity pressure spurred rapid HaH expansion, with several hundred hospitals joining the initiative in its first

years (Appelbaum and Batt, 2023).[4] Estimates of potential eligibility suggest scope for further expansion. An analysis using 2019 Medicare claims estimated approximately 825,000 potentially eligible annual HaH episodes across Medicare fee-for-service and Medicare Advantage beneficiaries (Pelizzari et al., 2022).[5]

Access to HaH is determined through a sequential and highly selective process. Patients must first satisfy condition, payor, and geographic eligibility rules. In the program studied here, geographic eligibility is based on residence within an operational service area constructed to meet CMS's 30-minute emergency-response requirement.[6] Hospital staff then assess clinical stability, anticipated procedures, and monitoring needs to determine whether the patient's condition can be managed safely outside the hospital. A subsequent social-stability assessment considers the safety of the residence, access to utilities and communication, caregiver or household support, and the patient's ability to participate in home treatment. Operational constraints, including mobile staffing and current program capacity, can further affect whether an otherwise suitable patient receives an offer. Only after the hospital makes HaH available does the patient or caregiver decide whether to accept. Patient choices may depend on perceived safety, trust in home-based care, prior experience, caregiver availability, housing conditions, and the convenience or comfort of recovering at home (Saenger et al., 2020; Centers for Medicare & Medicaid Services, 2024; Wang et al., 2024).

Clinical evidence generally finds that HaH can provide safe care for appropriately selected patients. An updated review of admission-avoidance HaH finds little or no difference in mortality or readmissions among older patients, alongside reductions in residential-care use and health-service costs (Edgar et al., 2024). Evidence from randomized trials in the U.S. also documents lower treatment costs in some settings (Levine et al., 2020).[7] A recent randomized trial in a rural setting found no statistically significant cost difference alongside improvements in patient experience and physical activity (Levine et al., 2025).[8] Observational evidence from the AHCAH period similarly reports favorable short-run outcomes, although comparisons remain vulnerable to patient and provider selection (Adams et al., 2023; Centers for Medicare & Medicaid Services, 2024; Vakkalanka et al., 2026). Results therefore require attention to program design, patient populations,

[4] In the study setting, all payors reimburse HaH at parity with B&M inpatient care.

[5] Potential use also depends on patients' circumstances and preferences. In Singapore, Ko et al. (2023) combined an administrative eligibility estimate of 53% with estimates of functional suitability and patient agreement to propose a substitution target of approximately 18% of medical admissions.

[6] CMS requires appropriate emergency personnel to reach the patient's home within 30 minutes, rather than requiring residence within 30 minutes of the admitting hospital. Programs may meet this standard through local emergency medical services (EMS). The program studied here uses a fixed ZIP-code eligibility rule based on its emergency response resources.

[7] Approximately 63% of potentially eligible patients declined trial enrollment, illustrating that clinical eligibility and provider decisions do not translate automatically to treatment at home.

[8] A separate pragmatic trial found that HaH combining virtual physician care with in-person home visits met a non-inferiority criterion for the composite of 30-day mortality or unplanned readmission (Maniaci et al., 2025).

and the outcomes and costs being measured. Section 3 formalizes how patient benefits, treatment costs, and released capacity can generate differences between the private and social value of an HaH offer.

3 Economic Framework and Welfare

The central economic object of interest in my study is the shadow value of B&M capacity released by an accepted HaH offer. This section describes how patient demand, treatment returns, and the value of freed capacity shape hospital and social planner incentives. Three wedges govern their potential disagreement: patient and health benefits, resource costs and profitability, and the value assigned to released capacity.

Patient choice and the consequences of HaH. Consider an HaH-eligible patient i admitted when hospital congestion is c. If offered HaH, the patient chooses between home treatment, h, and B&M care, b. Let

$$\Delta u_i = u_{ih} - u_{ib} + \varepsilon_i, \qquad \varepsilon_i \equiv \varepsilon_{ih} - \varepsilon_{ib},$$

denote the patient's total money-metric value of HaH relative to B&M. Here, u_{ih} and u_{ib} are the systematic utility components, while ε_{ih} and ε_{ib} are setting-specific idiosyncratic components observed only by the patient. Acceptance occurs when $\Delta u_i \geq 0$. Given the hospital's pre-offer information, the acceptance probability and expected patient surplus from an offer are

$$\rho_i = \mathbb{P}(\Delta u_i \geq 0), \qquad \mathcal{Q}_i = \mathbb{E}\left[\max\{\Delta u_i, 0\}\right].$$

The probability ρ_i determines whether an offer reallocates treatment, while $\mathcal{Q}_i$ measures the value of the additional treatment option.

Patient surplus captures both the health benefits and household burdens that patients take into account when choosing a treatment setting. Let $\Delta\mathcal{H}_i$ denote any additional monetized health benefits or harms, such as consequences patients do not fully anticipate. Let $\Delta\mathcal{C}_i$ denote real resource savings relative to B&M care, including provider resources and household inputs not already counted through patient valuation.[9]

An accepted transfer also makes inpatient bed space available for other uses. Let $r_i^S(c)$ denote the social shadow value of B&M capacity released by an accepted transfer over the period the patient would otherwise occupy a B&M bed. It captures the net benefits of additional admissions and reductions in waiting and crowding for other patients. Its value depends on the duration of the

[9]Caregiver roles vary across programs, and burden assessments must distinguish pre-existing caregiving demands from duties attributable to HaH (Leff et al., 2026).

release and the capacity pressure it relieves.[10] Its expected contribution at the offer stage is $\rho_i r_i^S(c)$.

Social value of an HaH offer. A social planner chooses whether to offer HaH while respecting the patient's treatment choice. Let k_i denote the real incremental cost of preparing and extending an offer, borne by the hospital regardless of acceptance. Transfer and treatment costs incurred upon acceptance enter $\Delta\mathcal{C}_i$. The expected social gain from offering HaH is

$$G_i^S(c) = \mathcal{Q}_i + \rho_i \left[\Delta\mathcal{H}_i + \Delta\mathcal{C}_i + r_i^S(c)\right] - k_i. \tag{1}$$

The planner offers when $G_i^S(c) > 0$. Q_i already accounts for the patient's option to decline. Health, resource, and capacity gains arise only after acceptance and are therefore weighted by ρ_i.

Hospital incentives. Let

$$\Delta\Pi_i = \Pi_{ih} - \Pi_{ib}$$

denote incremental hospital accounting profit from an accepted HaH placement. Under payment parity, this reflects savings in costs borne by the hospital. These savings coincide with $\Delta\mathcal{C}_i$ when hospital profits capture all relevant resource changes not already reflected in patient valuation. They can differ when households or other providers bear resource costs that are not fully reflected in hospital profits. Reimbursement changes create a further difference by changing hospital returns without directly changing the resources used to provide care.

Let $r_i^F(c)$ denote the hospital's private shadow value of the released B&M capacity. Freed bed space gives the hospital the option to accommodate another admission or retain operating slack. Its value combines the returns from these uses over the remaining period of the release. Benefits can extend across departments because inpatient patient-flow protocols influence emergency-department bottlenecks and delays (Armony et al., 2015). The private and social shadow values differ to the extent that the hospital internalizes these consequences for patients, staff, and the broader healthcare system.

Hospitals may also value patient welfare, health, or access to home care. Let m_i denote the hospital's expected nonpecuniary benefit from extending an offer, expressed in profit-equivalent dollars. The hospital's expected gain is

$$G_i^F(c) = m_i + \rho_i \left[\Delta\Pi_i + r_i^F(c)\right] - k_i. \tag{2}$$

[10] Research on hospital reserve capacity emphasizes the role of uncertain demand (Joskow, 1980; Gaynor and Anderson, 1995). Evidence on ambulance diversion and ICU strain illustrates the potential health consequences of restricted access and congestion (Shen and Hsia, 2011; Gowrisankaran et al., 2026).

The hospital offers when $G_i^F(c) > 0$. Capacity relief can justify an offer even when home treatment lowers accounting profit. Greater congestion strengthens this incentive when it raises the private value of released capacity.

Private and social incentives. The difference between social and private offer values is

$$\begin{aligned}\Delta G_i(c) &= G_i^S(c) - G_i^F(c) \\ &= \underbrace{\mathcal{Q}_i + \rho_i \Delta\mathcal{H}_i - m_i}_{\text{demand/health wedge}} \quad + \quad \underbrace{\rho_i\left(\Delta\mathcal{C}_i - \Delta\Pi_i\right)}_{\text{resource/profitability wedge}} \quad + \quad \underbrace{\rho_i\left[r_i^S(c) - r_i^F(c)\right]}_{\text{capacity wedge}}. \end{aligned} \tag{3}$$

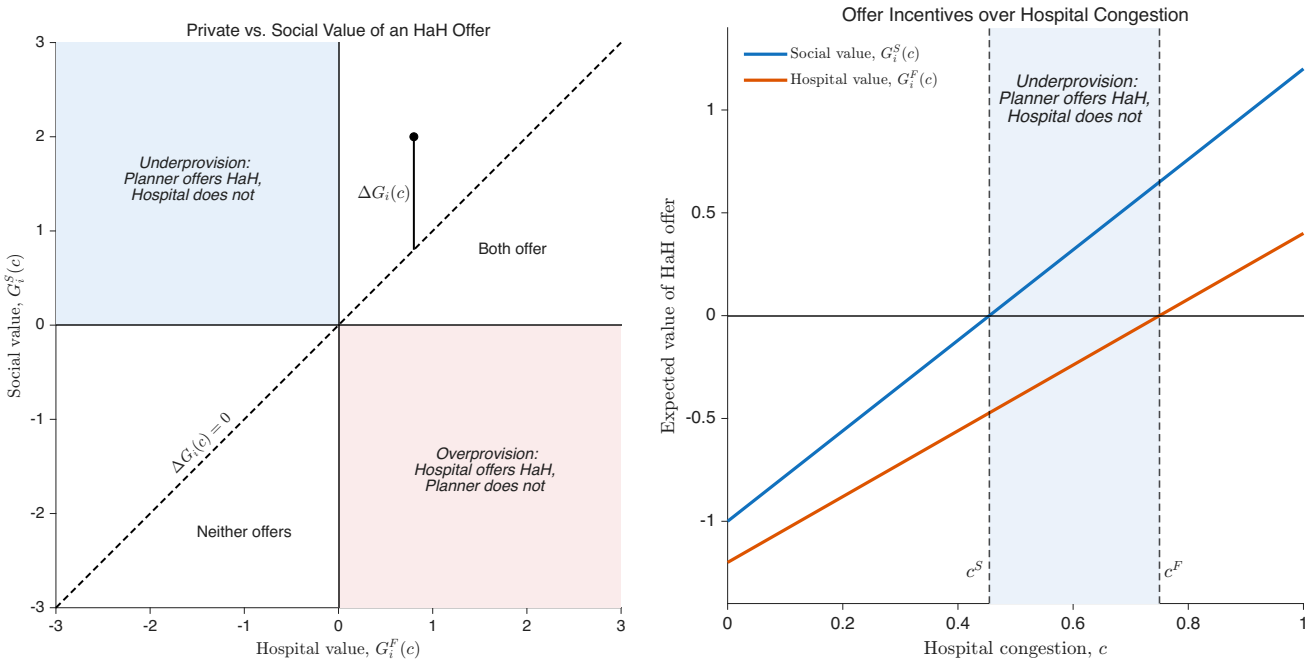
The first wedge compares patient and health benefits with the hospital's nonpecuniary valuation. The second compares social resource savings with accounting returns. Payment rules and the allocation of costs between the hospital and households can therefore make a transfer privately attractive even when its resource savings are small, or financially unattractive despite substantial social savings.

The capacity wedge compares the two shadow values of freed bed space. Benefits to other patients that the hospital does not internalize can make the social shadow value exceed its private counterpart, particularly during congestion peaks. Patient and health benefits or real resource savings not fully reflected in hospital returns can also make HaH socially worthwhile when capacity pressure is modest. In such cases, the planner may favor an offer at lower occupancy than the hospital, whose private incentive to release capacity remains weak. HaH is underprovided for patient i when $G_i^S(c) > 0$ but $G_i^F(c) < 0$, and overprovided when $G_i^F(c) > 0$ but $G_i^S(c) < 0$. Patient acceptance governs the capacity, health, and resource gains realized under either policy. Valuable candidates combine willingness to receive home care with substantial gains from the resulting transfer.

Figure 1 illustrates these allocation differences. Panel (a) compares private and social offer values: the upper-left region denotes underprovision and the lower-right region denotes overprovision. The 45-degree line represents agreement in valuations. Panel (b) illustrates a case in which social offer value becomes positive at a lower congestion level than private offer value. The interval between these thresholds identifies states in which the planner offers HaH but the hospital does not. This possibility motivates the Section 8 comparison of reimbursement supplements at lower and higher occupancy, examining how payment design shapes patient access, patient option value, capacity relief, and hospital returns.

Empirical scope. The empirical analysis uses this framework to connect hospital offer incentives, patient choices, and the shadow value of freed capacity. The joint model uses variation in

Figure 1: Private and Social Incentives for Hospital-at-Home Offers



(a) Private versus social offer value

(b) Congestion and offer thresholds

Notes: Values and thresholds are illustrative. Panel (b) varies congestion while holding other determinants fixed.

accounting returns and operating conditions, conditional on patient characteristics and acceptance beliefs, to recover the hospital's valuation of capacity relief in profit-equivalent units. A separate expected consumer-surplus term captures the hospital's valuation of making HaH available to patients. Accounting data measure hospital costs, while a novel patient survey provides evidence on household preferences and non-medical spending. Counterfactuals report hospital returns, potential bed-days released, and patient option value in normalized utility units. These outcomes quantify the principal allocation mechanisms, with the three wedges organizing their interpretation from private and social perspectives.

4 Data and Reduced-Form Evidence

4.1 Data, Assessment Sample, and Offer Documentation

The introduction of hospital-at-home care at a large southeastern health system provides a unique opportunity to study hospital capacity allocation and patient treatment choices from the program's inception to current maturation. To do so, I construct a novel proprietary dataset combining information from four sources. From the health system, I obtain (i) rich electronic health records

(EHR), (*ii*) administrative data on hourly bed census, staffed B&M capacity, and HaH staffing, (*iii*) negotiated payor contracts and accounting data on costs and revenues, spanning September 2021 through February 2026. The data cover two large urban medical centers and two satellite hospitals in less densely populated cities. I supplement these data with (*iv*) a novel survey I administered to HaH-eligible patients and their caregivers. The survey elicits expected household expenses under each treatment setting, beliefs about care, and caregiver income and employment.[11] In addition to data on patient diagnoses and procedures, the EHR data provide information on patient demographics, primary and secondary payors, post-stay outcomes, and patients' progression through the HaH screening process (see Figure A.1 for a detailed overview of the screening stages). These comprehensive data allow me to model the complex process of patient admission to a HaH program, while accounting for the various trade-offs hospitals and patients face in choosing in-hospital or at-home care.

Table 1: Hospital admissions and HaH selection, September 2021–February 2026

	All	Location & Payor eligible	Screened	Screened, not offered	Offered	Declined	Accepted
Encounters	76,481	41,595	34,563	28,234	6,329	2,938	3,391
Unique patients	49,830	26,613	22,881	19,535	5,219	2,597	2,899
Age (SD)	66.2 (17.1)	70.0 (16.3)	71.1 (15.8)	71.0 (16.0)	71.4 (14.8)	71.8 (14.3)	71.1 (15.1)
Male	0.468	0.461	0.464	0.468	0.447	0.432	0.461
White	0.653	0.682	0.681	0.672	0.724	0.708	0.737
Black	0.257	0.233	0.233	0.248	0.167	0.199	0.139
Latino/Hispanic	0.042	0.041	0.042	0.038	0.059	0.048	0.069
Commercial	0.135	0.072	0.054	0.052	0.063	0.055	0.069
Medicare	0.680	0.797	0.815	0.815	0.815	0.834	0.798
Medicaid	0.108	0.069	0.080	0.083	0.066	0.071	0.062
Dual Medicaid coverage	0.148	0.169	0.167	0.177	0.123	0.143	0.106
Multiple payors	0.364	0.484	0.482	0.487	0.460	0.452	0.468
Total LOS, days (SD)	6.4 (5.7)	6.2 (5.3)	6.4 (5.4)	6.5 (5.6)	6.0 (4.2)	5.4 (4.2)	6.5 (4.1)
DRG weight (SD)	1.5 (0.9)	1.4 (0.8)	1.4 (0.8)	1.4 (0.8)	1.2 (0.6)	1.3 (0.6)	1.2 (0.6)
ED entry	0.826	0.915	0.945	0.938	0.975	0.972	0.978
Nearest hospital, mins. (SD)	38.5 (35.4)	23.5 (14.0)	22.7 (12.6)	22.8 (12.8)	22.4 (11.8)	22.7 (11.5)	22.2 (12.0)
Net revenue, $000 (SD)	18.8 (26.0)	15.7 (18.3)	15.2 (16.8)	15.8 (17.8)	12.3 (10.9)	12.6 (10.7)	12.0 (11.0)
Neighborhood traits							
Household-size (SD)	2.6 (1.2)	2.5 (1.0)	2.5 (0.9)	2.5 (1.0)	2.5 (0.6)	2.5 (0.5)	2.5 (0.6)
College/graduate share	0.426	0.496	0.500	0.499	0.506	0.491	0.518
Log per-capita income (SD)	10.6 (0.5)	10.7 (0.5)	10.7 (0.5)	10.7 (0.5)	10.8 (0.4)	10.7 (0.4)	10.8 (0.4)
Share below 150% poverty line	0.197	0.169	0.168	0.170	0.159	0.165	0.154

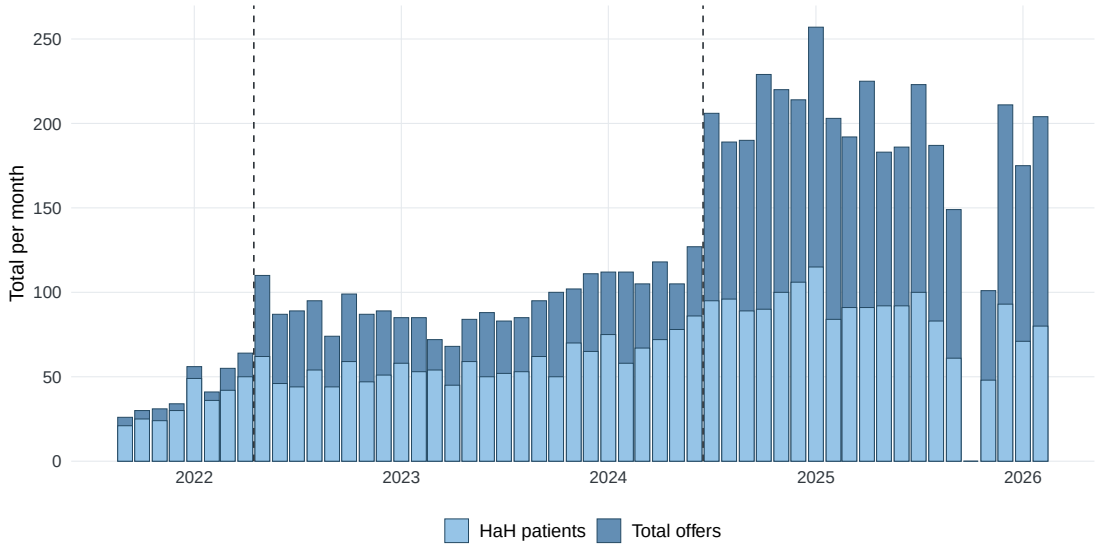
Notes: Subset to residents in the study state and inpatient conditions with 3+ HaH discharges. Numbers reflect shares unless SD is indicated. Screened means evidence of human-review for HaH suitability or evidence of HaH treatment. Offered includes recorded offers or accepted HaH; declined means recorded offer without acceptance. ACS education status is bachelor plus graduate share, income is log(1 + per-capita income), and poverty is the source share below 150% of the poverty line. Household-size is winsorized at the 99.9th percentile of the block-group distribution to limit the influence of extreme values. All monetary values described in this manuscript are expressed in February 2026 U.S. dollars ($).

[11] Appendix E describes the survey and its role in characterizing patient preferences.

The selection stages are summarized in Table 1. Roughly 80% of condition-location-payor eligible encounters are screened, about 18% of screened encounters have a recorded offer, and approximately half of these observed offers are accepted. These recorded frequencies describe how patients progress from eligibility to treatment, although incomplete documentation understates offers in earlier periods. The program expanded substantially during the study period. Figure 2 plots monthly observed offers and accepted placements. Accepted placements are observed throughout and therefore provide a consistent measure of treatment activity. Recorded offers also reflect changes in EHR procedures that improved documentation of patient declinations.[12]

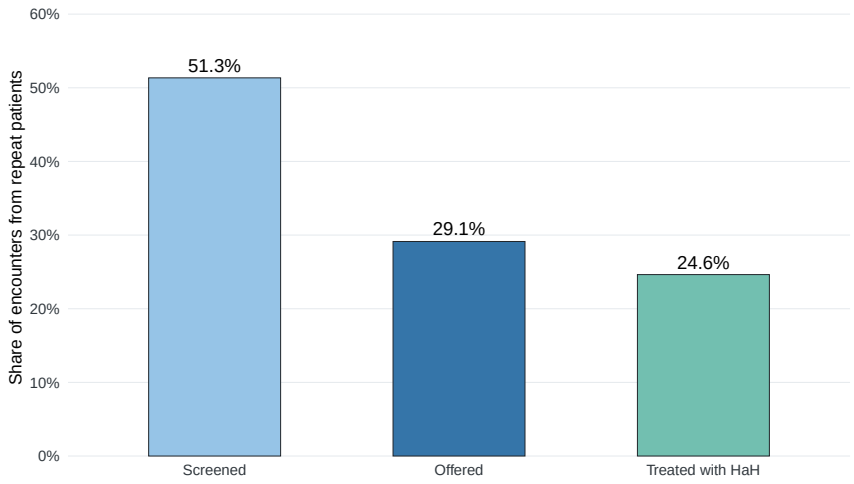
To account for undocumented declined offers, I use overlapping reporting indicators to construct hospital-regime documentation probabilities, q_i, treating the most recent "modern regime" documentation as complete. These probabilities and the number of assessments inform a separate reconstruction of offer totals for the reduced-form evidence presented in this section. Appendix C.1 describes this adjustment and its assumptions. The subsequent analysis examines how hospitals choose whom to offer HaH and how patients respond. It therefore focuses on encounters with an observed HaH assessment, taking the preceding screening decision as given.

Figure 2: Hospital-at-Home Offers and Admissions Over Time



Notes: Monthly observed offers and accepted placements across all four hospitals. Dashed lines mark documentation changes. Earlier offer totals omit undocumented declinations.

Figure 3: Share of Encounters Accounted for by Repeat Patients



Notes: Bars report the share of screened, offered, and HaH-treated encounters contributed by repeat patients. Each percentage uses all encounters at that selection stage as its denominator, with each encounter counted separately.

4.2 Selection and Repeated Patient Decisions

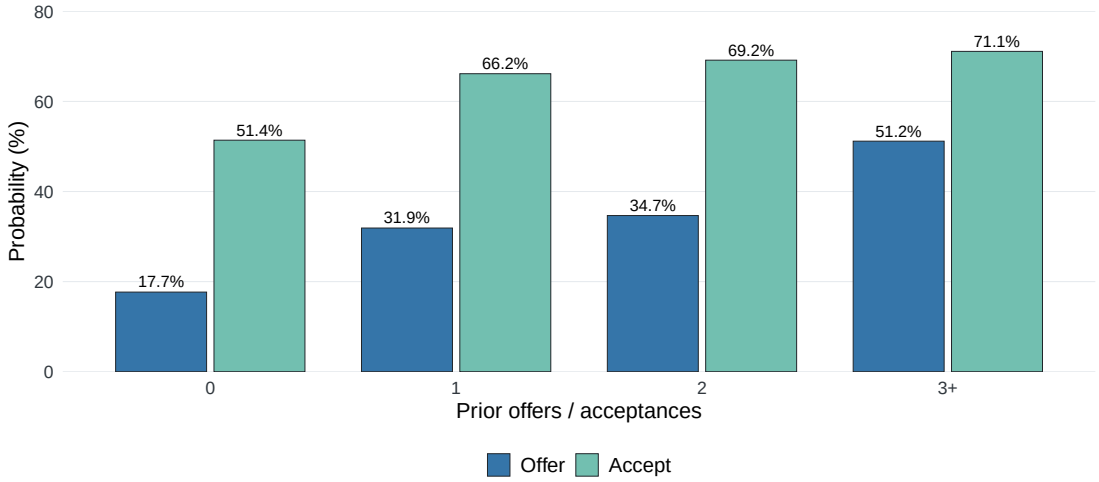
Patients who receive HaH offers differ from other screened patients, and those who accept differ further from those who decline. Relative to screened patients without an observed offer, offered patients have lower clinical severity, are more likely to be White and less likely to be Black or Medicaid insured, and generate lower reimbursed revenues. Among patients with an observed offer, those who accept are somewhat younger and more likely to be male, White, and commercially insured than those who decline. These differences motivate examining hospital offers and patient acceptance as separate decision margins. The hospital may also use information about patient suitability or preferences that is unobserved by the econometrician, thus accounting for observed patient characteristics alone may not fully explain which patients receive offers.

The repeated nature of inpatient encounters provides information on persistent willingness to receive HaH. Figure 3 shows that repeat patients account for approximately 50% of screened encounters, 30% of offered encounters, and 25% of accepted placements. Repeated choices can therefore help distinguish time-invariant patient preferences from history dependence, in which acceptance varies with prior HaH offers and treatment.

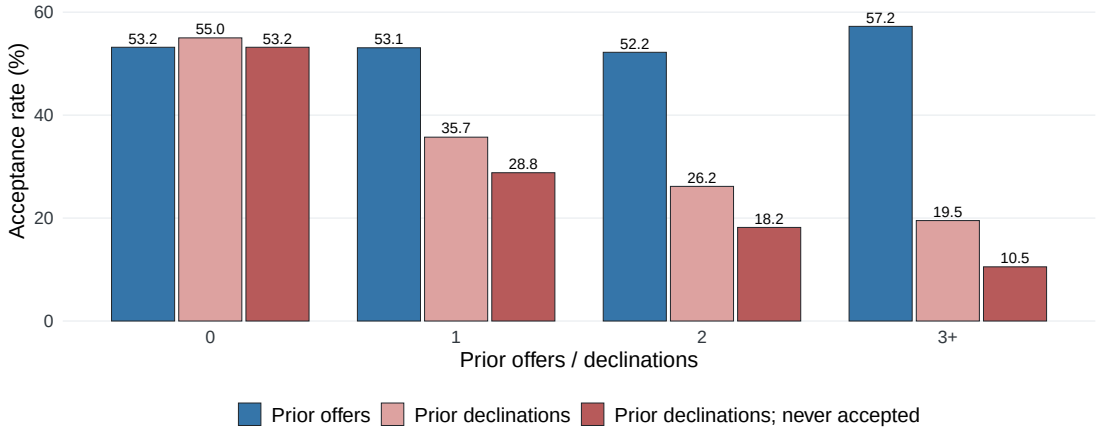
Figure 4 shows that acceptance conditional on an observed offer rises by 14.8 percentage points between zero and one prior acceptance, then increases more modestly before leveling off. Con-

[12]The October 2025 federal government shutdown interrupted HaH activity through a temporary pause of the AHCAH waiver, followed by a shorter partial shutdown in January 2026. These operational interruptions are distinguished from the documentation changes marked in Figure 2.

Figure 4: Probability of Outcome Across Offer and Acceptance History



(a) Offer and Acceptance Rates by Prior Interactions



(b) Acceptance Rates by Prior Offers and Declinations

Notes: Offer rates condition on screening and acceptance rates on an observed offer. Panel (a) groups offer rates by prior offers and acceptance rates by prior acceptances. Panel (b) groups acceptance rates by prior offers and prior declinations, showing the latter comparison both overall and among patients with no prior acceptance. All histories refer to earlier encounters.

versely, offer rates continue to increase with prior HaH offers. Repeated declinations, especially without any prior acceptance, predict much lower subsequent acceptance. These patterns motivate an initial treatment-associated shift alongside time-invariant patient preferences and hospital targeting based on prior interactions. They do not by themselves distinguish stable tastes from the effects of earlier treatment or outreach, since selection into treatment and subsequent offers also shapes the observed rates (Heckman, 1981; Dube et al., 2010). The estimated model that follows therefore combines a time-invariant patient effect with rich measures of prior HaH interactions to disentangle these channels, while accounting for endogenous offer selection.

4.3 Capacity Pressure and Hospital Offers

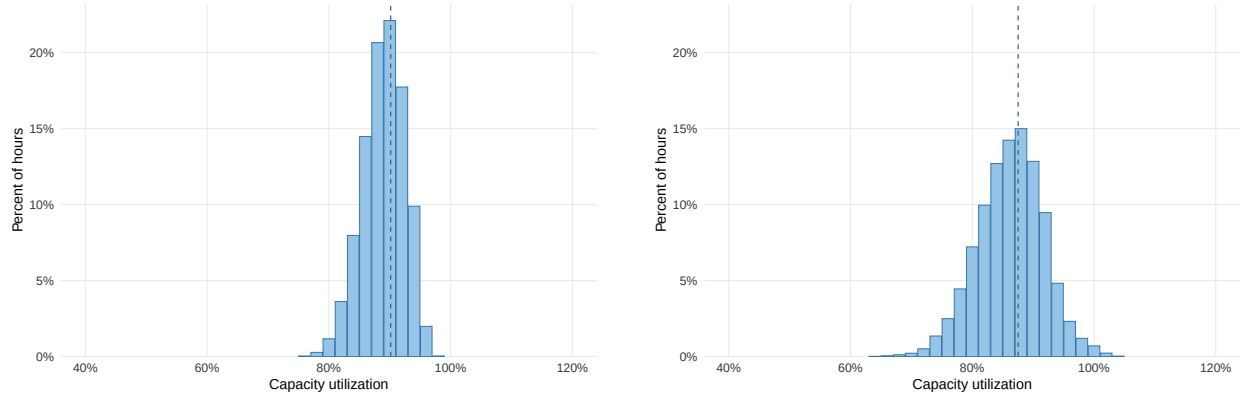
Administrative census data capture the persistently high inpatient utilization in this market. Figure 5 plots the distribution of hourly capacity utilization separately by hospital. The two large medical centers, Hospitals A and B, operate at high utilization for a large share of hours with mass concentrated around or above 90% utilization, while the two satellite facilities exhibit lower and more dispersed utilization. These levels are operationally meaningful because hospitals generally require slack bed capacity to move patients and resources throughout the system; utilization above roughly 85% is often associated with regular bed shortages and periodic capacity crises (Bagust et al., 1999; National Guideline Centre, 2018; Pratt and Wood, 2021). Thus, the same HaH technology is embedded in hospitals facing different degrees of capacity pressure, with the strongest congestion motive arising at the two urban facilities.[13]

Figure 6 relates empirical offer rates to B&M occupancy. Offer rates at the two large hospitals rise most clearly with occupancy, particularly toward the upper end of their empirical support. The satellite facilities have higher offer rates over much of their support but flatter occupancy profiles. These level differences need not represent differences in capacity value alone: patient composition, operational differences, and persistent hospital policy also vary across facilities. The relevant comparison I shall examine is how offer rates change with capacity pressure within a given hospital.

Figure 7 shows how HaH screening and offer probabilities vary with B&M capacity pressure. I estimate linear probability models for two sequential margins. The first outcome is whether an HaH-eligible patient is screened for home-care. The second is whether a reviewed patient receives an offer. In each specification, the slope on B&M occupancy varies by hospital. The figure reports the hospital-specific marginal effect of a 10 percentage-point increase in B&M occupancy.

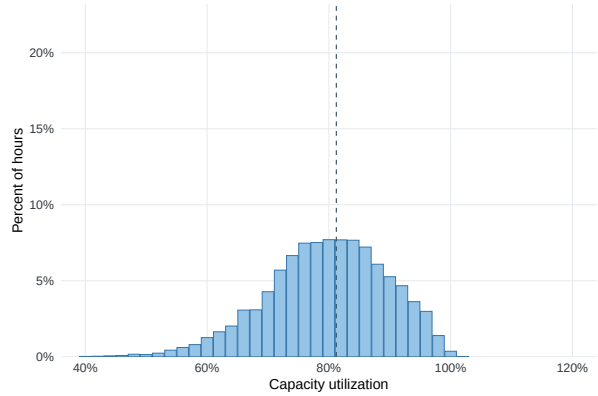
[13]The time-series analogues are reported in Figure A.2. These plots show substantial within-hospital variation in utilization over time, which provides within-hospital variation for estimating the conditional congestion response in screening and offer behavior in the analyses to come.

Figure 5: Distribution of Hourly Brick-and-Mortar Capacity Utilization

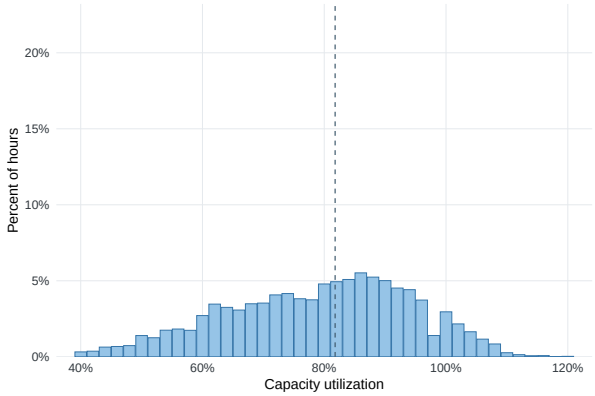


(a) Hospital A

(b) Hospital B



(c) Hospital C



(d) Hospital D

Notes: Utilization is hourly B&M census divided by contemporaneous staffed capacity. Each observed hospital-hour receives equal weight.

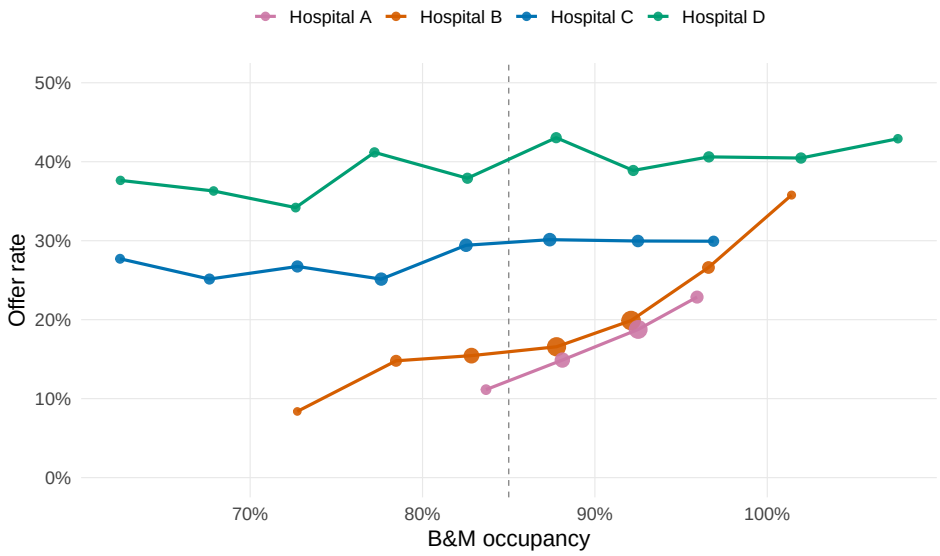
I estimate

$$Screened_{ih} = \alpha^S + \beta_h^S Occ^{BM}_{h,t_i^A} + \theta_h^S Stock_{h,t_i^A} + X_i'\gamma^S + \delta_h + \tau^S_{ym(t_i^A)} + \eta^S_{dow(t_i^A)} + \varepsilon^S_{ih},$$

$$Offer_{ih} = \alpha^O + \beta_h^O Occ^{BM}_{h,t_i^O} + \theta_h^O Stock_{h,t_i^O} + X_i'\gamma^O + \delta_h + \tau^O_{ym(t_i^O)} + \eta^O_{dow(t_i^O)} + \varepsilon^O_{ih}.$$

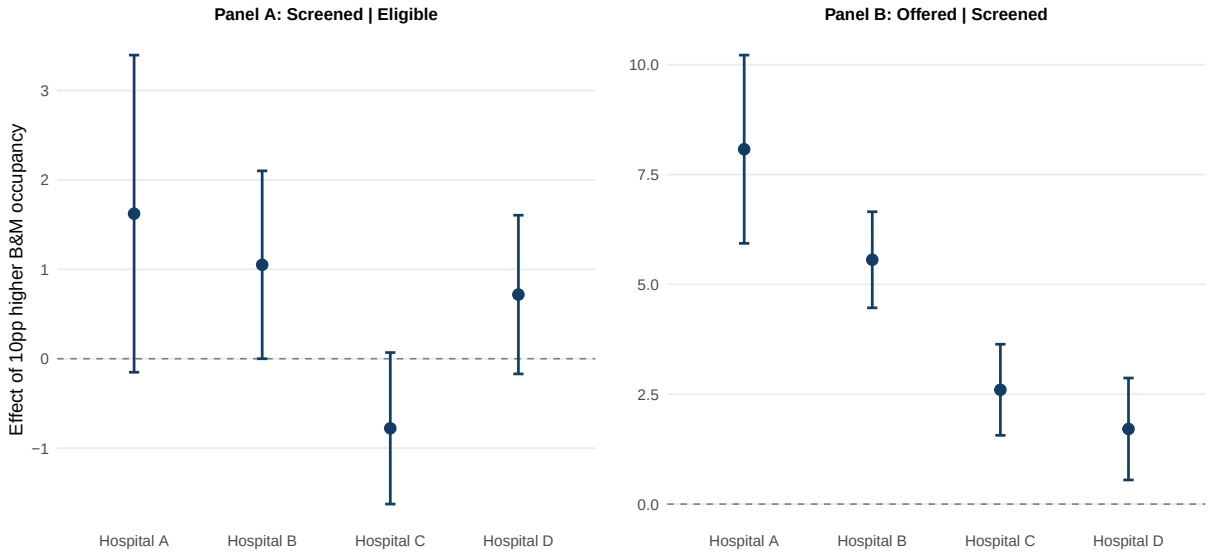
Here, t_i^A is the admission time and t_i^O is the offer-decision time. $Occ^{BM}_{h,t}$ is B&M occupancy, and $Stock_{h,t}$ contains both the HaH census and B&M emergency-department patient stock. The vector X_i contains patient demographics, payor, condition, and illness severity. The association with screening is weak overall, whereas offer probabilities are positively and significantly associated with occupancy at all four hospitals, with the largest associations at the two large urban medical

Figure 6: Documentation-Adjusted Offer Rates by B&M Occupancy



Notes: Bubble sizes correspond to the share of total hospital-hours spent in the corresponding B&M occupancy range.

Figure 7: Marginal Effects of Capacity Utilization by Decision Margin



(a) Marginal screening response

(b) Marginal offer response

Notes: Points report hospital-specific percentage-point differences in screening and offer probabilities associated with a 10 percentage point increase in B&M occupancy. Screening is estimated among eligible encounters and offers among screened encounters. Regressions control for HaH census, B&M emergency-department patient stock, patient demographics, payor, condition, and illness severity, with hospital, year-month, and day-of-week fixed effects. Standard errors are clustered at the hospital-day level. Offer outcomes incorporate the documentation adjustment described in Appendix C.1.

centers. While these estimates do not directly identify the value of released B&M bed capacity, they show that observed offer behavior changes with temporal capacity pressure.

Health outcomes and modeling scope. A fuzzy spatial regression discontinuity design exploits the program's geographic eligibility boundary to study health and utilization effects; Appendix B presents the design and summarizes the analysis developed in Capron et al. (2026). The estimates point to mixed effects, including lower mortality and fewer falls during the acute-care episode but more subsequent inpatient readmissions, with limited evidence of differences in several other health- and resource-related outcomes. These local estimates provide context for the allocation analysis. The baseline model retains escalation risk and acute-care needs as determinants of offers without directly estimating health effects.

5 Empirical Model of Hospital-at-Home Offers and Patient Acceptance

In this section, I specify and estimate a tractable joint model of demand for and supply of hospital-at-home care. Jointly modeling these decisions characterizes heterogeneity in patient willingness to receive inpatient-level care at home while recovering hospital offer incentives in a framework that accounts for endogenous offer selection and the dependence of realized capacity relief on patient acceptance. Because access to HaH occurs only after a patient presents to a hospital or its ED, I condition on the admitting hospital and abstract from hospital choice and competition across hospitals. The model instead focuses on the subsequent treatment-location margin: whether the hospital offers HaH and, conditional on an offer, whether the patient accepts care at home rather than remaining in the B&M setting.

The model conditions on an observed HaH assessment and does not separately estimate the preceding screening decision. An assessment produces two sequential decisions. The hospital first decides whether to offer at-home care to an eligible patient. If an offer is made, the patient decides whether to accept HaH or remain in B&M. Given most consequences of an offer are realized only after an accepted transfer, the hospital anticipates and internalizes the probability that the patient will accept. The model has three observed outcomes: an accepted HaH placement, a documented rejection, and no observed offer. The panel structure of the model allows willingness to accept to vary across patients through a time-invariant preference component, with repeat encounters exploited to distinguish this heterogeneity from encounter-specific choice variation. The hospital values the capacity released by an accepted transfer while balancing this benefit against the corresponding exposure to HaH staffing demands, location profitability, and patient care needs.

5.1 Setup and patient acceptance

Let i index HaH assessments and let $j(i)$ denote the patient associated with assessment i admitted to hospital $b(i)$. Let $O_i \in \{0,1\}$ indicate the hospital makes an actual HaH offer and let $A_i \in \{0,1\}$ indicate the patient accepts. Define

$$p_i = \mathbb{P}(O_i = 1 \mid \mathcal{I}_i^O), \qquad \rho_i = \mathbb{P}(A_i = 1 \mid O_i = 1, \mathcal{I}_i^O),$$

where $\mathcal{I}_i^O$ represents information observed by the econometrician prior to an offer decision.

The repeated-choice patterns in Section 4 motivate a specification that separates time-invariant preference heterogeneity from history dependence. Conditional on an offer, the patient chooses between HaH (h) and B&M ($b(i)$) care. Let ΔU_{ij} denote the patient's utility of h relative to $b(i)$ at assessment i for patient $j(i)$ with time-invariant HaH preference ξ_j. I parameterize this indirect utility difference as

$$\Delta U_{ij} = X_i'\beta_U + h(\mathcal{H}_i)'\theta - \alpha P_i + \sigma_\xi \xi_j + \varepsilon_i, \tag{4}$$

where $\varepsilon_i \sim N(0,1)$ is a mean-zero normal utility-difference shock. The vector X_i contains race, payor, age, sex, distance to the admitting hospital, condition severity, neighborhood characteristics, and condition and hospital-by-quarter fixed effects. Let $\mathcal{H}_i$ denote the patient's complete HaH interaction history prior to assessment i, and let $h(\mathcal{H}_i)$ summarize this history through variables such as prior offers and acceptances. The scalar P_i contains out-of-pocket price differences borne by the patient.[14] The patient-specific term $\xi_j \sim N(0,1)$ captures time-invariant unobserved willingness to receive acute care at home. The patient accepts HaH when $\Delta U_{ij} \geq 0$.

5.2 Hospital Offer Decision

While patient acceptance determines whether a release actually occurs, hospital offer selection first determines which transfers are attempted. The hospital may value consequences beyond contemporaneous accounting profit. The model therefore uses observed and latent offers to recover the private capacity value of scarce bed stock together with the hospital's valuations on accounting returns, HaH program strain, patient preferences, and clinical risk.[15]

The hospital observes the assessed patient, their clinical condition, and the current operating state before choosing whether to offer HaH. Let ρ_i^H denote the hospital's acceptance belief that

[14] Age, condition severity weight, home-to-hospital distance, household size, neighborhood college share, log per-capita income, and the share below 150% of the poverty line are standardized using fixed reference means and standard deviations. Prior interaction histories are raw cumulative counts. Indicators are not standardized. Appendix C.2 provides a discussion of the relevant transformations.

[15] Consistent with health-economics models in which providers place weight on mission-driven payoffs as well as financial incentives (Ellis and McGuire, 1986; Jack, 2005; Brekke et al., 2011).

patient $j(i)$ would choose HaH if offered, and let Q_i^H denote the corresponding expected patient surplus from access to HaH. Both are conditional on the hospital's information at the offer decision. The deterministic component of the hospital's value from offering rather than not offering is

$$\Gamma_i = Z_i'\kappa + \rho_i^H \Delta W_i, \tag{5}$$

where Z_i contains determinants of the offer decision whose effects do not require acceptance and ΔW_i is the incremental value of an accepted HaH placement. The hospital offers HaH when $\Gamma_i + \nu_i \geq 0$, where $\nu_i \sim N(0,1)$ is a mean-zero normal offer payoff shock. This rule determines the offer probability p_i. I specify the value of an accepted placement as

$$\Delta W_i = \beta_\Pi \Delta\Pi_i + \beta_H H_i + \beta_E E_i^B + \lambda_B' M_i^B + \lambda_H M_i^H. \tag{6}$$

These terms measure the expected profits difference ($\Delta\Pi$), escalatory care risks (H), expected B&M hospitalization duration (E^B), released B&M capacity (M^B), and HaH program strain (M^H), respectively. The vector

$$\Theta_S = (\kappa, \beta_\Pi, \beta_H, \beta_E, \lambda_B, \lambda_H)$$

collects all supply-side parameters. The coefficients in κ enter the offer index directly, capturing baseline offering frictions, prior interactions, clinical needs, and the patient value of access. Within κ, let ω collect coefficients on fixed effects and auxiliary controls.

The components of ΔW_i summarize the predicted consequences of an accepted transfer. Let E_i^B and E_i^H denote expected discounted remaining days in B&M and HaH care, respectively. These measures are constructed from assessment-origin discharge models that condition on hospital, condition, treatment location, current length of stay, and day of week, using a daily discount factor of $\delta = 0.99$. Appendix C.3 details their construction. Expected remaining B&M duration enters separately and affects the firm's payoff through a conditional duration effect, meaning hospitals' willingness to offer a patient can vary with the anticipated remaining stay conditional on accounting returns, clinical risk, and capacity consequences. This allows the capacity terms to fully capture how the value of releasing those bed-days varies with hospital utilization.

Let $\Delta\Pi_i^{\$} = \pi_i^H - \pi_i^B$ denote the conditional HaH-minus-B&M dollar profit difference, where $\pi_i^o = R_i^o - C_i^o$ is expected remaining revenue minus direct treatment costs under option $o \in \{H, B\}$. Reimbursement parity across treatment locations implies $\Delta\Pi_i^{\$} = C_i^B - C_i^H$. The offer index uses the centered difference in \$1,000 units, denoted by $\Delta\Pi_i$ and defined in equation (C.13). Figures C.3 and C.4 illustrate the fitted discharge hazards and remaining cost profiles by condition and treatment location. Appendix C.4 explains how episode-level cost estimates yield these conditional predictions of remaining direct treatment costs.

Let H_i denote the probability that the patient would require return transfer from HaH back to B&M care due to escalation in acute care needs. Because this risk arises only under HaH treatment, it enters ΔW_i and is therefore scaled by the probability of acceptance. To construct HaH program strain, let N_i^H denote HaH census and L_i denote HaH staffing at the time of the assessment. Define HaH utilization as

$$u^H(N^H, L) = \frac{N^H}{L}. \tag{7}$$

The program-strain function is

$$m^H(N^H, L) = L\left[u^H(N^H, L)\right]^2 = \frac{(N^H)^2}{L}. \tag{8}$$

The expected strain generated by adding the assessed patient is therefore

$$\begin{aligned} M_i^H &= E_i^H \left[m^H(N_i^H + 1, L_i) - m^H(N_i^H, L_i)\right] \\ &= E_i^H \ \frac{2N_i^H + 1}{L_i}. \end{aligned} \tag{9}$$

This quadratic specification implies that the contemporaneous strain from adding a patient is greater when the HaH program is more highly utilized. Multiplication by L_i allows the level of the strain function to scale with staffing levels.

The B&M term M_i^B is of central interest to the paper's primary research question. It measures occupancy-weighted bed-days released by an accepted HaH transfer. Let N_i^B denote B&M census and K_i denote staffed B&M capacity. Similarly, define B&M utilization as

$$u^B(N^B, K) = \frac{N^B}{K}. \tag{10}$$

The capacity-pressure function is

$$m^B(N^B, K) = K\left[u^B(N^B, K)\right]^2 = \frac{(N^B)^2}{K}. \tag{11}$$

Holding staffed capacity fixed at its assessment value, the corresponding hospital-specific released-capacity component is[16]

$$\begin{aligned} M_{ib}^B &= \mathbb{1}\{b(i) = b\} \ E_i^B \left[m^B(N_i^B, K_i) - m^B(N_i^B - 1, K_i)\right] \\ &= \mathbb{1}\{b(i) = b\} \ E_i^B \ \frac{2N_i^B - 1}{K_i}. \end{aligned} \tag{12}$$

[16]The indicator activates hospital b's capacity component only for assessments at that hospital.

This term assigns greater weight to bed releases when occupancy is higher and the patient's expected remaining B&M stay is longer. With E_i^B entering the payoff separately, the capacity coefficients capture how the value of releasing those bed-days varies with occupancy, conditional on other payoff inputs.

Let M_i^B collect the hospital-specific components $\{M_{ib}^B\}_{b\in\mathcal{B}}$, with corresponding coefficient vector λ_B. Consequently,

$$\lambda_B' M_i^B = \sum_{b\in\mathcal{B}} \lambda_{B,b} M_{ib}^B \tag{13}$$

is the measured B&M capacity contribution to the hospital's offer index, conditional on acceptance. Each hospital has its own coefficient on the smooth released-capacity component. Released capacity may be retained as operating slack or used to admit another patient. The model therefore identifies the composite private value of making additional capacity available. It does not separately identify the portion attributable to reduced operating pressure, improved patient throughput, or additional pecuniary profits from backfill (i.e., admitting patients waiting in the queue).

The vector Z_i contains the offer-stage determinants that do not require patient acceptance. To distinguish the components of structural interest from nuisance controls, let

$$Z_i'\kappa = \kappa_O + \kappa_H f(\mathcal{H}_i) + \kappa_S' S_i + \kappa_Q Q_i^H + \zeta_i'\omega. \tag{14}$$

The parameter κ_O is the common net offer term. A negative value is consistent with outreach, vetting, and administrative effort incurred before acceptance, but its level also absorbs baseline benefits and depends on the reference categories and regressor centering. It is therefore interpreted as a baseline offer friction rather than a separately measured financial cost. The variable $f(\mathcal{H}_i)$ is the strictly prior expected number of true offers made to the patient. It includes recorded offers and the posterior probabilities of undocumented declined offers. Its coefficient κ_H allows prior interactions to affect current offer value through repeated outreach, familiarity with the patient and caregivers, or persistent targeting.

The vector S_i contains acute-severity and clinical-care-burden indices measured at the assessment. Care burden combines nursing-assessment (NWASS) and fall-risk (Morse) scores, while acute severity uses organ-failure-risk (SOFA) scores. These baseline markers summarize the patient's health and care needs at assessment and enter directly as determinants of offer likelihood. The model does not predict their subsequent evolution under either treatment location. Predicted escalation risk, by contrast, describes a consequence specific to HaH treatment and enters conditional on acceptance. The term $\kappa_Q Q_i^H$ allows the hospital to value the expected patient surplus from having the option to choose HaH, separately from the acceptance-weighted consequences of an actual transfer. This is consistent with a mission-driven preference for providing specific services

to patients who value them more. The final vector ζ_i contains condition type indicators, hospital and department fixed effects, and indicators for missing clinical EHR inputs and the partial federal government shutdown described in Section 4.[17]

5.3 Dynamic Model Counterpart

Appendix F develops the dynamic counterpart to the static specification. Both use the same measured payoff components and distinguish offer-stage consequences from those contingent on acceptance. The static model already values expected discounted profits and capacity exposure over the remaining stay; the dynamic version additionally accounts for how today's transfer changes future operating states and subsequent offer decisions. The static specification permits joint maximum likelihood estimation, which is asymptotically efficient under correct specification and standard regularity conditions (Newey and McFadden, 1994). The dynamic implementation uses two-step GMM, whose efficiency depends on the moments and their weighting (Hansen, 1982); appropriately constructed two-step estimators can attain the efficiency of maximum likelihood (Aguirregabiria et al., 2021). The practical advantage of the static baseline is that it avoids the additional transition estimation and continuation-value approximations required by the dynamic implementation. The dynamic estimates indicate that continuation values contribute little to the hospital's offer incentives, providing further support for the static specification. Unless otherwise explicitly stated, the main-text estimates and counterfactuals therefore refer to the static model.

6 Estimation and Identification

6.1 Offer Documentation

Estimation must account for incomplete documentation of HaH offers. As described in Section 4.1, the hospital-regime rejection-capture probabilities, q_i, are constructed from reporting patterns and fixed before estimation. Accepted placements are always observed; a rejected offer is recorded with probability q_i. These inputs link actual offers to the three observed EHR outcomes.

Let $D_i \in \{0,1\}$ indicate whether a rejected offer is documented. The observed outcome $Y_i \in \{a,d,n\}$ denotes an accepted placement ($O_i = 1$, $A_i = 1$), a documented rejection ($O_i = 1$, $A_i = 0$, $D_i = 1$), or no observed offer ($O_i = 0 \vee O_i = 1$, $A_i = 0$, $D_i = 0$), respectively. Conditional on

[17] Appendix C.2 details the centering, scaling, and other transformations applied to the offer-model inputs.

p_i and ρ_i, the observed outcome probabilities are

$$\mathbb{P}(Y_i = a) = p_i \rho_i, \tag{15}$$

$$\mathbb{P}(Y_i = d) = p_i(1-\rho_i)q_i, \tag{16}$$

$$\mathbb{P}(Y_i = n) = 1 - p_i + p_i(1-\rho_i)(1-q_i). \tag{17}$$

The last outcome combines actual non-offers with latent rejected offers that were not documented. The probabilities in equations (15)–(17) sum to one for every assessment.

The maintained measurement assumption is that, conditional on hospital and documentation regime, q_i describes capture among rejected offers and does not independently shift offer behavior or acceptance utility after conditioning on the included controls. Appendix C.1 describes this adjustment. The empirical offer frequencies in Section 4.1 do not use fitted p_i or ρ_i; the posterior probabilities below are separate model objects used to update patient histories.

Let $\widehat{O}_i \equiv \mathbb{P}(O_i = 1 \mid Y_i)$ denote the posterior probability that a true offer occurred. Accepted placements and documented rejections reveal an offer, so $\widehat{O}_i = 1$ for these outcomes. When no offer is recorded, either no offer occurred or an offer was declined and not documented. Therefore, the posterior probability of a true offer when no offer is documented is

$$\widehat{O}_i = \mathbb{P}(O_i = 1 \mid Y_i = n) = \frac{p_i(1-\rho_i)(1-q_i)}{1 - p_i + p_i(1-\rho_i)(1-q_i)}. \tag{18}$$

These posterior probabilities update the patient's prior HaH interaction histories, which enter the demand and supply specifications below. The update occurs only after evaluating the current outcome and therefore affects future assessments.

6.2 Selection on Unobservables

Hospitals may condition offer decisions on information about patient willingness that is unavailable to the econometrician. For example, staff may observe home support, patient or caregiver reluctance, or aspects of clinical suitability that are related to the patient's willingness to accept HaH. As a result, patients who receive offers may be selected on unobserved acceptance utility, so estimating acceptance only among observed offers would confound patient preferences with offer selection. Following binary-outcome models with sample selection (Van de Ven and Van Praag, 1981), I jointly model offer and acceptance decisions by allowing the patient utility shock (ε_i) and hospital offer-payoff shock (ν_i) to be jointly normally distributed:

$$\begin{pmatrix} \varepsilon_i \\ \nu_i \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \right], \qquad -1 < r < 1. \tag{19}$$

Conditional on the modeled information, these shock pairs are independent across assessments and independent of the time-invariant patient type $\xi_j(i)$. The hospital observes ν_i when making the offer and therefore uses

$$\varepsilon_i \mid \nu_i = \nu \sim N\big(r\nu, 1-r^2\big)$$

to form its beliefs about patient acceptance and surplus. The hospital therefore forms $\rho_i^H(\nu)$ and $Q_i^H(\nu)$ conditional on its realized private encounter information.

Acceptance among offers is instead $\rho_i = J_i/p_i$, with the joint placement probability J_i defined below. Because acceptance beliefs and expected patient surplus depend on ν, I write the offer index as $\Gamma_i(\nu)$. The offer region and corresponding offer probability are

$$\mathcal{O}_i = \{\nu : \Gamma_i(\nu) + \nu \geq 0\}, \qquad p_i = \int_{\mathcal{O}_i} \phi(\nu)\, d\nu, \tag{20}$$

where ϕ is the standard-normal density. Let $\Theta_D = (\beta_U, \theta, \alpha, \sigma_\xi)$ collect the demand parameters and let $\Theta = (\Theta_D, \Theta_S, r)$ collect all jointly estimated parameters. Beliefs about time-invariant preferences use observed assessment histories, with earlier private disturbances not retained in that history.

6.3 Joint Likelihood and Estimation

Forecast capacity inputs. In estimation, I replace contemporaneous census in equations (9) and (12) with forecasts of B&M and HaH census over the patient's remaining stay. Separate hospital-level B&M and program-wide HaH forecasts are estimated at multiple future horizons using only census and calendar information available at the assessment, with the forecasting models updated monthly on an expanding historical window. Let $\widehat{N}_i^o(\tau)$ denote forecast census at horizon τ under setting $o \in \{B,H\}$ and let $S_i^o(\tau)$ denote remaining-stay survival. The exposure-weighted forecast census is

$$\overline{N}_i^o = \frac{\int_0^\infty \delta^{\lfloor \tau \rfloor}, S_i^o(\tau)\widehat{N}_i^o(\tau)\, d\tau}{E_i^o}, \qquad o \in \{B,H\}, \tag{21}$$

with $\delta = 0.99$. Staffed B&M capacity and HaH staffing are held fixed at their assessment values. These forecasts capture expected pressure during the remaining stay without specifying a structural model of future admissions or discharges. This is the preferred specification because the value of a capacity release depends on congestion over the period in which the patient would otherwise occupy the bed, rather than only on the operating state at the assessment. Appendix D compares

these estimates with an alternative that holds B&M and HaH census fixed at their assessment-time values over the remaining stay. Appendix C.3 provides the forecasting and exposure-construction details.

Conditional beliefs and offer-region integration. Let ξ_g denote the preference value at support point g, and let w_{ig}^- denote its probability weight based on patient $j(i)$'s observed history before assessment i. Conditional on $\xi_{j(i)} = \xi_g$ and the hospital's private shock $\nu_i = \nu$, the patient's utility difference is normally distributed with mean

$$\mu_{ig}(\nu) = X_i'\beta_U + h(\mathcal{H}_i)'\theta - \alpha P_i + \sigma_\xi \xi_g + r\nu$$

and variance $1 - r^2$. Write $s_\varepsilon = \sqrt{1-r^2}$ for its conditional standard deviation. Conditional acceptance at node g and the hospital's acceptance belief are therefore

$$\rho_{ig}^H(\nu) = \Phi\left(\frac{\mu_{ig}(\nu)}{s_\varepsilon}\right), \tag{22}$$

$$\rho_i^H(\nu) = \sum_g w_{ig}^- \rho_{ig}^H(\nu). \tag{23}$$

Averaging the expected gains from access to HaH across preference nodes gives the hospital's expectation of patient surplus,

$$Q_i^H(\nu) = \sum_g w_{ig}^- \left[\mu_{ig}(\nu)\Phi\left(\frac{\mu_{ig}(\nu)}{s_\varepsilon}\right) + s_\varepsilon \phi\left(\frac{\mu_{ig}(\nu)}{s_\varepsilon}\right)\right], \tag{24}$$

where Φ and ϕ are the standard-normal distribution and density functions. The weights w_{ig}^- depend only on the patient's strictly prior observed history and are updated as described below. Because the hospital does not observe ξ_j, the offer region is common across preference nodes.

At each trial parameter vector, I locate all zero crossings of $\Gamma_i(\nu) + \nu$ and integrate over the union of offer intervals. Define

$$J_{ig} = \int_{\mathcal{O}_i} \rho_{ig}^H(\nu)\phi(\nu)\,d\nu, \qquad J_i = \sum_g w_{ig}^- J_{ig}, \qquad \rho_i = \frac{J_i}{p_i}. \tag{25}$$

Here J_i is the joint probability of an offer and acceptance. Because ν_i also informs acceptance and patient surplus, the offer region can contain multiple intervals; the calculation does not impose a unique cutoff.

Observed-outcome likelihood. Recall $Y_i \in \{a, d, n\}$ denotes an accepted offer, a documented rejection, or no observed offer, respectively. Conditional on preference node g, the probability of the observed outcome at assessment i is

$$\begin{aligned}\ell_{ig}(Y_i;\Theta) = J_{ig}^{\mathbb{1}\{Y_i=a\}} \left[q_i(p_i - J_{ig})\right]^{\mathbb{1}\{Y_i=d\}} \\ \times \left[1 - p_i + (1-q_i)(p_i - J_{ig})\right]^{\mathbb{1}\{Y_i=n\}}.\end{aligned} \tag{26}$$

Let $\mathcal{T}_j$ denote the chronologically ordered assessments for patient j that enter the likelihood, and let F_j^0 denote the distribution of the time-invariant patient taste immediately before the first assessment in $\mathcal{T}_j$. Conditional on the time-invariant patient effect, the patient-level likelihood contribution is the product of equation (26) across assessments. Integrating over the patient effect gives

$$\begin{aligned}L_j(\Theta) &= \int \prod_{i \in \mathcal{T}_j} \ell_i(Y_i;\Theta,\xi)\, dF_j^0(\xi) \\ &\approx \sum_{g=1}^{G} w_{jg}^0 \prod_{i \in \mathcal{T}_j} \ell_{ig}(Y_i;\Theta),\end{aligned} \tag{27}$$

where w_{jg}^0 denotes the node weight immediately before the first likelihood assessment for patient j. Because the data cover the HaH program from inception, there is no unobserved pre-program HaH interaction history to initialize. At a patient's first program assessment, $\mathcal{H}_i = \varnothing$ and the type weights equal the population quadrature weights w_g. Any history-only assessments preceding the patient's first likelihood observation subsequently update these weights, yielding w_{jg}^0. The joint log likelihood that is maximized is

$$\mathcal{L}(\Theta) = \sum_j \log L_j(\Theta). \tag{28}$$

Sequential integration over time-invariant heterogeneity. I approximate the distribution of time-invariant patient preferences using Gauss–Hermite quadrature, with transformed standard-normal nodes ξ_g and prior weights w_g. The estimator evaluates equation (27) recursively. Immediately before assessment i, the predictive likelihood contribution and subsequent posterior update are

$$L_i(\Theta) = \sum_{g=1}^{G} w_{ig}^- \ell_{ig}(Y_i;\Theta), \tag{29}$$

$$w_{ig}^+ = \frac{w_{ig}^- \ell_{ig}(Y_i;\Theta)}{\sum_{k=1}^{G} w_{ik}^- \ell_{ik}(Y_i;\Theta)}. \tag{30}$$

The updated weights become the prior weights for the patient's next assessment. At the beginning of the full observed patient history, the weights equal the Gaussian quadrature weights w_g. Assessments used only for history updating contribute no direct likelihood term, but their posterior updates are retained before evaluating subsequent assessments. Because the offer probability and predictive likelihood are evaluated using w_{ig}^{-} before the current outcome updates the weights, the current outcome does not enter its own acceptance utility or offer index. Appendix C provides additional details on computation and estimation. Standard errors use the observed information matrix, conditional on the generated inputs, and do not adjust for clustering or first-stage estimation uncertainty.

6.4 Identification

The fixed documentation adjustment and complete reporting in the modern regime distinguish the probabilities of actual non-offers and incompletely recorded rejections. Conditional on the fixed documentation probabilities, variation in acceptances, documented rejections, and no observed offers identifies the latent offer and acceptance probabilities. Repeated observations for the same patient identify time-invariant preference heterogeneity through serial dependence in acceptance and rejection outcomes.

Observed patient characteristics, prior interaction histories, and repeated decisions identify systematic and time-invariant heterogeneity in acceptance utility. The patient out-of-pocket price coefficient additionally requires variation in patient prices conditional on these characteristics. Because such variation is limited in the empirical application due to parity reimbursement and cost-sharing, patient utility is reported in its normalized choice scale rather than converted to dollars. The household beliefs and burden survey provides variation in the expected difference in non-medical spending between HaH and B&M care, but coverage over the full sample is scarce. Appendix E provides an in-depth discussion of the survey-augmented demand model and its results.

Separating type updating from direct history dependence relies on a sequential-independence assumption. Conditional on the patient's time-invariant type, current observed demand covariates, and the contemporaneous hospital shock,

$$\varepsilon_i \perp\!\!\!\perp \mathcal{H}_i \mid \xi_j, X_i, P_i, \nu_i. \tag{31}$$

Thus, prior HaH outcomes can predict future acceptance by changing the posterior distribution over ξ_j or through the direct utility shift $h(\mathcal{H}_i)'\theta$, but not through additional serial dependence in the encounter-specific patient shock.

Supply-side variables excluded from patient utility shift the offer probability and help iden-

tify selection into offers. These variables include forecast B&M capacity pressure, HaH staffing exposure, direct-variable savings, assessment department, and an indicator for the partial federal-shutdown. Identification requires these variables to be conditionally independent of unobserved acceptance utility after controlling for observed patient characteristics, clinical condition, hospital-by-quarter demand effects, and prior patient history. Under this exclusion restriction and the maintained joint-normal specification, variation in these offer shifters helps identify r through changes in acceptance among selected offers. The offer intercept is a net baseline term under the stated normalizations that captures the non-pecuniary marginal cost of offering the program to an eligible assessment.

The hospital payoff coefficients are identified in offer-index units because the variance of the normal offer disturbance is normalized to one. The direct-variable savings coefficient provides a dollar normalization. Ratios to β_Π express other offer-index components in profit-equivalent units when combined with a \$1,000 scaling conversion. The duration coefficient is identified from variation in expected remaining B&M days conditional on the other payoff inputs. The hospital-specific capacity coefficients use within-hospital variation in forecast occupancy and staffed capacity interacting with that duration, separately from the duration effect itself. The patient-surplus coefficient uses variation in expected patient surplus beyond acceptance-weighted placement consequences. The interpretation is conditional on the cost, duration, and census predictions, the functional form of the capacity terms, and the included clinical and operational controls. The resulting capacity contribution is composite and does not separately identify the financial return from admitting another patient. Normalization and centering affect component levels, while changes in capacity value between two states are invariant to additive centering constants.

7 Results

7.1 Patient Acceptance

Table 2 shows that willingness to accept HaH differs persistently across patients and also varies with prior HaH interactions. The preferred specification includes time-invariant patient preferences, previously completed HaH treatment, and counts of prior acceptances and expected declinations. The estimated standard deviation of the persistent preference component is 0.232. To illustrate the magnitude of these differences, consider an assessment at which a patient with the mean persistent preference would accept an offer with probability 50%. Holding other circumstances fixed, patients one preference standard deviation below and above the mean would accept with probabilities of approximately 37% and 63%, respectively.[18]

[18]These illustrations start from zero conditional mean utility and hold hospital private information and all other utility determinants fixed. A direct utility change $\Delta\mu$ gives an acceptance probability $\Phi(\Delta\mu/\sqrt{1-\widehat{r}^2})$, where $\widehat{r}=$

Table 2: Patient Acceptance Estimates

Description	Parameter	Estimate	Standard error
Asian	$\beta_{U,\text{Asian}}$	0.128^{***}	(0.041)
Black/African American	$\beta_{U,\text{Black}}$	-0.042^{**}	(0.020)
Latino/Hispanic	$\beta_{U,\text{Latino}}$	0.149^{***}	(0.028)
Medicaid	$\beta_{U,\text{Medicaid}}$	-0.021	(0.034)
Medicare	$\beta_{U,\text{Medicare}}$	-0.073^{***}	(0.027)
Home-to-hospital distance	$\beta_{U,\text{distance}}$	0.007	(0.007)
Age	$\beta_{U,\text{age}}$	0.050^{***}	(0.009)
Male	$\beta_{U,\text{male}}$	0.014	(0.012)
Neighborhood average household size	$\beta_{U,\text{household}}$	-0.001	(0.008)
Neighborhood college education share	$\beta_{U,\text{college}}$	0.019^{*}	(0.010)
Neighborhood log(per-capita income)	$\beta_{U,\text{income}}$	-0.015	(0.011)
Neighborhood share below 150% FPL	$\beta_{U,\text{poverty}}$	-0.019^{**}	(0.008)
Condition severity weight	$\beta_{U,\text{severity}}$	-0.034^{**}	(0.015)
Prior expected HaH declinations	θ_{decline}	-0.152^{***}	(0.032)
Ever previously treated with HaH	θ_{treated}	0.151^{***}	(0.034)
Count of prior HaH acceptances	θ_{accept}	-0.050^{*}	(0.030)
Encounter-shock correlation	r	-0.723^{***}	(0.032)
Time-invariant HaH preference SD	σ_{ξ}	0.232^{***}	(0.074)
Condition fixed effects		Yes	
Hospital-by-quarter fixed effects		Yes	
Assessments		31,944	
Patients		21,354	

Notes: Descriptive subscripts identify components of the patient-characteristic coefficient vector β_U and history coefficient vector θ. Age, condition weight, home-to-hospital distance, neighborhood household size, college share, log per-capita income, and neighborhood poverty share use fixed reference means and standard deviations. Prior acceptances and expected declinations are raw cumulative counts. Previously treated with HaH indicates a distinct prior HaH-treated encounter from this specific health system. Indicators are unstandardized. Omitted categories are White for race and Commercial for payor. The specification also includes condition and hospital-by-quarter fixed effects, and "other" race and payor categories. Standard errors are conditional on the generated inputs and use the observed information matrix. The preference-SD standard error uses the delta method. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Prior treatment and declinations predict subsequent choices beyond these persistent preference differences. At the same illustrative reference, one previously accepted and completed HaH episode corresponds to an acceptance probability of approximately 56%, compared with 50% without prior treatment. This comparison combines the prior-treatment indicator and acceptance count,

-0.723. The history comparisons isolate direct utility terms without updating preference beliefs or hospital offer selection. These are conditional illustrations rather than sample-average probability effects.

giving a direct utility difference of $0.151 - 0.050 = 0.101$. Using the same reference, one additional expected prior declination corresponds to an acceptance probability of approximately 41%. The coefficient on additional prior acceptances is negative (-0.050) and significant at the 10% level, providing no evidence of progressively greater willingness after each acceptance. Because predicted acceptance enters the hospital's offer decision, I check whether the hospital estimates are robust to alternative models of patient preferences and history. Appendix Tables D.1 and D.2 show that the main offer estimates remain broadly similar across alternative history specifications and when normal preference heterogeneity is replaced by a discrete type distribution.

These findings distinguish two margins relevant to expansion. Time-invariant preference differences create scope for more effective targeting toward patients already willing to receive home care. The additional association with first treatment suggests that willingness may also change after experience, although the estimates do not establish a causal learning effect. Better information about existing preferences and interventions that change willingness therefore represent distinct approaches to expanding participation. The information counterfactuals I conduct below examine this targeting channel in more detail.

Holding other modeled factors fixed, willingness to accept increases with age and is higher among Asian and Latino patients than among White patients. It is lower among Black patients than among White patients and among Medicare patients than among commercially insured patients. Greater clinical severity and neighborhood poverty are also associated with lower willingness. These patterns may reflect differences in household support, housing conditions, or preferences for home care, but the baseline model does not separately identify these channels. Appendix E provides supplementary evidence from patient and caregiver responses.

7.2 Hospital Offers

Table 3 shows which factors make HaH offers more or less attractive to hospitals. Higher financial returns, patient benefits, and B&M capacity relief favor offers, while greater clinical needs, escalation risk, and HaH program strain discourage them. Coefficient magnitudes depend on variable units and, for placement consequences, the hospital's acceptance belief.

Financial returns and patient surplus. Higher incremental accounting profit increases offer value ($\widehat{\beta}_{\Pi} = 0.057$ per \$1,000). Hospitals also value providing patients with access to home treatment ($\widehat{\kappa}_{Q} = 0.876$). To compare the financial and patient-benefit motives in common units, consider an acceptance probability of 47.7%, matching the model-implied average among offers. Holding this probability and other payoff inputs fixed, a 0.10-unit increase in expected patient surplus has the same effect on offer value as approximately \$3,220 in additional profit from an

Table 3: Hospital Offer Estimates

Description	Parameter	Estimate	Standard error
HaH–B&M profit difference (per $1,000)	β_{Π}	0.057^{***}	(0.013)
HaH escalation risk	β_H	-0.940^{***}	(0.317)
B&M stay-duration exposure	β_E	-0.326^{***}	(0.051)
Hospital A capacity relief	$\lambda_{B,A}$	0.145^{***}	(0.025)
Hospital B capacity relief	$\lambda_{B,B}$	0.109^{***}	(0.026)
Hospital C capacity relief	$\lambda_{B,C}$	0.123^{***}	(0.026)
Hospital D capacity relief	$\lambda_{B,D}$	0.047^{*}	(0.027)
HaH program strain	λ_H	-0.004^{***}	(0.001)
Care burden	$\kappa_{S,\text{care}}$	-0.098^{***}	(0.007)
Acute severity	$\kappa_{S,\text{acute}}$	-0.043^{***}	(0.006)
Strictly prior expected offers	κ_H	0.170^{***}	(0.015)
Patient option value	κ_Q	0.876^{***}	(0.055)
Baseline net offer term	κ_O	-0.928^{***}	(0.043)
Supply-condition fixed effects		Yes	
Hospital fixed effects		Yes	
Department fixed effects		Yes	
Assessments		31,944	

Notes: All reported coefficients enter the offer equation additively. The parameters $\kappa_{S,\text{care}}$ and $\kappa_{S,\text{acute}}$ denote the care-burden and acute-severity components of κ_S, respectively. Acceptance-contingent variables are multiplied by $\rho_i^H(v)$. The HaH–B&M profit difference is centered within hospital-quarter-condition cells and expressed in $1,000 units. B&M relief, B&M stay-duration exposure, HaH strain, and escalation risk subtract fixed global reference means without rescaling. B&M relief and HaH strain use forecast census weighted by discounted remaining exposure, holding staffed capacity and HaH staffing fixed at assessment. Care burden and acute severity are standardized indices with zero-imputed underlying standardized scores. Prior expected offers are raw cumulative counts. The specification includes supply-condition, hospital, and department fixed effects as well as indicators for missing clinical measures. Standard errors are conditional on the generated inputs and use the observed information matrix. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

accepted placement.[19]

This comparison expresses the hospital's valuation of patient benefits; it does not convert patient utility into willingness to pay. The positive surplus coefficient indicates targeting toward patients who value the HaH option more, beyond how acceptance beliefs weight the financial, clinical, and capacity consequences of placement. This is consistent with a mission-related ben-

[19]The reference belief equals the model-implied aggregate acceptance rate among offered patients. An additional $1,000 of placement profit increases the offer index by $0.477(0.057) \approx 0.027$, while a 0.10-unit increase in expected patient surplus, measured in normalized utility units, increases it by $0.10(0.876) \approx 0.088$. These comparisons hold acceptance beliefs and other payoff inputs fixed. Index changes are in normalized hospital payoffs, rather than percentage-point changes in offer probabilities.

efit from providing valued treatment options. The positive profit coefficient describes a marginal financial incentive and does not imply that HaH has positive absolute accounting returns or lower treatment costs.

Clinical concerns and HaH program strain. Hospitals are less inclined to offer HaH to patients with greater care needs or a higher predicted risk of requiring transfer back to B&M care. Care burden and acute severity enter directly as standardized indices, while escalation risk is weighted by the hospital's predicted probability of acceptance. Holding other inputs fixed, offsetting a ten-percentage-point increase in escalation risk would require approximately $1,650 in additional profit from an accepted placement.

Hospitals are also less inclined to offer placements that generate greater predicted HaH program strain, which combines forecast census, available staffing, and expected remaining HaH duration. Offsetting an increase in this measure from its 25th to its 75th percentile would require approximately $1,100 in additional profit from an accepted placement, holding other inputs fixed.[20] These estimates describe incentives over the observed range of program operations; substantial expansion may change staffing requirements and other constraints.

Prior interactions and offering frictions. Patients with more prior expected offers are more likely to receive another offer, consistent with repeated outreach, familiarity with patients and caregivers, and persistent targeting. The negative baseline net offer term is consistent with offering frictions at the reference values of the regressors. These may include staff time spent preparing offers, contacting patients and caregivers, and coordinating potential transfers. This preparatory effort begins before acceptance is known.

Capacity pressure and remaining duration. The hospital-specific B&M capacity-relief coefficients are positive and statistically significant at the 10% level or better at all four hospitals. Holding other inputs fixed, higher forecast occupancy therefore strengthens the incentive to release a bed. The negative remaining-duration coefficient ($\widehat{\beta}_E = -0.326$) captures an association between longer expected stays and lower offer value, holding the other measured payoff components fixed. The positive capacity coefficients capture the offsetting benefit of releasing those bed-days, which grows with forecast occupancy. The following section expresses this capacity benefit in profit-equivalent dollars.

[20] At the reference acceptance belief, the escalation comparison changes the offer index by $0.477(-0.940)(0.10) \approx -0.045$. The strain comparison uses the uncentered forecast-weighted measure, increasing from 26.7 to 41.7, and its unrounded coefficient of approximately -0.0042. Its index effect is $0.477(-0.0042)(41.7-26.7) \approx -0.030$. The additional placement profit needed to offset each change equals its absolute index effect divided by $0.477(0.057)$ and multiplied by $1,000.

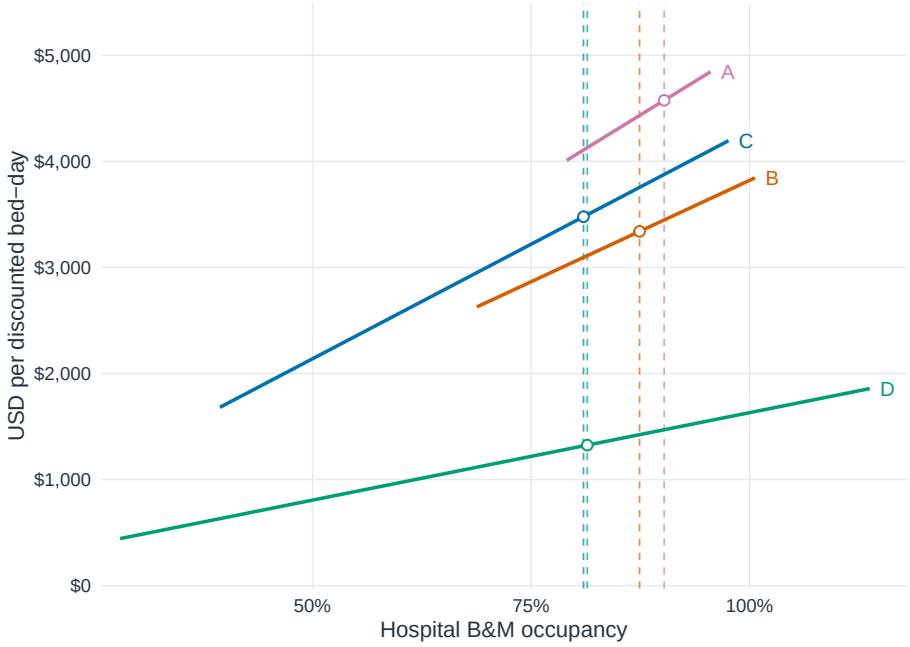
7.3 Profit-Equivalent Capacity Value

Recovering the firm's weight on accounting profits permits its valuation of capacity relief to be expressed in profit-equivalent dollars. The resulting measure is the additional accounting profit that would generate the same offer incentive as the released capacity. For assessment i, the model-implied gross private value of B&M capacity released by an accepted HaH placement is

$$\widehat{CV}_i^B = \frac{1{,}000}{\widehat{\beta}_\Pi} \widehat{\lambda}_{B,b(i)} E_i^B \frac{2\overline{N}_i^B - 1}{K_i}. \tag{32}$$

This measures the hospital's value of having the patient's bed available for another admission or operating slack over the expected remaining stay. Because both capacity relief and accounting profits enter conditional on acceptance, their common acceptance multiplier cancels in the coefficient ratio. Multiplying $\widehat{CV}_i^B$ by ρ_i gives the expected capacity value per offer.

Figure 8: Capacity Value per Discounted Bed-Day



Notes: Lines show estimated private B&M capacity value per discounted bed-day across each hospital's displayed occupancy range, holding staffed capacity at its median among estimation-sample assessments. Matching dashed vertical lines denote hospital-specific median operational occupancy.

The point estimates imply positive capacity value at all four hospitals over observed operating conditions, with all four relief coefficients statistically distinguishable from zero. Dividing equation (32) by E_i^B gives the implied value per discounted bed-day released. Figure 8 plots this value against B&M occupancy for each hospital, holding staffed capacity at its hospital-specific me-

dian.[21] Evaluating this capacity schedule at each hospital's median operational occupancy yields illustrative values of approximately $4,660, $3,410, $3,540, and $1,450 per discounted bed-day at hospitals A through D, respectively.[22] Holding staffing and other inputs fixed, a ten-percentage-point increase in exposure-weighted forecast occupancy raises this value per discounted bed-day by approximately $507, $382, $432, and $165, respectively.

7.4 Model Fit

Table 4 shows that the model closely reproduces aggregate outcome rates. It predicts accepted HaH placements in 10.2% of assessments, compared with 10.3% observed, and matches each of the three observed outcomes within 0.3 percentage points. The predicted true-offer rate is 21.4%, approximately 0.7 percentage points below the separately reconstructed, documentation-adjusted reference of 22.1%.

Figure D.3 and Figure D.4 show that the model also captures broad differences in offer rates across hospitals, occupancy levels, and calendar time. Figure D.5 shows that predicted outcome probabilities generally match observed frequencies across groups of assessments. Some localized departures remain, particularly at distributional extremes and during the early program period. All comparisons are in sample, with patient histories based on strictly prior outcomes. Table D.3 reports outcome log loss and accepted-placement Brier scores for the baseline and alternative specifications, both overall and for selected patient-history subgroups.

Table 4: Observed and Predicted Outcome Rates

Outcome	Empirical (%)	Predicted (%)	Difference (pp)
Accepted HaH placement	10.28	10.22	−0.06
Documented rejection	9.03	8.81	−0.22
No observed offer	80.69	80.96	+0.27
True HaH offer	22.13	21.41	−0.72

Notes: Empirical rates are observed frequencies for the first three rows and a separately reconstructed, documentation-adjusted offer rate for the fourth (Appendix C.1). Predicted rates average the corresponding model probabilities. Differences are predicted minus empirical rates, expressed in percentage points.

[21] Appendix Figure D.2 complements these schedules by showing the occupancy distribution of realized HaH placements weighted by estimated private capacity value per discounted bed-day. For example, at Hospital A, days above median occupancy account for 59.7% of these weighted placements, days in the top quartile account for 31.4%, and the highest occupancy decile accounts for 16.0%.

[22] This calculation evaluates the capacity schedule at hospital-specific median operational occupancy among estimation-sample assessments: 91.8%, 89.3%, 82.4%, and 89.1% at hospitals A–D, respectively. Staffed capacity is set to the corresponding hospital-specific sample median: 632, 500, 108, and 45 beds.

8 Counterfactual Analysis

The counterfactual analysis distinguishes three potential limits on HaH deployment: financial incentives to offer home treatment, information about patients' willingness to accept, and staffing pressure within the HaH program. Reimbursement experiments address payment design and who gains access, while a matched-offer information comparison isolates allocation at a fixed offer volume. Supplementary payment and staffing experiments appear in Appendix D.3. I hold the estimated demand and supply parameters fixed, modify the affected patient or hospital inputs, and evaluate the offer policy in the new environment. Except in the information comparison described below, offer and joint offer–acceptance probabilities are computed from the conditional beliefs and offer-region integrals in equations (23) and (25).

Outcome accounting. Let $\mathcal{A}$ denote the set of assessment opportunities being evaluated. The mean true-offer rate is $|\mathcal{A}|^{-1}\sum_{i\in\mathcal{A}} p_i$, expected accepted placements are $\sum_{i\in\mathcal{A}} p_i\rho_i$, and acceptance among offers is $\sum_i p_i\rho_i/\sum_i p_i$. The latter weights acceptance by the probability that each patient receives an offer. Let T_i^B denote remaining B&M duration at assessment. Mean potential B&M bed-days released per assessment are

$$\frac{1}{|\mathcal{A}|}\sum_{i\in\mathcal{A}} p_i\rho_i\mathbb{E}(T_i^B \mid \mathcal{I}_i^O). \tag{33}$$

These are undiscounted days of focal B&M occupancy avoided under the model. They are not additional admissions or a prediction that beds remain empty. HaH exposure is calculated analogously using the remaining HaH duration.

Patient option value. Using the expected patient surplus $Q_i^H(v)$ defined in equation (24), expected option value created by access is

$$\mathcal{V}_i = \int_{\mathcal{O}_i} Q_i^H(v)\phi(v)\,dv, \tag{34}$$

where $\mathcal{O}_i$ is the offer region in the counterfactual environment. Aggregate expected patient option value is $\sum_{i\in\mathcal{A}}\mathcal{V}_i$. No further acceptance multiplier is needed because $Q_i^H(v)$ already accounts for the option to decline. I report this normalized option value separately from hospital returns and bed-days, with its welfare interpretation conditional on the maintained demand specification and information structure. All reference means and standard deviations remain fixed at their estimation values. Counterfactual profit, capacity, duration, and strain inputs are transformed using those same references rather than re-centered after the shock. A change in supply can affect acceptance

among offers by changing both the composition of offered patients and the hospital-payoff shock values at which offers are made. With demand inputs and pre-assessment preference weights held fixed, these changes reflect selection under the estimated correlated-shock distribution.

8.1 Hospital-at-Home Reimbursement

Whether HaH will continue to be reimbursed at parity with B&M care is a central issue for its future financing. Medicare Payment Advisory Commission (2024) questions whether equal payments are appropriate when treatment costs may differ across settings and notes that parity provides no mechanism for Medicare to share in potential savings from home treatment. However, reducing reimbursement can also discourage hospitals from offering HaH, limiting access to a treatment option that many patients value and reducing potential capacity relief.[23] The first experiment therefore scales HaH reimbursement from 90% to 110% of the admission-specific payment received by the hospital, holding B&M reimbursement fixed.

Under baseline payment parity, let R_i denote encounter i's observed total-episode reimbursement under either treatment location. A payment multiplier $m_P \in [0.90, 1.10]$ changes reimbursement for an accepted HaH placement to $m_P R_i$. The counterfactual profit input is therefore

$$\Delta\Pi_i(m_P) = \Delta\Pi_i + \frac{(m_P - 1)R_i}{1{,}000}, \tag{35}$$

where $m_P = 1$ reproduces the baseline and the estimation centering reference remains fixed. The adjustment applies to encounters eligible for HaH reimbursement at admission. This experiment changes provider returns without directly changing patient prices or other observed demand characteristics.

Table 5 shows that reimbursement changes affect both potential capacity release and patient benefits from access to HaH. When the payment change applies across eligible payors, a 10% reimbursement reduction decreases potential B&M bed-days released by 1,778, or 12.5%, and lowers aggregate expected patient option value by 13.1%. A 5% reduction decreases these outcomes by 957 bed-days and 7.0%, respectively. Conversely, a 5% reimbursement increase generates 1,181 additional potential bed-days and raises patient option value by 8.7%; a 10% increase generates 2,640 additional days and raises option value by 19.6%. These responses reflect changes in hospital offers and selection into treatment, holding patient prices and modeled treatment preferences fixed.[24] Higher reimbursement expands access to home treatment while preserving patients' option

[23] In a related setting, Gruber et al. (2023) find that England's four-hour emergency-department target reduced time in the ED and mortality while increasing inpatient admissions. Their findings illustrate how provider incentives can affect both patient flow and health outcomes, sometimes in surprising ways.

[24] The simulations recompute offer and joint offer–acceptance probabilities under the estimated correlated-shock

to remain in B&M. The increase in expected patient option value captures the benefit of expanded treatment choice at unchanged patient prices.

Table 5: Reimbursement Changes, Released Capacity, and Hospital Contribution

HaH reimbursement change	−10%	−5%	+5%	+10%
Additional B&M bed-days	−1,778	−957	1,181	2,640
Hospital contribution ($ millions)				
(1) Focal contribution: (1.1)+(1.2)+(1.3)	−0.24	−0.17	0.46	1.56
(1.1) Induced transfers treatment cost effect	3.08	1.65	−2.01	−4.42
(1.2) Reimbursement: baseline placements	−4.10	−2.05	2.05	4.10
(1.3) Reimbursement: induced placements	0.77	0.22	0.42	1.89
(2) Potential backfill contribution	[−3.07, −1.14]	[−1.66, −0.62]	[0.77, 2.07]	[1.72, 4.66]
(3) Combined cash contribution: (1)+(2)	[−3.30, −1.38]	[−1.83, −0.79]	[1.22, 2.53]	[3.28, 6.23]
Structural profit equivalents ($ millions)				
Value of B&M occupancy relief	−6.06	−3.27	4.08	9.13
Patient option value (firm valuation)	−4.94	−2.65	3.30	7.41
Patient option value				
Change relative to baseline (%)	−13.06	−7.02	8.72	19.60

Notes: All changes are relative to baseline reimbursement over the full assessment cohort, in February 2026 U.S. dollars ($). Reimbursement changes apply to eligible HaH episodes across all payors, with B&M payment unchanged. Estimated parameters, factual histories, pre-assessment preference weights, staffing, census, and forecast occupancy exposures are held fixed. Backfill ranges combine 50–100% fill with revenue including or excluding estimated settlements and direct or all reported variable costs. Combined cash contribution adds focal and potential backfill contribution within each accounting scenario. Components and totals are calculated before rounding. Structural occupancy-relief values use forecast-weighted capacity inputs and the model's $\delta = 0.99$ discount factor. Patient option value integrates expected patient surplus over the offer region, with no additional acceptance multiplier. Its percentage change is measured relative to aggregate baseline option value. The firm's valuation converts the change in option-value levels into profit-equivalent dollars using $1{,}000\widehat{\kappa}_Q/\widehat{\beta}_\Pi$ and does not measure patient willingness to pay. Both structural valuations are distinct from cash revenue and are reported separately, rather than added to combined cash contribution.

To value the additional throughput that released capacity could support, I apply hospital-specific contribution margins from the observed backfill population. For hospital b, let $\Delta BD_b(m_P)$ denote additional bed-days released, f_b the fraction filled, and c_b contribution dollars per occupied

distribution, holding parameters, factual histories, pre-assessment preference weights, staffing, census, and forecast occupancy exposures fixed. These conditional assessment-level totals measure potential capacity release. The exercise does not propagate policy-induced changes into subsequent histories, occupancy forecasts, or admissions. All counterfactual results are point estimates conditional on the estimated model and generated inputs.

day.[25] Potential additional hospital contribution is

$$\Delta V^{\$}(m_P) = \sum_{b \in \mathcal{B}} \Delta BD_b(m_P) f_b c_b. \tag{36}$$

Under sustained excess demand, I consider scenarios in which 50–100% of released bed-days are filled, allowing for delays, staffing constraints, and bed compatibility. A 10% reimbursement increase generates $1.7–$4.7 million in potential additional backfill contribution across fill rates and accounting definitions. Additional HaH payments total approximately $6.0 million, including $4.1 million for baseline placements, while induced transfers increase remaining direct-variable treatment costs by approximately $4.4 million. The resulting $1.6 million increase in focal contribution leaves a combined hospital cash gain of $3.3–$6.2 million. Appendix D.3 reports sensitivity to backfill assumptions.

Table 5 separately reports $9.1 million in additional gross private capacity value. This structural component measures the hospital's estimated value of having released capacity available, conditional on the other modeled determinants of offers. The cash calculation instead applies observed contribution margins to assumed backfill.

Occupancy-contingent reimbursement. The uniform experiment changes the return to HaH at every occupancy level. I next compare supplements at high occupancy, where capacity relief may be particularly valuable, with supplements at lower occupancy, when the hospital's private capacity-relief incentives may be weaker. The latter may expand access to an alternative treatment modality that patients value even when contemporaneous congestion relief is less valuable. I restrict this exercise to Medicare fee-for-service encounters (excluding Medicare Advantage), given Medicare's central role in shaping HaH reimbursement and the scope for implementing a coordinated payment policy through a single public payor.

Separate policies target the hospital-specific bottom or top quartile of reference daily occupancy, or occupancy below 80% or above 90% of staffed capacity. Each changes reimbursement only in its targeted states, retaining parity otherwise. Quartiles identify relatively low and high utilization within each hospital and need not represent the same absolute congestion level. Table 6 reports 10% supplements, while Appendix D.3 provides additional details on policy support, the full rate grid, and alternative thresholds.

Low-occupancy supplements generate additional placements and patient option value. Patient option value per $1 million of additional payments varies modestly across policies. Private capac-

[25] The financial value of shorter inpatient stays depends on treatment costs, reimbursement, and the additional throughput released capacity can support (Gregory et al., 2003). Related HaH calculations value potential backfill from diverting post-operative readmissions, excluding HaH treatment costs (Foley et al., 2021).

ity value per payment dollar varies more across the selected patients and operating states: targeting above 90% generates $1.68 per supplemental dollar, compared with $1.00 below 80%, while patient option-value gains are approximately 85 and 82 units per $1 million, respectively. Low-state subsidies do not deliver larger option-value gains per additional placement or payment dollar than the corresponding high-occupancy rules.

Table 6: Medicare FFS Supplements Across Occupancy States

	Uniform	Bottom quartile	Top quartile	$< 80\%$	$> 90\%$
Panel A: Changes from parity under a 10% supplement					
Expected HaH placements	251.6	39.9	122.5	31.7	130.0
Potential B&M bed-days	1,198	183	603	128	656
Patient option value	226.7	33.6	114.1	27.5	121.0
Admission payments ($ millions)	2.70	0.42	1.38	0.33	1.43
Focal cash contribution ($ millions)	0.73	0.11	0.41	0.08	0.40
Panel B: Incremental benefits and payment incidence					
Option value per additional placement	0.90	0.84	0.93	0.87	0.93
Option value per $1 million of payments	83.8	80.5	82.9	82.3	84.9
Private capacity value per $1 of payments	1.53	1.41	1.57	1.00	1.68
Payments on baseline placements (%)	72.8	73.0	70.4	83.5	70.1

Notes: Changes are relative to parity over the full assessment cohort. Each policy pays a 10% Medicare FFS supplement in the stated occupancy states and parity otherwise. Quartiles use fixed hospital-specific reference daily occupancy. Admission payments include supplements on baseline and induced placements. Panel B divides aggregate incremental benefits by additional placements or payments. Patient option value is normalized utility. Private capacity value is the structural profit-equivalent component, distinct from cash receipts and social benefits.

Most supplemental payments accrue to placements already made under parity. Increasing the uniform supplement from 5% to 10% generates less option value per additional payment dollar than increasing it from parity to 5%, while option value per induced placement rises slightly and treatment cost per induced placement changes little. Appendix D.3 examines these distinctions.

Who gains or loses access? Across the reported Medicare FFS payment scenarios, the measured characteristics of patients whose access changes differ modestly. Additional top-quartile recipients have 0.32 more expected B&M days remaining and 1.03 percentage points higher predicted return-transfer risk than additional bottom-quartile recipients. Nursing burden, neighborhood characteristics, and prior HaH experience differ relatively little. Uniform rate changes also affect access unevenly across patient A 10% Medicare FFS supplement generates 3.74 additional placements per 100 assessed opportunities among patients with prior HaH treatment, compared

with 1.54 among those without prior treatment, consistent with previously treated patients being more likely to receive and accept HaH offers. Rate cuts likewise produce larger losses for this group. Rate cuts produce larger losses for previously treated patients as well. Gains are slightly larger in higher-income and lower-poverty neighborhoods and smaller in the highest predicted escalation-risk quartile, while differing little between the highest and lowest remaining-stay quartiles. Appendix D.3 reports the recipient profiles and subgroup access changes in greater detail.

Table 7: Patient-Preference Information at a Matched Offer Volume

	Population preference weights	Current	Perfect persistent preference information
Expected offers	6,841	6,841	6,841
Expected accepted placements	3,258	3,266	3,486
Potential B&M bed-days released	14,238	14,285	15,201
Acceptance among offers (%)	47.6	47.7	51.0
Patient option value: change (%)	−0.6	0.0	16.7

Notes: Outcomes use the same assessment opportunities and common factual predictive weights. Expected offers are matched to the current-information baseline within 72 hospital-quarter cells. All regimes retain observed characteristics, direct history effects, and the hospital's private payoff shock; perfect information additionally reveals the persistent preference type. The option-value row reports the percentage change from current information in normalized expected patient surplus.

8.2 Information About Patient Demand

This information experiment compares pooled, current, and perfect knowledge of time-invariant patient preferences. The pooled benchmark uses population preference weights; current information uses patient-specific predictive weights. All regimes retain observed characteristics, direct history effects, and the hospital's private payoff shock ν_i. Perfect information additionally reveals the persistent preference type, but not the patient's encounter-specific choice shock. To distinguish allocation from expansion, I hold expected true offers fixed at the baseline level within each hospital-quarter cell. Appendix D.3 describes the information weights, common evaluation distribution, and offer matching.

Table 7 shows scope to improve access through better targeting of existing offers. Replacing history-based preference weights with population weights changes outcomes little. Perfect knowledge of time-invariant preferences increases accepted placements by approximately 6.8% and potential B&M bed-days released by 6.4% at the same expected offer volume, in turn raising patient option value by 16.7%. These results benchmark the potential gains from learning about preferences beyond observed histories, abstracting from information-acquisition costs.

9 Conclusion

Hospital-at-Home illustrates how consumer participation can shape effective supply when production is capacity constrained. Hospitals decide which patients receive an offer, while patient choices determine whether that offer translates into home treatment and released B&M capacity. This co-production of care creates scope for capital avoidance by allowing hospitals to accommodate fluctuations in inpatient demand without permanent B&M expansion. Realizing this flexibility depends on the alignment between hospital incentives and patients' willingness to receive care in alternative settings.

The model estimates show that hospitals value patient access alongside financial returns, clinical needs, and capacity relief. The private value of released B&M capacity increases with occupancy and is equivalent to approximately \$3,410–\$4,660 per discounted bed-day at the two largest hospitals' median occupancy levels. This value captures the option to accommodate another patient or preserve operating slack and can encourage HaH offers even when home treatment raises treatment costs.

The reimbursement experiments show that payment policy affects patient access and potential capacity relief. Higher HaH reimbursement expands access to home treatment and increases the expected value patients receive from having this option. In the uniform reimbursement increase scenario, induced transfers raise treatment costs by more than their additional payments. Most additional payments go to baseline placements, offsetting the financial loss on induced transfers. Filling released beds through additional admissions can generate further hospital contribution. Across the Medicare fee-for-service policies examined, patient option-value gains per additional payment dollar vary relatively little across occupancy rules. High-occupancy supplements generate greater estimated private capacity value per dollar spent.

HaH makes patients' choices over care location part of the hospital's capacity decision. Its future will depend on hospitals' ability to internalize the benefits of home treatment and released capacity, while aligning offers with patients' preferences and clinical needs. Hospitals must be able to capture enough of these gains to finance the clinical teams and logistical infrastructure that home care requires. Efficient growth will require the benefits of additional home placements to justify their labor, coordination, and household resource costs. Better information can align offers with patient preferences, while staffing and care delivery choices shape the feasible scale of treatment. The public policy challenge is to align hospitals' private returns with the social value of these decisions. Payment arrangements should enable hospitals to capture gains from valuable home treatment and congestion relief spillovers to other patients while accounting for the true resource costs borne across care settings and households.

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Appendices

A Supplementary Descriptive Evidence

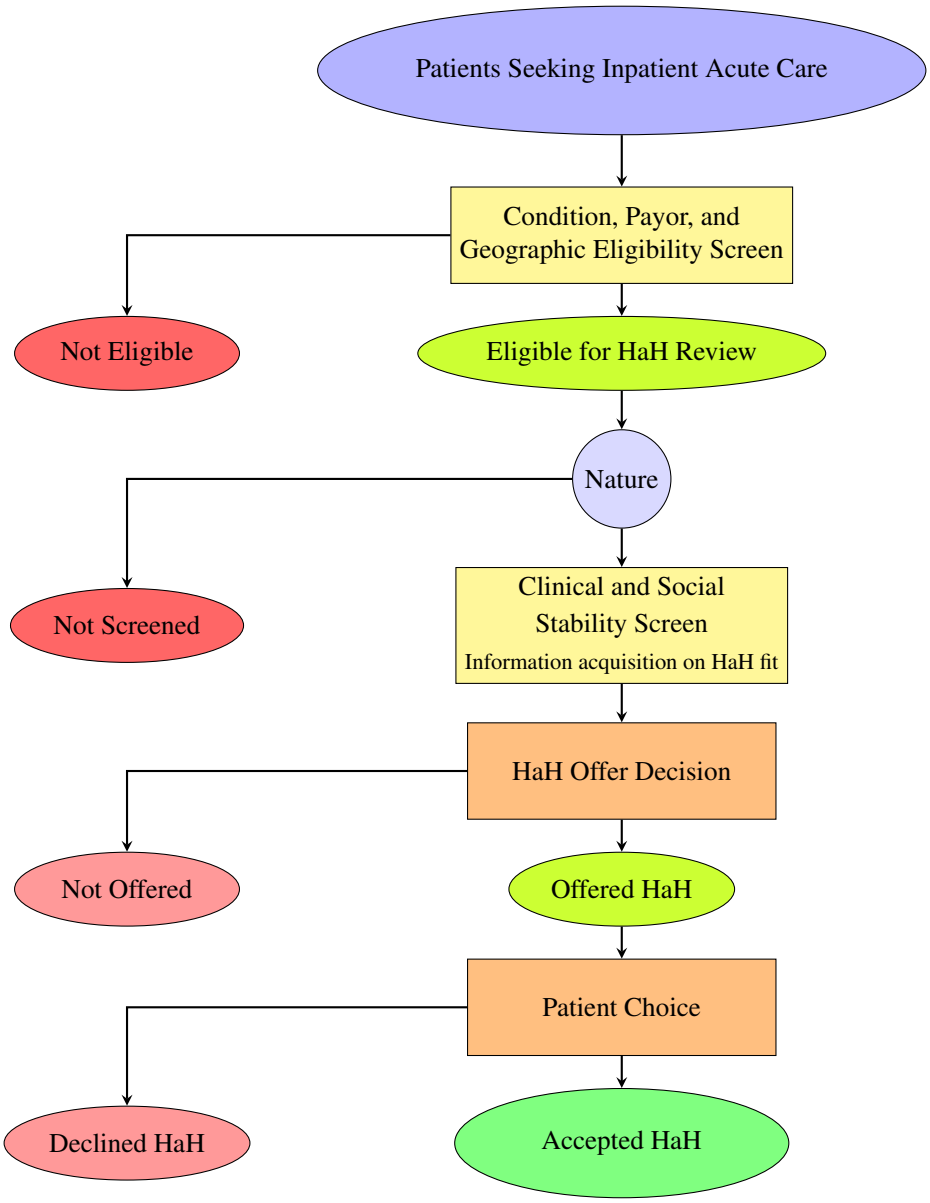
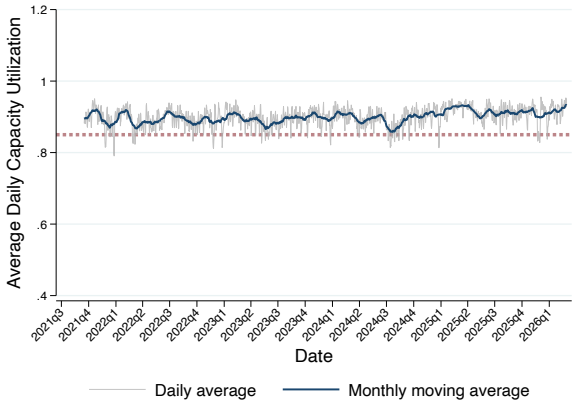


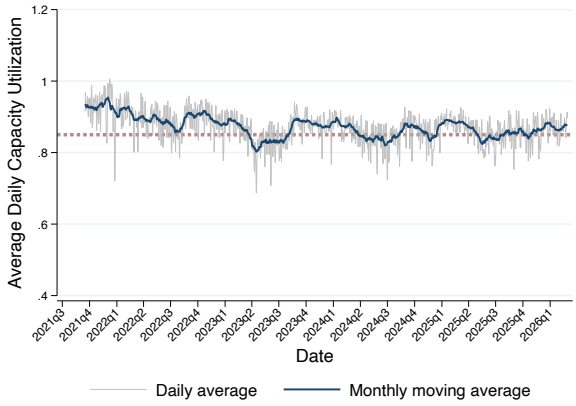
Figure A.1: Hospital-at-Home Patient Screening Process

Notes: Hospital-at-home programs allocate patients through sequential stages. First, hospitals apply condition, payor, and geographic eligibility criteria, although specific rules vary across systems and some programs are payor agnostic. Nature represents whether eligible patients are reviewed, allowing for "leakage" or operational oversight that are not due to active decisions. Clinical and social review may assess clinical stability, treatment and monitoring needs, home safety, caregiver support, and access to utilities and communication. Review findings inform the hospital's HaH offer decision. Offered patients choose whether to accept home-based care or remain in brick-and-mortar care. Dark red denotes eligibility or review exclusions, while light red denotes hospital or patient decisions.

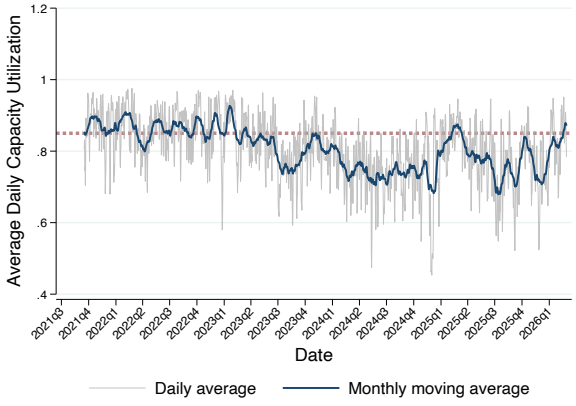
Figure A.2: Brick-and-Mortar Capacity Utilization Over Time



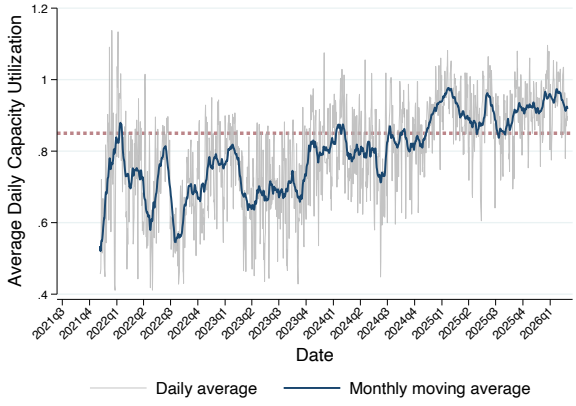
(a) Hospital A



(b) Hospital B



(c) Hospital C



(d) Hospital D

B Evidence on the Causal Health Effects of HaH

To estimate the causal effect of HaH on health outcomes, I exploit the program's geographic eligibility rule and implement a fuzzy spatial regression discontinuity design (RDD) (Lee and Lemieux, 2010; Keele and Titiunik, 2015). The complete design and analysis are developed in Capron et al. (2026). I summarize the econometric method, identifying variation, and estimates relevant to the demand and supply models here.

The program served patients residing within a fixed list of ZIP codes constructed to satisfy the 30-minute emergency-response requirement mandated by CMS. Using the operational ZIP-code list, rather than a reconstructed hospital isochrone, allows me to recover the eligibility boundary actually faced by patients. Panel (a) of Figure B.1 illustrates this boundary and the spatial distribution of the analytic sample and HaH receipt.

Let r_i denote patient i's signed distance in kilometers from the eligibility boundary, with $r_i < 0$ indicating residence inside the eligible area and $r_i \geq 0$ indicating residence outside. Geographic eligibility is therefore $E_i = \mathbb{1}\{r_i < 0\}$. Within bandwidth h around $r_i = 0$, the outcome equation follows the standard local-polynomial fuzzy-RD specification (Hahn et al., 2001; Calonico et al., 2014):[26]

$$Y_i = \alpha + \tau HaH_i + \gamma' X_i + \sum_{k=1}^{K} \left[\theta_k^- r_i^k E_i + \theta_k^+ r_i^k (1 - E_i) \right] + \varepsilon_i, \tag{B.1}$$

where $HaH_i = \mathbb{1}\{\text{treated with HaH}\}$ and X_i contains prespecified patient, admission, neighborhood, and hospital characteristics.

Because HaH receipt is endogenously determined, I instrument for HaH_i using geographic eligibility, E_i. Identification requires that potential outcomes and treatment receipt vary smoothly through the boundary absent the eligibility rule, that eligibility affects outcomes only through HaH receipt, and that eligibility weakly increases the probability of treatment. The discontinuity in the probability of receiving an HaH offer is shown in Panel (b) of Figure B.1. This offer discontinuity illustrates the access mechanism; the treatment first stage uses HaH receipt itself. The corresponding first-stage equation is

$$HaH_i = \pi_0 + \pi_1 E_i + \rho' X_i + \sum_{k=1}^{K} \left[\kappa_k^- r_i^k E_i + \kappa_k^+ r_i^k (1 - E_i) \right] + \upsilon_i. \tag{B.2}$$

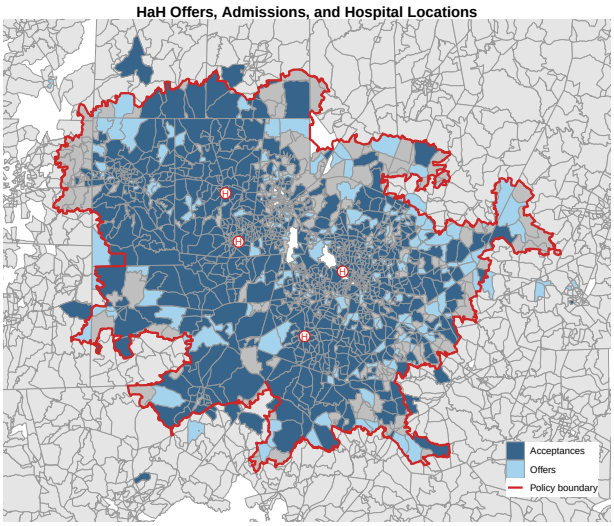
Letting ϕ_1 denote the reduced-form discontinuity in Y_i, the local average treatment effect for compliers at the boundary is (Imbens and Angrist, 1994)

$$\tau = \frac{\phi_1}{\pi_1} = \frac{\lim_{r \uparrow 0} \mathbb{E}[Y_i \mid r_i = r] - \lim_{r \downarrow 0} \mathbb{E}[Y_i \mid r_i = r]}{\lim_{r \uparrow 0} \mathbb{E}[HaH_i \mid r_i = r] - \lim_{r \downarrow 0} \mathbb{E}[HaH_i \mid r_i = r]}. \tag{B.3}$$

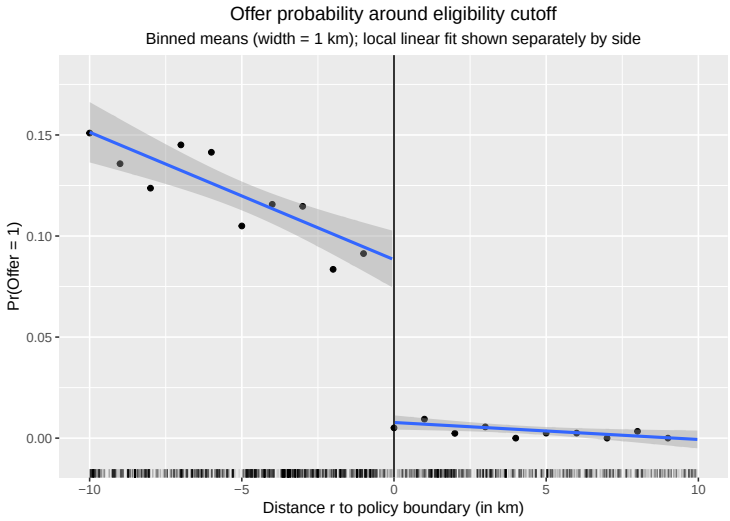
The full set of results can be found in Table 1 of Capron et al. (2026). Residence inside the eligibility boundary increased HaH receipt by approximately 4 percentage points at the discontinuity, with first-stage F-statistics ranging from 45 to 159 across studied outcomes. The estimates point to mixed effects across margins: lower mortality and fewer falls during the acute-care episode, but more subsequent inpatient readmissions. Differences in longer-run mortality, length of stay, discharge destination, and several other outcomes were not statistically distinguishable from zero. HaH was also associated with fewer clinical orders, particularly imaging and consultations, without evidence that these services were simply shifted earlier in the inpatient stay prior to transfer home. Taken together, the results suggest broadly similar health outcomes across treatment set-

[26] Flexible polynomials of order K are allowed to vary on either side of the cutoff. These polynomials are not required for identification but allow the relationship between distance and the outcome to differ across sides of the boundary.

Figure B.1: Program Treatment Boundary and Discontinuous Offer Probability



(a) Offers and Acceptances



(b) Discontinuity in HaH Offer Rate

tings, with evidence of both short-run benefits and downstream utilization costs. These local estimates motivate the baseline abstraction from direct health effects in Section 5, while escalation risk and acute-care needs remain determinants of HaH use. They do not establish equivalence across treatment settings or identify health effects for patients newly reached under expanded eligibility.

C Model Inputs and Estimation

C.1 Offer Documentation and Empirical Frequencies

Evidence from electronic health records indicates that flowsheet data miss some declined HaH offers, while accepted placements are observed throughout. I first use overlapping reporting fields to construct hospital-regime declination-capture probabilities. These probabilities are fixed inputs to both the joint demand–supply likelihood and a separate binomial model for reconstructing offers. Only the latter produces the bounded counts and rates used in the reduced-form evidence; those reconstructed outcomes are not used as inputs to the joint likelihood.

Capture of declinations across reporting regimes. New procedures for documenting declined offers were introduced over the program's tenure. Figure C.1 shows that older indicators remain in use after new fields appear. Their continued availability supports comparing capture across regimes. The adjustment assumes that capture by the older fields is transportable across adjacent regimes and that modern documentation is complete.

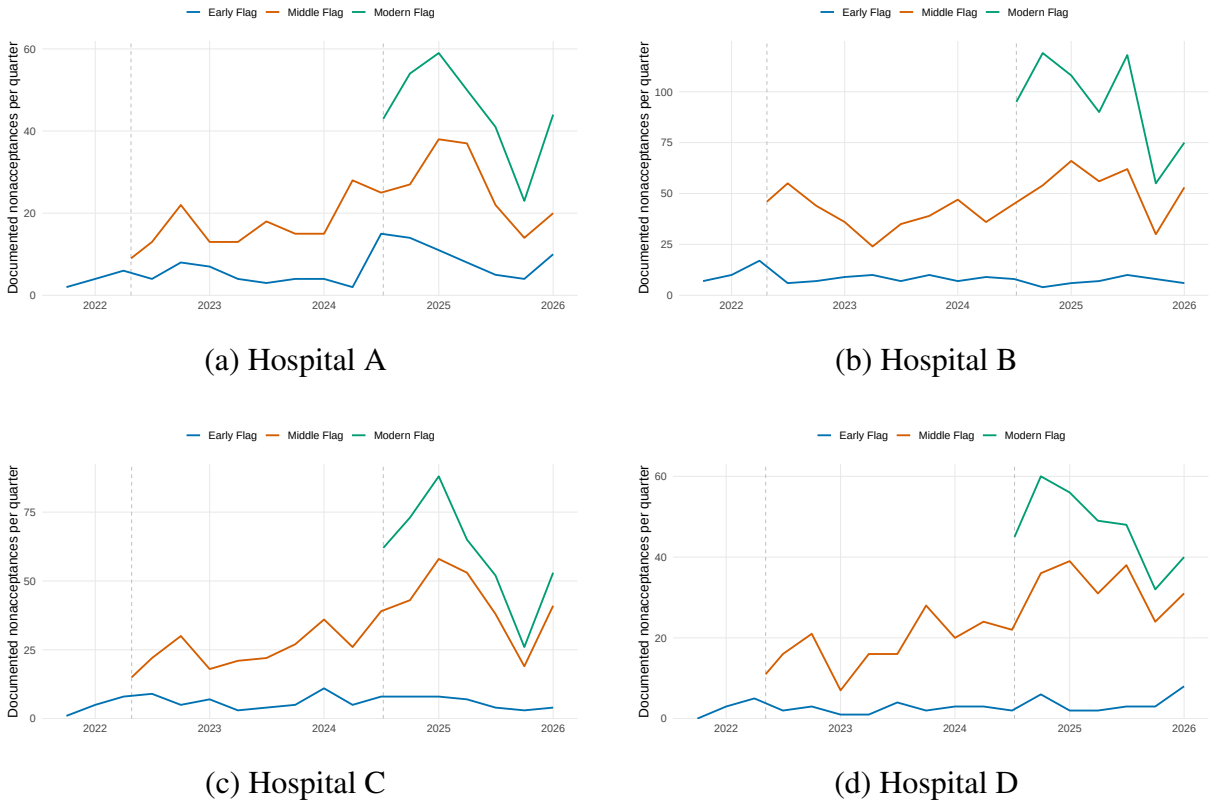
Let q_{br} be the probability that a declined offer is documented at hospital b in regime r, with $q_i = q_{b(i),r(i)}$. Modern capture is normalized to $q_{b,\text{modern}} = 1$. Within each hospital,

$$q_{b,\text{middle}} = \frac{\text{modern-regime declines with either older flag}}{\text{all modern-regime declines}},$$

$$q_{b,\text{early}} = q_{b,\text{middle}} \times \frac{\text{middle-regime declines with social-stability evidence}}{\text{all observed middle-regime declines}}.$$

Table C.1 reports the declination-capture probabilities used in joint maximum likelihood estimation. Each assessment is assigned $q_i = q_{b(i),r(i)}$ before estimation. These values are then held fixed at every likelihood evaluation and are not estimated with the demand and supply parameters. The joint model fits the observed acceptances, documented declines, and no-observed-offer outcomes using these capture probabilities. The table also reports implied overall offer capture, including acceptances, but only as a descriptive summary rather than an additional likelihood input.

Figure C.1: Offer Documentation Across Reporting Regimes



(a) Hospital A

(b) Hospital B

(c) Hospital C

(d) Hospital D

Notes: Quarterly documented non-acceptances carrying each reporting flag. An assessment may carry multiple flags. Dashed vertical lines mark reporting transitions.

Table C.1: Reporting Regimes and Capture Probabilities

	Declination capture, q_{br}			Implied overall offer capture		
Hospital	Early	Middle	Modern	Early	Middle	Modern
A	0.163	0.675	1.000	0.521	0.867	1.000
B	0.108	0.572	1.000	0.421	0.750	1.000
C	0.156	0.665	1.000	0.553	0.829	1.000
D	0.088	0.665	1.000	0.423	0.833	1.000

Notes: Overall capture is $(A+D)/(A+D/q)$ within hospital-regime, where A and D count acceptances and documented declines.

Reconstructing reduced-form offers. Directly scaling documented declines by $1/q_i$ can imply more offers than assessments for certain hospital-regime cells. I therefore fit a separate binomial model that uses the number of non-accepted assessments to limit possible hidden offers and pools information across months. Let c index hospital–month–regime cells, split at reporting transitions, with M_c total assessments, A_c acceptances, D_c documented declines, and $Z_c = M_c - A_c$ denoting total non-acceptances. If θ_c is the probability of a declined offer among non-accepted assessments,

$$D_c \mid Z_c, \theta_c \sim \text{Binomial}(Z_c, q_c\theta_c), \qquad \text{logit}(\theta_c) = \mu_{b(c),r(c)} + u_c, \quad u_c \sim N(0,\tau^2).$$

Hospital-regime means and a shared monthly variance are estimated by marginal maximum likelihood with normal quadrature, using the same fixed capture rates, $q_c = q_{b(c),r(c)}$. This empirical-Bayes fit is performed separately, outside the joint demand–supply estimator. The reconstructed offer count is

$$\widetilde{O}_c = A_c + D_c + (Z_c - D_c)\widehat{\mathbb{E}}\left[\frac{(1-q_c)\theta_c}{1-q_c\theta_c}\,\middle|\, D_c, Z_c, q_c\right].$$

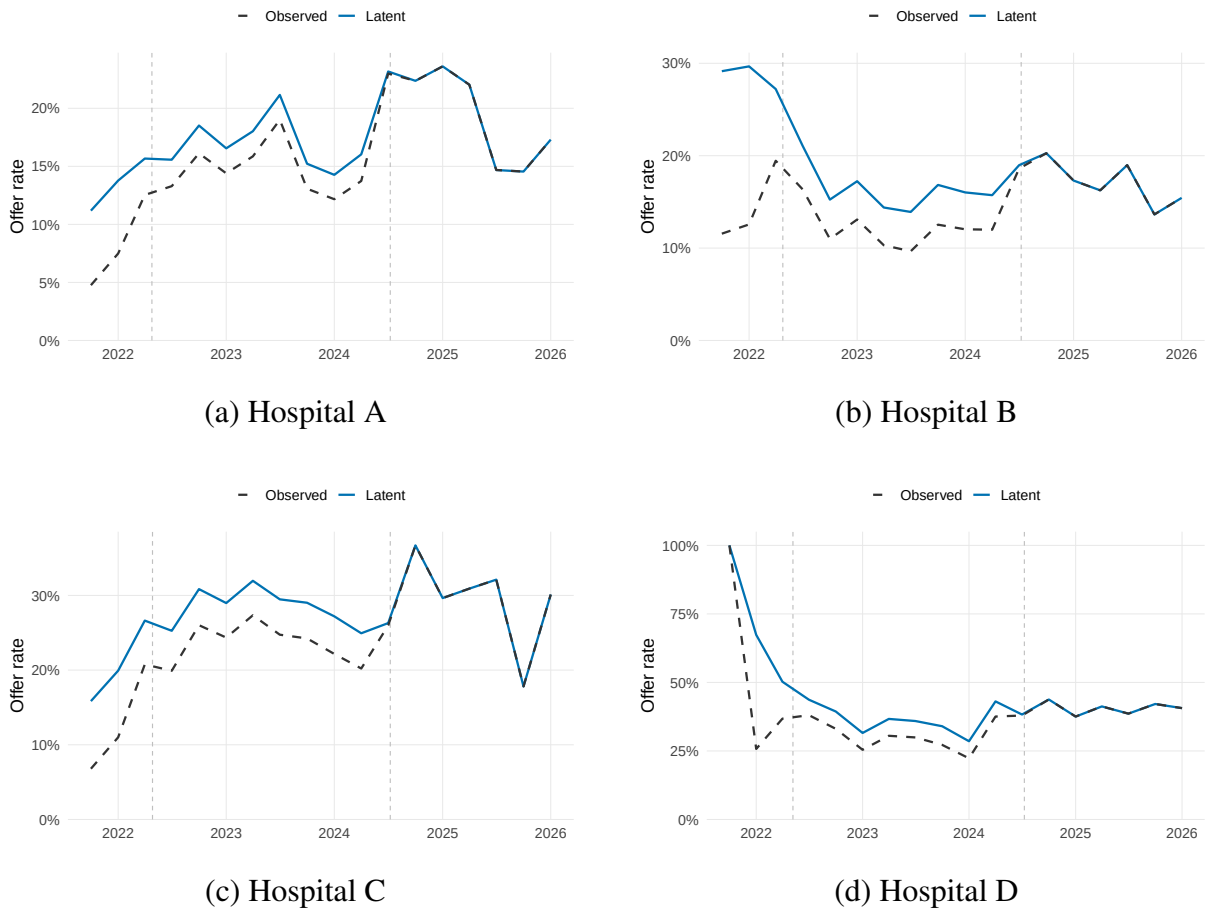
The expectation averages the hidden-offer probability over the cell posterior at the fitted hyperparameters. Thus known offers count as one, while each no-observed-offer assessment receives this posterior probability. Counts lie between observed offers and assessments; modern counts equal observed offers. Group rates divide summed reconstructed counts by summed assessments. These are the reduced-form outcomes, replacing raw inverse-capture totals in the empirical evidence presented throughout. Table C.2 summarizes the reconstruction. Partial pooling stabilizes cell counts with little change in the aggregate total. The resulting latent offers are plotted in Figure C.2.

Table C.2: Descriptive Offer Reconstruction

Assessments	31,944
Hospital–month–regime cells	211
Monthly heterogeneity SD, $\widehat{\tau}$	0.214
Observed offers	6,169
Raw inverse-capture total (benchmark)	7,071.0
Bounded reconstructed offers	7,070.7
Marginal log likelihood	−558.1
Complete-pooling log likelihood	−571.7
Early-cell held-out log-score difference	−0.052

Notes: Complete pooling sets $\tau = 0$ within hospital-regime. Log likelihoods include binomial constants. The held-out comparison leaves out one early-period cell at a time, with the negative difference indicating no predictive gain over complete pooling in those cells.

Figure C.2: Recovered Latent Offer Rates



(a) Hospital A

(b) Hospital B

(c) Hospital C

(d) Hospital D

Notes: Dashed vertical lines mark reporting transitions.

C.2 Input Transformations

Selected regressors are centered or standardized to express their effects relative to a common baseline and in interpretable units. The reference means and standard deviations used for these transformations are fixed before estimation and retained in counterfactuals, so changes reflect altered inputs rather than a changing benchmark. Census, staffing, and duration enter their physical schedules in raw units; the transformations apply to the resulting regressors. The offer-index equations suppress this transformation in the notation used in the main text.

Standardized covariates. Demand covariates standardized by a reference mean and SD are age, condition weight, home-to-hospital distance, and census block group characteristics (household size, college share, log per-capita income, and share below 150% federal poverty line). Supply inputs use the latest valid NWASS, Morse, and SOFA records strictly before the actual assessment timing. Care burden averages standardized NWASS and Morse scores and then rescales the resulting index; acute severity uses rescaled SOFA. Missing component scores contribute zero on the standardized scale, with a separate indicator for each of the three measures. No additional clinical scores enter demand, and missing scores do not remove assessments. A standardized coefficient refers to one reference SD, and dividing it by that SD gives the coefficient per original unit.

Centered payoff inputs. The profit regressor is the conditional HaH-minus-B&M dollar return, less its fixed hospital-quarter-condition reference, divided by \$1,000, with equation (C.13) providing the mapping. Hospital-specific B&M relief, B&M stay-duration exposure, HaH strain, and escalation risk subtract fixed global means without rescaling. Their coefficients retain the original units.

Acceptance, histories, and indicators. Payoffs that are conditional on acceptance use uncentered $\rho_i^H(v)$. Prior accepted placements, expected rejections, and expected true offers enter as raw cumulative counts. Categorical variables, fixed-effect indicators, flags for missing fields, and the federal shutdown indicator remain untransformed.

References and component values. For clarity here, write $M_i^{B,\text{raw}}$ for the physical capacity vector and $\overline{M}^B$ for its fixed reference. Then

$$\rho_i^H(v)\lambda_B'(M_i^{B,\text{raw}} - \overline{M}^B) = \rho_i^H(v)\lambda_B' M_i^{B,\text{raw}} - \rho_i^H(v)\lambda_B'\overline{M}^B.$$

Centered component levels therefore depend on the reference. Section 5 suppresses this distinction, and the gross capacity-value measure in equation (32) uses the uncentered input. Differences in that measure across states retain their interpretation under a common reference.

C.3 Exposure Inputs and Numerical Estimation

Remaining treatment exposure. Let T_i^o denote counterfactual remaining duration in days under location $o \in \{B,H\}$, with its distribution predicted from information $\mathcal{I}_i^O$ at assessment. Superscripts B and H refer to B&M and HaH, as in Section 5.2. Let ℓ_i^{LOS} denote elapsed LOS, distinguishing it from the likelihood contribution $\ell_{ig}(Y_i;\Theta)$. The discharge models vary with hospital, condition, location, elapsed LOS, and day of week. Under the baseline zero transfer delay, the HaH path begins at the assessment.

For duration $T \geq 0$, the discounted exposure schedule is

$$A_\delta(T) = \sum_{t=0}^{\lfloor T \rfloor - 1} \delta^t + (T - \lfloor T \rfloor)\delta^{\lfloor T \rfloor}, \qquad \delta = 0.99, \tag{C.1}$$

where an empty sum is zero. The last term prorates a fractional final day. The exposure inputs used in the supply model are

$$E_i^o = \mathbb{E}\left[A_\delta(T_i^o) \mid \mathcal{I}_i^O\right], \qquad o \in \{B,H\}. \tag{C.2}$$

Thus discounting is applied to each possible remaining duration before averaging over its predictive distribution.

To connect these inputs to discharge predictions, write $\mathcal{S}_i^o(u) = \mathbb{P}(T_i^o > u \mid \mathcal{I}_i^O)$ for remaining-stay survival. Then

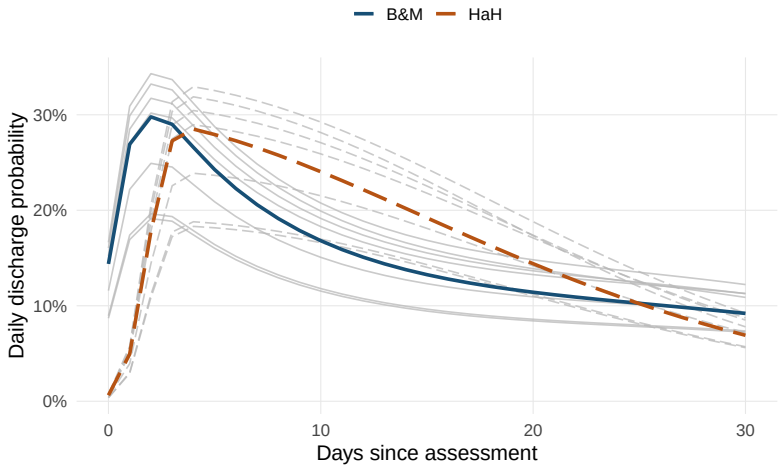
$$E_i^o = \sum_{t=0}^{\infty} \delta^t \int_t^{t+1} \mathcal{S}_i^o(u)\, du. \tag{C.3}$$

For integer-valued durations this reduces to $\sum_{t\geq 0} \delta^t \mathcal{S}_i^o(t)$. If $\mu_i^o(t)$ is the exit probability for this duration during $(t,t+1]$ conditional on remaining in care at t, survival on the daily grid satisfies

$$\mathcal{S}_i^o(t) = \prod_{v=0}^{t-1} \{1 - \mu_i^o(v)\}, \qquad \mathcal{S}_i^o(0) = 1.$$

Remaining B&M and HaH exposure therefore use their respective discharge paths. Section 5.2 multiplies these discounted days by capacity increments evaluated at exposure-weighted forecast census and the assessment's staffing state. The undiscounted bed-days in Section 8 instead use $\mathbb{E}[T_i^o \mid \mathcal{I}_i^O] = \int_0^\infty \mathcal{S}_i^o(u)\, du$.

Figure C.3: Predicted Discharge Hazards



Histories and sequential likelihood. Assessments are processed chronologically within patient. An acceptance increments prior accepted placements by one; a documented rejection increments prior rejections by one. A no-observed-offer outcome increments expected rejections by the posterior $\widehat{O}_i$ in equation (18). Expected true offers increase by $\widehat{O}_i$ for every outcome, with $\widehat{O}_i = 1$ for accepted placements and documented rejections. All increments occur after evaluating the current predictive likelihood, and all parameter-dependent histories are reconstructed at each trial vector $\Theta = (\Theta_D, \Theta_S, r)$. Using the likelihood notation of Section 6.3, the objective is

$$\mathcal{L}(\Theta) = \sum_j \sum_{i \in \mathcal{T}_j} \log L_i(\Theta), \qquad L_i(\Theta) = \sum_g w_{ig}^- \ell_{ig}(Y_i; \Theta). \tag{C.4}$$

The recursion updates node weights and histories at every retained assessment, including observations with zero direct objective weight. If all such observations precede the included history, this objective equals the integrated patient-level likelihood in equation (27). If they are interspersed, equation (C.4) specifies the selected predictive contributions with the intervening updates retained.

Gauss–Hermite quadrature uses 20 standard-normal nodes ξ_g and normalized prior weights w_g. The posterior mean of the time-invariant utility component before assessment i is $\sigma_\xi \sum_g w_{ig}^- \xi_g$; the mean of the standardized latent type is $\sum_g w_{ig}^- \xi_g$.

Optimization and inference. I maximize the joint log likelihood using a quasi-Newton algorithm with analytic gradients, locating all zero crossings of $\Gamma_i(v) + v$ and integrating over the resulting offer intervals. Derivatives account for the conditional beliefs, offer-region boundaries, and sequential updates to patient histories and preference weights. Parameters are internally rescaled

for numerical stability and returned to their original units for reporting. Standard errors use the observed information matrix and the delta method where needed.

C.4 Treatment Cost Model

The profit input in equation (6) uses predicted direct-variable costs remaining at assessment. Its construction combines a common LOS schedule, a HaH transfer increment, and a HaH-day cost difference. These are conditional accounting predictions rather than causal treatment effects.

Episode-level cost specification. For the episode associated with assessment i, let C_i^{ep} denote total direct treatment cost, L_i^{ep} total LOS, and h_i^{ep} HaH days. Let $D_i^H = \mathbb{1}\{h_i^{\text{ep}} > 0\}$ and let $c(i)$ denote the condition group. The specification is

$$C_i^{\text{ep}} = a_{c(i)} + g_{c(i)}(L_i^{\text{ep}}) + D_i^H s_{c(i)} + r_{c(i)} h_i^{\text{ep}} + Z_i^{C\prime}\gamma_C + \varepsilon_i^C, \tag{C.5}$$

$$g_c(L) = b_{c0}L + \sum_{k\in\{3,7,14,30\}} b_{ck}(L-k)_+, \tag{C.6}$$

where s_c captures the condition-specific component of the episode cost increment associated with entering HaH, while r_c measures the difference in cost per HaH day relative to the common LOS schedule. Neither coefficient is constrained by its label to be positive. The controls Z_i^C include DRG-family and hospital fixed effects, quarter-by-treatment-setting fixed effects, and age, sex, ED-entry, and condition-weight variables. Treatment setting distinguishes B&M-only episodes ($D_i^H = 0$) from episodes including HaH ($D_i^H = 1$). The quarter-by-setting effects allow episode costs to evolve differently across these groups.

The model is estimated by penalized least squares on episode totals, selecting the ridge penalty by held-out root mean squared error across five calendar blocks. The common LOS schedule comes from the full screened sample used in the joint demand-supply estimation model. The HaH transfer and daily-rate increments come from the subsample of documented-offer encounters. Therefore, I impose the assumption that their conditional predictive relationship extends beyond observed offers in order to form predictions for the full assessment sample.

Allocation to the remaining stay. Using the elapsed LOS and remaining durations defined in Appendix C.3, the common cost schedule for an integer remaining duration T is

$$G_c(\ell^{\text{LOS}}, T) = \sum_{t=0}^{T-1} \delta^t \left[g_c(\ell^{\text{LOS}} + t + 1) - g_c(\ell^{\text{LOS}} + t) \right]. \tag{C.7}$$

Let $a(i)$ denote the episode's admission quarter, and let $\psi_{a(i)}$ denote the HaH-minus-B&M difference in the quarter-by-setting fixed effects included in $Z_i^{C\prime}\gamma_C$. The full fixed episode increment associated with HaH is $s_{c(i)} + \psi_{a(i)}$. With zero transfer delay and A_δ from equation (C.1), conditional remaining costs are

$$C_i^B(T) = G_{c(i)}(\ell_i^{\text{LOS}}, T), \tag{C.8}$$

$$C_i^H(T) = G_{c(i)}(\ell_i^{\text{LOS}}, T) + s_{c(i)} + \psi_{a(i)} + r_{c(i)}A_\delta(T). \tag{C.9}$$

The condition-specific transfer increment and the HaH-specific quarter effect enter at transfer; the daily increment applies to remaining HaH exposure. Fractional final-day increments follow the same convention as E_i^H. Averaging over each setting's predicted remaining-duration distribution gives

$$\widehat{C}_i^B = \mathbb{E}\left[G_{c(i)}(\ell_i^{\text{LOS}}, T_i^B) \mid \mathscr{I}_i^O\right], \tag{C.10}$$

$$\widehat{C}_i^H = \mathbb{E}\left[G_{c(i)}(\ell_i^{\text{LOS}}, T_i^H) \mid \mathscr{I}_i^O\right] + s_{c(i)} + \psi_{a(i)} + r_{c(i)}E_i^H. \tag{C.11}$$

Already-incurred costs and episode controls common to the two settings, including the common quarter effect, cancel from the contrast. The HaH-specific quarter effect remains. The common LOS schedule contributes when predicted duration differs by location, separating this duration channel from the fixed HaH increment and daily-rate difference.

Profit units and counterfactual use. Let $\Delta\Pi_i^{\$}$ denote the uncentered dollar profit difference between HaH and B&M conditional on an accepted placement. Under the baseline reimbursement-parity assumption, as supported by negotiated contracts and observed reimbursements, the profit difference becomes

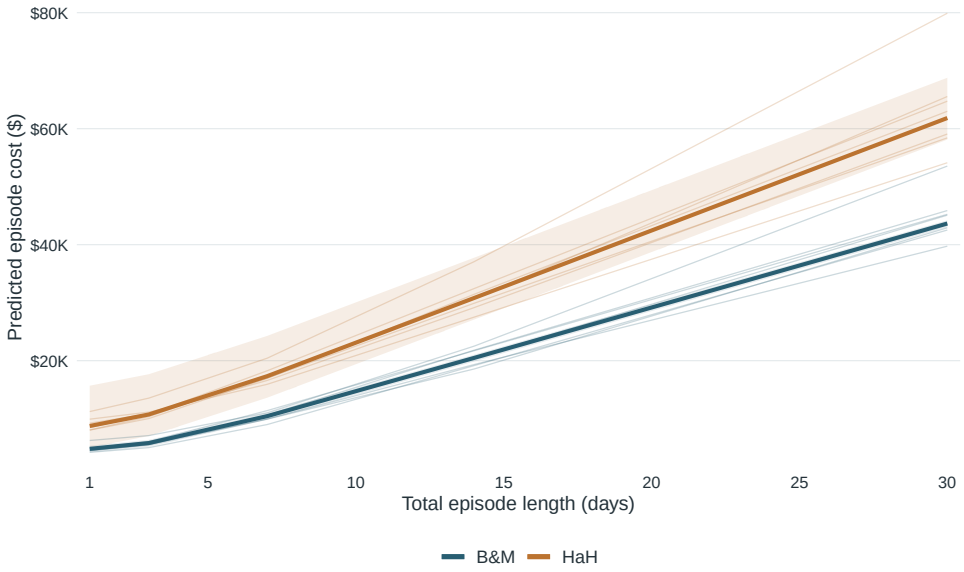
$$\Delta\Pi_i^{\$} = \widehat{C}_i^B - \widehat{C}_i^H. \tag{C.12}$$

For the offer equation, the input is

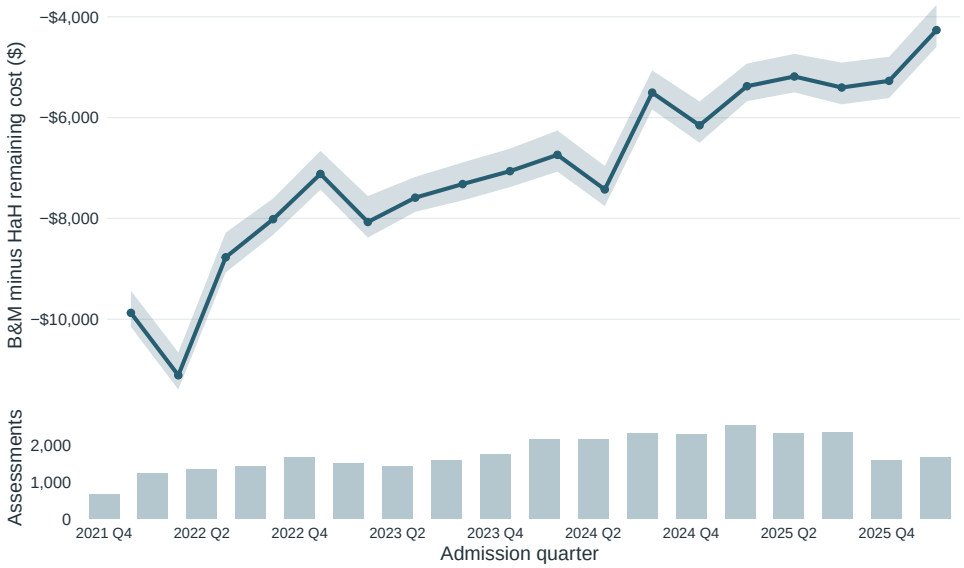
$$\Delta\Pi_i = \frac{\Delta\Pi_i^{\$} - \overline{\Delta\Pi}^{\$}_{b(i),\text{qtr}(i),c(i)}}{1{,}000}, \tag{C.13}$$

where $\text{qtr}(i)$ denotes the quarter assigned to assessment i for reference-cell construction. Acceptance weights this input once through $\rho_i^H \Delta W_i$. A positive coefficient favors higher conditional returns, holding other components fixed, rather than implying positive absolute HaH profits. Payment counterfactuals add the dollar reimbursement change before applying the original normalization. Hospital return outcomes use uncentered costs and revenues, keeping hospital accounting

Figure C.4: Predicted Treatment Costs



(a) Predicted Total Episode Cost by Length of Stay



(b) Expected Average Remaining Cost Difference by Admission Quarter

Notes: Panel (a) shows fitted, undiscounted episode costs for equal-length stays in both settings, immediate HaH transfer, and fixed controls. Thick lines use fixed condition weights; faint lines show individual conditions. Shading spans fitted values across 19 admission quarters. Panel (b) shows quarterly mean expected remaining B&M-minus-HaH costs across 31,944 assessments, using assessment-specific discharge survival, modeled transfer timing, and a daily discount factor of 0.99. Shading shows within-quarter 10th–90th percentiles; bars count assessments. Case mix and remaining duration vary across quarters. Negative differences indicate higher HaH costs. Neither shaded band is a confidence interval. Costs are in February 2026 dollars.

returns distinct from social resource savings.

Figure C.4 distinguishes fitted total-episode costs from the remaining-cost inputs used in the offer model. Panel (a) compares equal-length stays with immediate HaH transfer and fixed controls. Panel (b) summarizes assessment-specific discounted remaining-cost contrasts by admission quarter. The negative quarterly means indicate higher predicted remaining HaH costs for the assessed cohort. Their movement over time reflects both quarter effects and changes in case mix and predicted remaining duration.

D Additional Model Results, Counterfactual Analysis, and Alternative Specifications

D.1 Supplementary Baseline Results and Model Fit

Figure D.1 illustrates how prior choices inform the model's assessment of patient willingness. More prior acceptances are associated with higher posterior home preference, but substantial overlap indicates that acceptance counts alone do not fully summarize inferred preferences. This pattern illustrates the information available for patient targeting through posterior updating. Evidence on additional treatment-associated shifts comes from the joint estimates in Table 2.

Figure D.1: Posterior Home Preference by Prior HaH Acceptance History

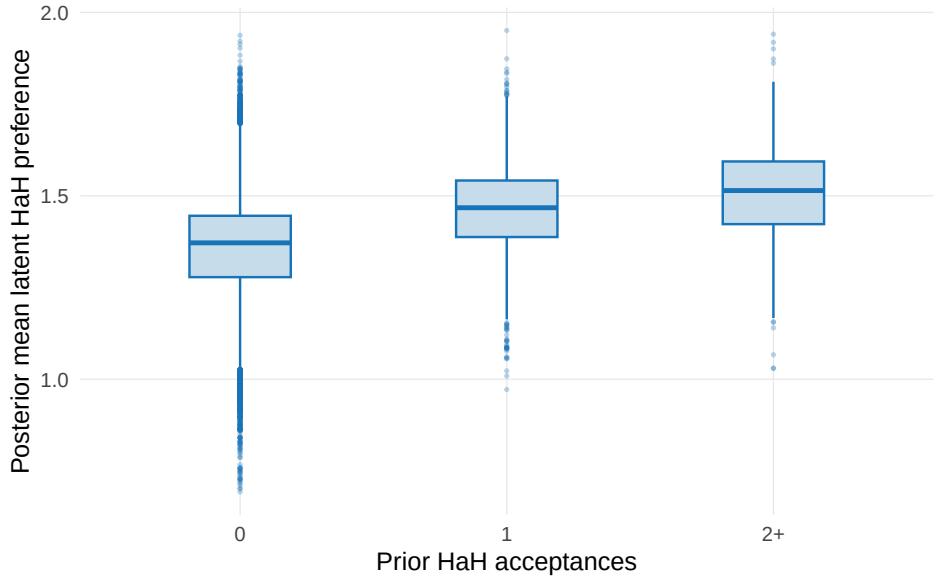
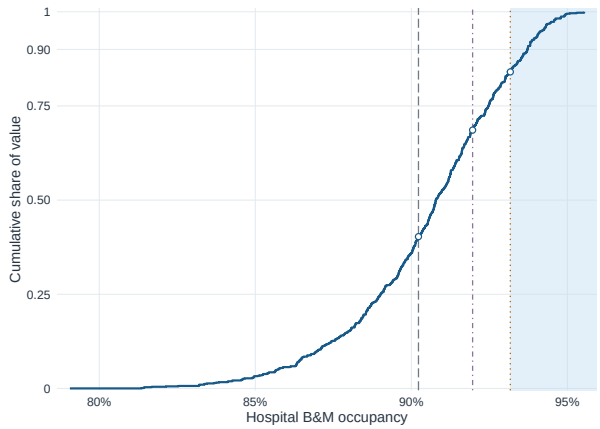
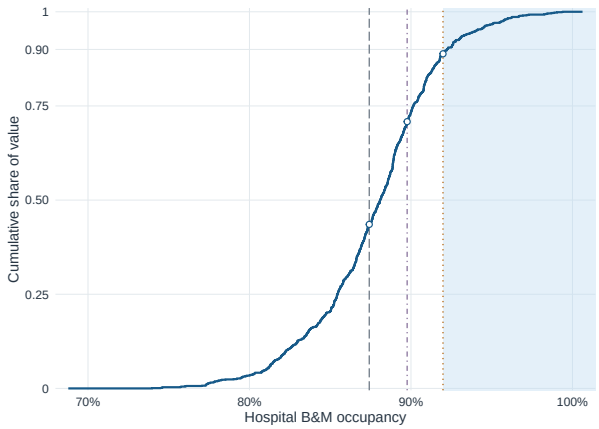


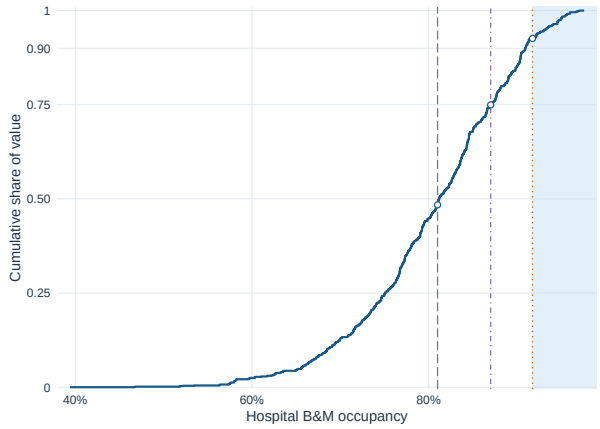
Figure D.2: Occupancy Distribution of Capacity-Value-Weighted HaH Placements



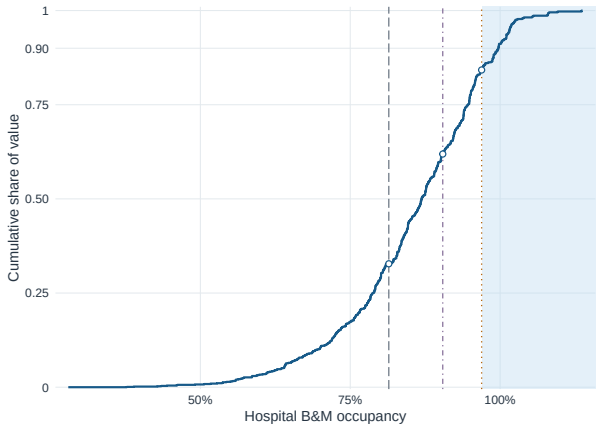
(a) Hospital A



(b) Hospital B



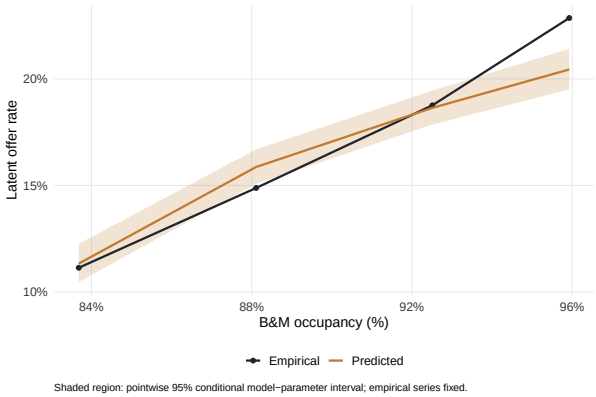
(c) Hospital C



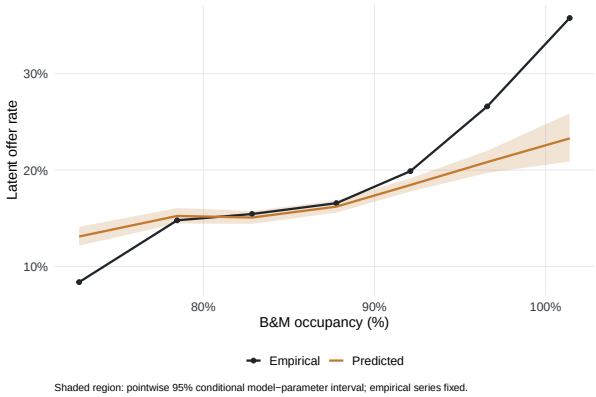
(d) Hospital D

Notes: Each realized HaH placement is weighted by $\widehat{CV}_i/E_i^B$, its estimated private capacity value per discounted B&M bed-day, evaluated at observed pre-offer occupancy and staffed capacity. Placements are grouped by assessment date and ranked by that date's mean hospital operational occupancy. The curves report the cumulative share of capacity-value-weighted placements associated with assessment dates whose mean hospital occupancy is at or below each level on the horizontal axis.

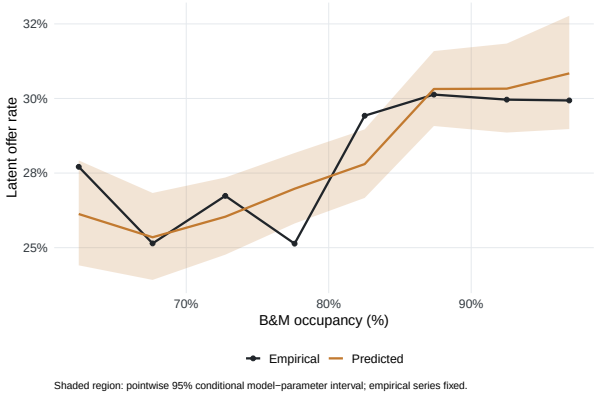
Figure D.3: Empirical and Predicted Offer Rates by B&M Occupancy



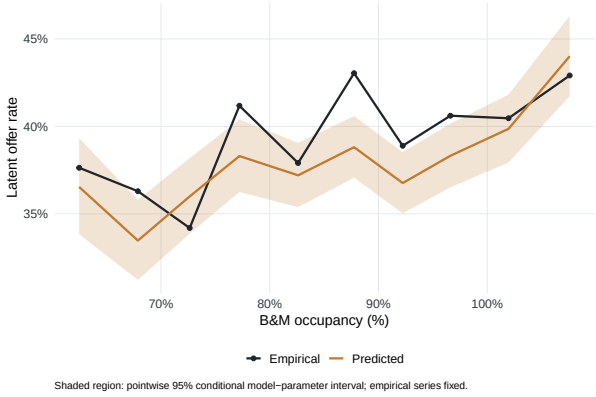
(a) Hospital A



(b) Hospital B



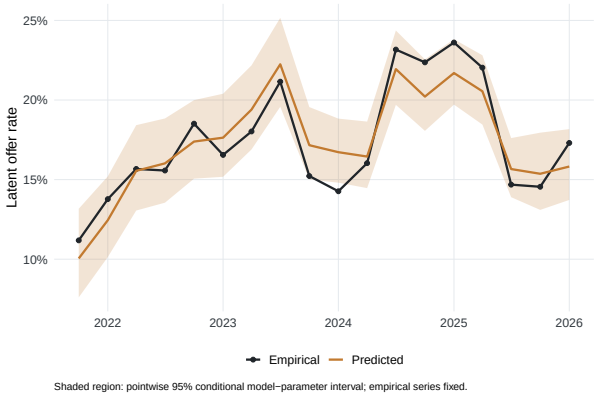
(c) Hospital C



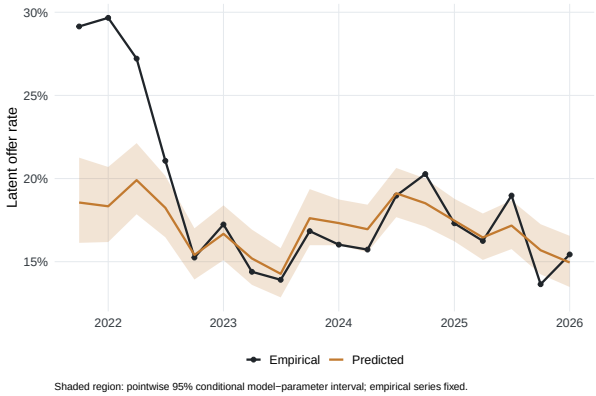
(d) Hospital D

Notes: Points show assessment-weighted mean offer rates and mean contemporaneous occupancy within each hospital's occupancy bins. Empirical offer rates use the documentation-adjusted reference; modeled rates average predicted offer probabilities. Shading indicates pointwise 95% confidence bands for the modeled mean offer rates, calculated by the delta method using the full estimated parameter covariance matrix, including posterior and history updating. Generated inputs and documentation parameters are held fixed.

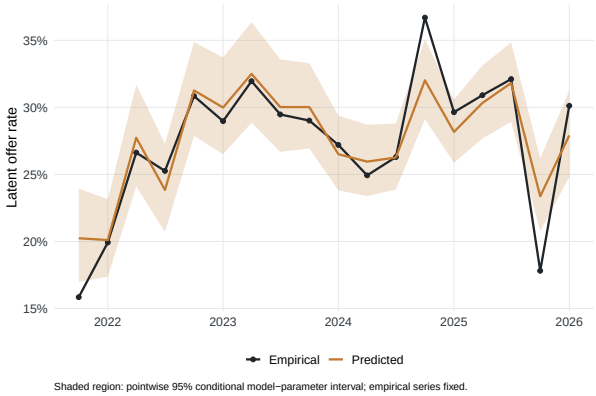
Figure D.4: Empirical and Predicted Offer Rates by Quarter



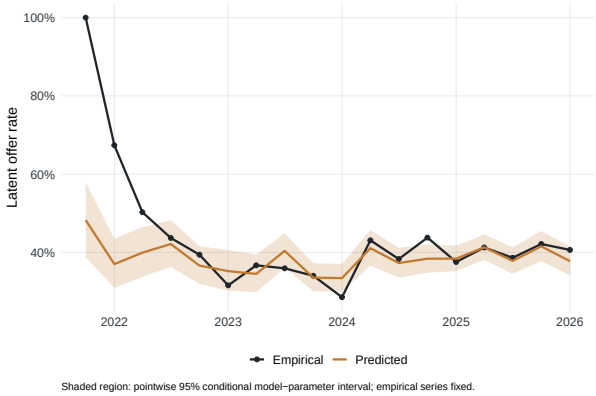
(a) Hospital A



(b) Hospital B



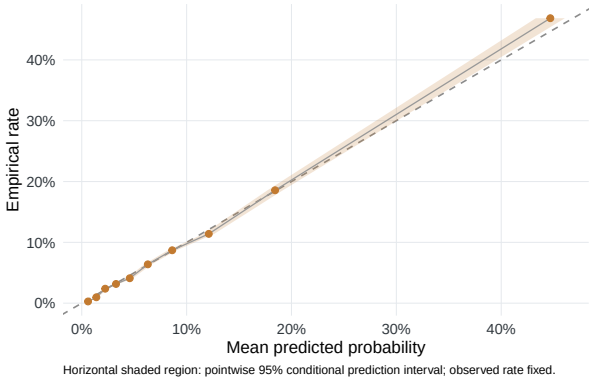
(c) Hospital C



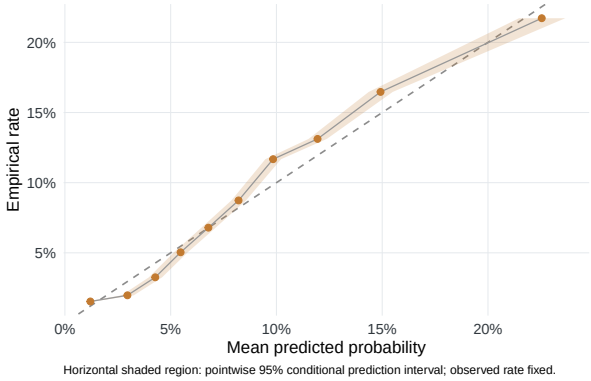
(d) Hospital D

Notes: Points show assessment-weighted mean offer rates within hospital-quarter cells. Empirical offer rates use the documentation-adjusted reference; modeled rates average predicted offer probabilities. Shading indicates pointwise 95% confidence bands for the modeled mean offer rates, calculated by the delta method using the full estimated parameter covariance matrix, including posterior and history updating. Generated inputs and documentation parameters are held fixed. Hospital D's first displayed quarter contains only two assessments, while the following quarter contains 101. The final quarter includes observations through February 2026.

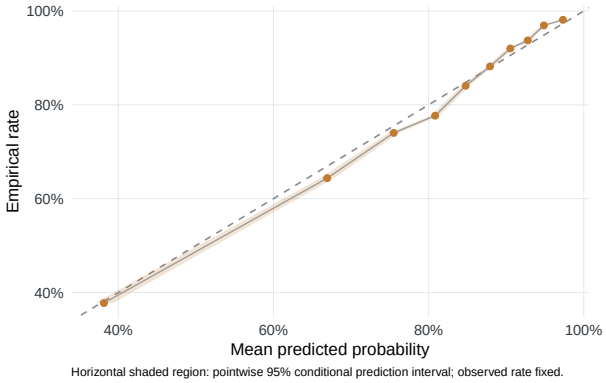
Figure D.5: Calibration of Observed Outcomes and Documentation-Adjusted Offers



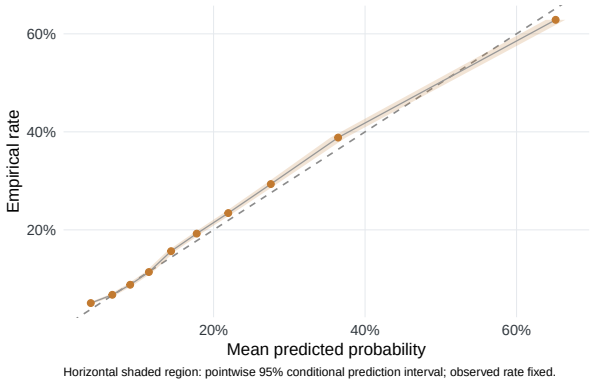
(a) Accepted placements



(b) Documented rejections



(c) No observed offers



(d) Latent offers

Notes: Each panel divides assessments into ten approximately equal-sized groups ranked by the corresponding predicted probability. Points compare the group mean predicted probability with its empirical rate; the dashed line indicates equality. Groups are formed separately for each panel. Horizontal shading shows pointwise 95% conditional intervals for mean predicted probabilities, with empirical rates held fixed. These intervals describe model-prediction uncertainty rather than patient-cluster bootstrap uncertainty in the empirical rates. Panels (a)–(c) use observed outcomes; panel (d) uses the separately constructed documentation-adjusted offer reference.

D.2 Alternative Model Specifications

Time-invariant preferences and treatment history. Tables D.1 and D.2 compare alternative descriptions of demand history. Every specification retains time-invariant patient heterogeneity, joint-normal encounter shocks, forecast capacity inputs, and posterior updating. Column (1) excludes direct demand-history terms, column (2) adds previously completed treatment, and column (3) instead includes prior acceptance and expected rejection counts. The baseline in column (4) combines these terms. Column (5) uses the same history terms as the baseline model but replaces normal heterogeneity with two discrete preference types. The baseline model has the lowest AIC among these specifications. Adding first treatment to the count-history model improves full-sample fit, with small reductions in in-sample terminal and first-post-treatment outcome log loss. Expected rejections retain a negative association with acceptance, while additional acceptances show no clear cumulative increase. Time-invariant heterogeneity remains, although its estimated magnitude falls and becomes less precise. The central supply coefficients remain broadly similar across specifications, including the two-type model. These results support the baseline's first-treatment flexibility.

Logit control-function specification. Column (3) of Table D.3 reports a static logit control-function alternative to the baseline model which uses the same forecast inputs, normal time-invariant patient heterogeneity, demand-history specification, and offer-documentation adjustment. This specification accounts for offer selection by including a term based on the offer probability in patient utility:

$$\Delta U_{ij}^{\Lambda} = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \sigma_\xi \xi_j + \varphi(1-p_i) + \varepsilon_i^{\Lambda}.$$

Here, $1-p_i$ is the generalized (score) residual from the logit offer equation, $O_i - p_i$, evaluated for an offered patient ($O_i = 1$). Including $\varphi(1-p_i)$ in acceptance utility serves as a control function for selection into offers, motivated by Petrin and Train (2010). Acceptance probabilities use the logistic distribution, and patient option value uses the corresponding logit inclusive value. The offer probability, acceptance belief, and option value are jointly determined through a fixed point: patient utility depends on the offer probability, while hospital offer incentives depend on anticipated acceptance and patient surplus. Parameters are jointly re-estimated under this specification. The coefficient φ captures selection through the utility adjustment, as opposed to the encounter-shock correlation r under the baseline. This specification also supplies the first-stage offer policy and demand estimates for the dynamic comparison in Appendix F, preserving the estimated relationships between patient preferences, prior interactions, and offer selection. These first-stage objects are held fixed while hospital payoff coefficients are re-estimated to incorporate continuation values. The logit structure permits a tractable conditional-choice-probability (CCP) inversion.

Table D.1: Demand History and Time-invariant Preferences

	Normally-distributed heterogeneity				Two types
	(1) No HaH history	(2) First treatment	(3) History counts	(4) Baseline specification	(5) Full history
Panel A: Patient acceptance					
Prior acceptance count	–	–	−0.013	−0.050	−0.052
			(0.032)	(0.030)	(0.030)
Expected prior rejection count	–	–	−0.123	−0.152	−0.149
			(0.035)	(0.032)	(0.034)
Previously completed treatment	–	0.082	–	0.151	0.152
		(0.033)		(0.034)	(0.037)
Encounter-shock correlation, r	−0.738	−0.712	−0.755	−0.723	−0.723
	(0.032)	(0.035)	(0.030)	(0.032)	(0.032)
Time-invariant preference SD, σ_ξ	0.435	0.372	0.348	0.232	0.233
	(0.033)	(0.041)	(0.063)	(0.074)	(0.071)
Panel B: Hospital offers					
Profit (per $1,000)	0.067	0.069	0.057	0.057	0.057
	(0.013)	(0.014)	(0.013)	(0.013)	(0.013)
Patient option value	0.836	0.886	0.815	0.876	0.874
	(0.055)	(0.062)	(0.051)	(0.055)	(0.055)
B&M duration exposure	−0.358	−0.344	−0.349	−0.326	−0.326
	(0.053)	(0.053)	(0.052)	(0.051)	(0.051)
Hospital A occupancy relief	0.158	0.153	0.153	0.145	0.145
	(0.026)	(0.026)	(0.026)	(0.025)	(0.025)
Hospital B occupancy relief	0.119	0.116	0.115	0.109	0.109
	(0.027)	(0.027)	(0.026)	(0.026)	(0.026)
Hospital C occupancy relief	0.129	0.127	0.127	0.123	0.123
	(0.028)	(0.027)	(0.027)	(0.026)	(0.026)
Hospital D occupancy relief	0.048	0.051	0.045	0.047	0.047
	(0.029)	(0.028)	(0.028)	(0.027)	(0.027)
Fit					
Parameter count	169	170	171	172	173
Log likelihood	−16,220.7	−16,217.8	−16,212.9	−16,203.7	−16,203.5
AIC	32,779.4	32,775.6	32,767.8	32,751.3	32,753.1

Notes: All parameters are jointly re-estimated with common controls, fixed forecast and documentation inputs, posterior updating, and supply-history terms. History headings refer only to direct demand terms. Treatment requires a distinct prior HaH-treated encounter discharged before assessment. Counts are cumulative prior acceptances and expected declinations. Standard errors in parentheses use the observed Hessian conditional on generated inputs. Column (5) uses two ordered, mean-zero support points with estimated masses; its implied preference SD and standard error are obtained from the estimated mass and support using the model-specific covariance matrix. Normal and two-type heterogeneity are non-nested. Supply inputs retain the baseline units and structure.

Table D.4 reports the joint-normal model using contemporaneous B&M and HaH census, rather than the forecast inputs used in the baseline. Patient-history and surplus estimates remain similar, while capacity coefficients are smaller. The log likelihood is $-16{,}207$, compared with $-16{,}204$ for the model in the main text.

Table D.2: Demand-History Restrictions and Sequential Prediction

Panel A: Nested likelihood-ratio tests

Expansion (columns)	Restrictions	LR statistic	p-value
First treatment: (2) vs. (1)	1	5.727	0.0167
History counts: (3) vs. (1)	2	15.543	0.000422
Baseline: (4) vs. (3)	1	18.484	0.0000171

Panel B: In-sample outcome log-loss differences

Expansion minus restriction	Terminal assessments (21,354 rows)	First post-treatment (915 rows)
First treatment: (2) − (1)	−0.00007	−0.00694
History counts: (3) − (1)	−0.00022	+0.00083
Baseline: (4) − (3)	−0.00023	−0.01209

Notes: Column numbers refer to Table D.1. Panel A tests zero first-treatment coefficients or jointly zero acceptance and rejection count coefficients. The number of restrictions gives the χ^2 degrees of freedom. In Panel B, negative differences favor the expansion. Scores evaluate the three observed outcomes using strictly prior histories. Differences are calculated from reported log losses.

Table D.3: Outcome Prediction Scores Across Static Specifications

Sample	Assessments	(1) Forecast census	(2) Contemporaneous census	(3) Control function (forecast census)
Panel A: Outcome log loss				
All assessments	31,944	0.50725	0.50736	0.50712
Repeat patients	16,402	0.50317	0.50337	0.50374
Terminal assessments	21,354	0.50137	0.50140	0.50091
First post-treatment	915	0.83136	0.83310	0.82985
Panel B: Accepted-placement Brier score				
All assessments	31,944	0.06942	0.06946	0.06924
Repeat patients	16,402	0.06828	0.06834	0.06818
Terminal assessments	21,354	0.06770	0.06771	0.06744
First post-treatment	915	0.16519	0.16590	0.16513

Notes: Columns (1)–(3) compare the baseline forecast model, contemporaneous-census alternative, and forecast-based logit control-function model (Appendix D.2). All use the same assessments and subgroups, normal time-invariant heterogeneity, and full demand-history specification. Panel A reports mean log loss for the three observed outcomes; Panel B reports the accepted-placement Brier score using joint placement probability J_i. Lower scores indicate better fit. Repeat patients includes all assessments of patients observed multiple times in the estimation sample; terminal assessments includes each patient's last assessment, including singletons. First post-treatment uses the first qualifying assessment after prior HaH treatment, when included in the estimation sample. Scores are in-sample, with parameters estimated on the full sample and histories using strictly prior outcomes.

Table D.4: Joint-Normal Estimates with Contemporaneous Census

	Estimate	Standard error
Panel A: Patient acceptance		
Asian	0.128^{***}	(0.041)
Black/African American	-0.040^{*}	(0.020)
Latino/Hispanic	0.152^{***}	(0.029)
Medicaid	-0.020	(0.034)
Medicare	-0.072^{***}	(0.027)
Home-to-hospital distance	0.007	(0.007)
Age	0.050^{***}	(0.009)
Male	0.015	(0.012)
Neighborhood average household size	-0.001	(0.008)
Neighborhood college education share	0.019^{*}	(0.010)
Neighborhood log(1 + per-capita income)	-0.015	(0.012)
Neighborhood share below 150% FPL	-0.019^{**}	(0.009)
Prior expected HaH declinations	-0.153^{***}	(0.032)
Ever previously treated with HaH	0.151^{***}	(0.034)
Count of prior HaH acceptances	-0.048	(0.030)
Condition severity weight	-0.034^{**}	(0.015)
Encounter-shock correlation, r	-0.724^{***}	(0.032)
Time-invariant HaH preference SD, σ_ξ	0.230^{***}	(0.075)
Panel B: Hospital offers		
Accounting profits (per $1,000)	0.056^{***}	(0.013)
HaH program strain	-0.003^{***}	(0.001)
HaH escalation risk	-0.985^{***}	(0.325)
Care-burden	-0.098^{***}	(0.007)
Acute-severity	-0.044^{***}	(0.006)
B&M stay-duration exposure	-0.276^{***}	(0.039)
Hospital A released-capacity	0.113^{***}	(0.018)
Hospital B released-capacity	0.081^{***}	(0.019)
Hospital C released-capacity	0.084^{***}	(0.017)
Hospital D released-capacity	0.017	(0.019)
Strictly prior expected offers	0.171^{***}	(0.015)
Patient option value	0.870^{***}	(0.055)
Baseline net offer term	-0.939^{***}	(0.043)
Assessments	31,944	
Patients	21,354	
Log likelihood	$-16{,}207.26$	
AIC	32,758.52	

Notes: The contemporaneous occupancy alternative retains the baseline controls, fixed effects, input transformations, documentation probabilities, patient-history terms, and joint-normal encounter shocks, but replaces forecast census with contemporaneous census. Standard errors use the observed information matrix conditional on generated inputs and are not clustered. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.3 Supplementary Counterfactual Analysis

This subsection reports supplementary payment scenarios, recipient and subgroup comparisons, the matched-offer information implementation, and finite HaH staffing expansions. The model, assessment cohort, histories, and operating states follow Section 8, except for the stated staffing change. Payment policies affect Medicare FFS encounters only. Cash uses remaining undiscounted direct-variable treatment costs; structural capacity value uses discounted forecast exposures. Results are conditional point estimates.

D.3.1 Reimbursement Rules, Support, and Alternative Rates

Let u_i be pre-offer operational B&M occupancy relative to staffed capacity, and let I_i^s indicate that an assessment falls in the occupancy states targeted by rule s. Operational occupancy includes occupied and unavailable beds. For Medicare FFS encounters, rule s and rate τ imply $m_i^s = 1 + \tau I_i^s$, with parity retained for other encounters. The uniform rule has $I_i^s = 1$ at every occupancy level; each low- or high-state rule is a separate policy. For J_i^s denoting the probability of an accepted placement, additional original-admission payments are

$$B_s(\tau) = \sum_{i \in \text{Medicare}-\text{FFS}} J_i^s(\tau)\, \tau I_i^s R_i. \tag{D.1}$$

Payments include rate changes on baseline and induced placements, excluding backfill admissions and downstream care. I evaluate $\tau \in \{-0.10, -0.05, 0, 0.05, 0.10\}$, with parity as the common comparator and no budget matching. Occupancy and staffed capacity remain fixed, excluding behavioral responses to the payment trigger.

Hospital-specific quartiles are fixed from the reference daily occupancy distribution rather than the distribution of assessment opportunities. Each hospital has 1,563 reference days. Table D.5 reports the cutoffs and the number of assessed Medicare FFS opportunities eligible for each payment rule. The bottom-quartile threshold ranges from 69.9% to 88.4%, illustrating why relatively low occupancy need not mean low absolute utilization. Common thresholds below 75% and above 85% provide additional sensitivity comparisons. The policies affect different numbers and hospital mixes of assessment opportunities; their total gains are not pure measures of responsiveness to occupancy.

Most additional payments accrue to baseline placements: 72.8% under the uniform 10% supplement (Table 6). Increasing the uniform rate from zero to 5% generates 89.1 option-value units per $1 million of additional payments; the further increase to 10% generates 80.0. Comparing each policy with parity, option-value gain per additional placement is 0.89 at a 5% supplement

Table D.5: Occupancy Cutoffs and Medicare FFS Policy Support

	Hospital A	Hospital B	Hospital C	Hospital D	Total
Reference 25th percentile (%)	88.4	84.6	75.0	69.9	—
Reference 75th percentile (%)	92.0	89.8	87.0	90.4	—
All FFS assessments	3,479	7,369	2,528	1,615	14,991
Bottom quartile	676	1,383	524	269	2,852
Top quartile	1,579	3,567	885	693	6,724
Below 75%	0	55	502	362	919
Below 80%	7	346	961	491	1,805
Above 85%	3,321	5,858	1,084	896	11,159
Above 90%	2,332	3,397	585	693	7,007

Notes: Cutoffs use hospital-specific reference daily operational occupancy and are held fixed in all simulations. The remaining rows count assessment opportunities, not unique patients or expected placements. Bottom and top quartile rules therefore need not each contain one quarter of assessments. Absolute lower and upper thresholds are strict. Medicare FFS is the affected payor group; aggregate counterfactual outcome levels also retain all unaffected encounters.

and 0.90 at a 10% supplement. Remaining treatment costs increase by approximately $7,900 per induced placement across these scenarios. The declining benefit per payment dollar therefore does not establish diminishing match quality or an increasing marginal resource-cost schedule.

Current occupancy determines payment eligibility, whereas the private capacity component uses encounter-specific forecast exposure. Differences in capacity value across rules reflect the patients and operating states selected, with census and forecasts fixed; they do not measure an endogenous decline in capacity value as HaH expands.

Backfill accounting. Equation (36) values potential throughput separately from structural private capacity value. For the uniform Medicare FFS 10% supplement, focal cash contribution increases by $0.73 million. Applying hospital-specific contribution margins raises the combined gain to $1.77 million at 50% backfill and $2.81 million at full backfill. The primary margins include estimated settlements in revenue and subtract direct variable costs. These are conditional hospital cash calculations; downstream payer savings are unmeasured.

D.3.2 Recipient Composition and Subgroup Access

Table D.7 describes the Medicare FFS placements induced by each supplement and foregone under a uniform reduction, weighting characteristics by the corresponding change in placement probability. Clinical-score means use observed values only. These are differences in recipient composition,

Table D.6: Payment Scenarios Across Occupancy Rules

Rule	Rate	Additional placements	Additional B&M days	Patient option value change	Payments ($ millions)	Private capacity value ($ millions)
Uniform	−10%	−184.4	−848	−159.2	−1.627	−2.874
	−5%	−97.9	−453	−84.9	−0.888	−1.539
	+5%	115.1	543	102.1	1.146	1.863
	+10%	251.6	1,198	226.7	2.704	4.133
Bottom quartile	−10%	−28.9	−129	−23.5	−0.253	−0.403
	−5%	−15.3	−69	−12.5	−0.138	−0.215
	+5%	18.0	82	14.9	0.175	0.258
	+10%	39.9	183	33.6	0.417	0.589
Top quartile	−10%	−88.8	−418	−79.1	−0.793	−1.475
	−5%	−47.3	−224	−42.3	−0.433	−0.792
	+5%	56.2	274	51.6	0.579	0.976
	+10%	122.5	603	114.1	1.376	2.154
$< 80\%$	−10%	−25.6	−103	−21.3	−0.238	−0.265
	−5%	−13.4	−54	−11.3	−0.129	−0.140
	+5%	15.0	60	12.8	0.152	0.157
	+10%	31.7	128	27.5	0.334	0.336
$> 90\%$	−10%	−93.3	−449	−83.4	−0.814	−1.624
	−5%	−49.8	−241	−44.6	−0.446	−0.874
	+5%	59.4	296	54.6	0.598	1.080
	+10%	130.0	656	121.0	1.426	2.397
$< 75\%$	−10%	−13.8	−54	−11.6	−0.132	−0.125
	−5%	−7.2	−28	−6.2	−0.071	−0.066
	+5%	8.0	31	6.9	0.084	0.073
	+10%	16.9	66	14.8	0.183	0.156
$> 85\%$	−10%	−138.1	−657	−121.3	−1.203	−2.343
	−5%	−73.5	−352	−64.8	−0.658	−1.258
	+5%	87.5	428	78.8	0.868	1.541
	+10%	192.6	953	176.4	2.082	3.439

Notes: Full-cohort changes from parity; each payment change applies only to Medicare FFS encounters in the stated occupancy states. Negative payments denote savings on original admissions. Each row is a separate fixed-rate policy. Payment budgets are not matched. Days are undiscounted potential B&M days; private capacity value uses discounted forecast exposures and the fitted hospital capacity valuation. Option value is normalized patient utility. The common zero-rate comparator has zero change in every column.

not policy effects on clinical risk. Patients induced by the uniform 10% supplement have somewhat longer expected remaining stays and higher predicted return-transfer risk than those whose placements are foregone under a 10% reduction.

Table D.7: Characteristics of Medicare FFS Placements at the Policy Margin

	Uniform -10% Foregone	Uniform $+10\%$ Induced	Bottom quartile $+10\%$ Induced	Top quartile $+10\%$ Induced
Expected placements	184.4	251.6	39.9	122.5
Expected remaining B&M stay (days)	4.60	4.76	4.60	4.92
Predicted return-transfer risk (%)	6.68	7.55	7.19	8.21
Nursing-assessment score (NWASS)	53.44	55.37	56.01	55.64
Neighborhood share below 150% FPL (%)	13.16	13.40	12.91	13.61
Prior completed HaH treatment (%)	14.92	13.70	13.59	14.85

Notes: Induced placements weight characteristics by $(J_i^s - J_i^0)_+$; foregone placements use $(J_i^0 - J_i^s)_+$, where J_i^s is the joint offer–acceptance probability under policy s. The opposite-sign placement masses are zero in these scenarios. These are expected placement weights over assessment opportunities, not counts of unique patients. All four margins concern Medicare FFS encounters. Clinical-score means use observed values only; weighted NWASS missingness is 0.8%, 0.8%, 0.6%, and 1.0%, respectively. FPL denotes the federal poverty level; poverty is a neighborhood measure. Quartile policies refer to hospital occupancy, not patient-characteristic quartiles.

Table D.8 reports changes in expected placements per 100 assessed Medicare FFS opportunities within each subgroup, $100\Delta J_g / N_g^{\text{FFS}}$. Patient-characteristic quartiles use observed baseline values over all 31,944 assessments, with equal assessment weights and fixed boundaries; ties and missing values can produce unequal group sizes. Positive prior rejection experience denotes a strictly prior documentation-adjusted expected-rejection count greater than zero.

The uniform 10% supplement generates 1.37 additional placements per 100 assessed FFS opportunities in the highest predicted-risk quartile, compared with 1.77 in the lowest; the remaining-stay extremes gain approximately 1.64 and 1.66. Higher predicted risk among marginal recipients therefore does not imply a larger access response among the highest-risk patients. Patients with prior completed HaH treatment have a larger per-assessment gain, 3.74 versus 1.54 without prior treatment, with histories held fixed.

D.3.3 Matched-Offer Preference Information

Let w_{ig}^k denote the hospital's preference weights under information regime k, and let $Q_{ig}^H(v)$ denote expected patient surplus conditional on type g and hospital shock v, as given by the node-specific expression in equation (24). The hospital's conditional acceptance belief and perceived patient

Table D.8: Access Changes Within Medicare FFS Subgroups

Characteristic	Group	Placement loss Uniform −10%	Placement gain Uniform +10%
Predicted return-transfer risk	Lowest quartile	1.457	1.770
	Highest quartile	0.826	1.368
Expected remaining B&M stay	Lowest quartile	1.359	1.657
	Highest quartile	1.032	1.640
Neighborhood log income	Lowest quartile	1.051	1.516
	Highest quartile	1.367	1.862
Neighborhood poverty share	Lowest quartile	1.324	1.787
	Highest quartile	1.030	1.469
Prior completed HaH treatment	No	1.115	1.543
	Yes	2.987	3.741
Prior expected rejection experience	No	1.295	1.779
	Yes	1.075	1.438

Notes: Entries are expected placement changes per 100 assessed FFS opportunities within the stated group. Losses under the reduction are reported as positive magnitudes. Patient-characteristic quartiles are fixed from the observed full-cohort distribution, not the occupancy quartiles used for payment eligibility. Neighborhood variables do not measure individual household income or poverty. Missing groups are retained when reconciling aggregate outcomes but omitted from this display.

surplus are

$$\rho_i^{H,k}(v) = \sum_g w_{ig}^k \rho_{ig}^H(v), \tag{D.2}$$

$$Q_i^{H,k}(v) = \sum_g w_{ig}^k Q_{ig}^H(v). \tag{D.3}$$

The pooled benchmark uses population quadrature weights w_g; the baseline uses predictive weights w_{ig}^-. Under perfect information, the hospital's weights concentrate on the revealed time-invariant type, yielding a type-specific offer region. Observed covariates, direct history effects, and the hospital's information about v_i remain unchanged; perfect information does not reveal the patient's encounter-specific shock.

Hospital beliefs determine offers, while common evaluation weights preserve the underlying preference distribution across regimes. Let $\mathcal{O}_{ig}^k$ denote the offer region for type g under regime k. This region is common across types under pooled and baseline information, but can differ by type under perfect information. Offer probabilities, joint offer–acceptance probabilities, and expected

patient option value created by access are

$$
\begin{aligned}
p_i^k &= \sum_g w_{ig}^- \int_{\mathcal{O}_{ig}^k} f_\nu(\nu)\, d\nu, \\
J_i^k &= \sum_g w_{ig}^- \int_{\mathcal{O}_{ig}^k} \rho_{ig}^H(\nu) f_\nu(\nu)\, d\nu, \\
V_i^k &= \sum_g w_{ig}^- \int_{\mathcal{O}_{ig}^k} Q_{ig}^H(\nu) f_\nu(\nu)\, d\nu.
\end{aligned}
\tag{D.4}
$$

To isolate allocation, let $c(i)$ index hospital $\times$ admission calendar quarter and let $G_{ig}^k(\nu) = \Gamma_{ig}^k(\nu) + \nu$ denote the full fitted offer payoff under regime k. I choose a common cell-specific threshold adjustment a_c^k so that offer regions of the form $\mathcal{O}_{ig}^k = \{\nu : G_{ig}^k(\nu) + a_{c(i)}^k \geq 0\}$ satisfy

$$
\sum_{i:c(i)=c} p_i^k(a_c^k) = \sum_{i:c(i)=c} p_i^0. \tag{D.5}
$$

These adjustments enforce an offer constraint without re-estimating payoff parameters. Common factual predictive weights in equation (D.4) preserve the evaluation population. Across the 72 hospital-quarter cells, the maximum absolute offer residual is below 10^{-8}. Holding offers fixed leaves placements and treatment costs free to change: perfect preference information increases remaining direct-variable treatment costs by approximately $1.81 million. The allocation results are reported in Table 7.

D.3.4 Finite HaH Staffing Expansion

The staffing experiment examines how additional HaH staffing capacity relaxes strain and changes offers and accepted placements. I increase effective HaH staffing from L_i to $\widetilde{L}_i = m_L L_i$, where $m_L \geq 1$. Holding forecast HaH census and remaining exposure fixed, the uncentered strain input becomes

$$
\widetilde{M}_i^H = E_i^H \frac{2\overline{N}_i^H + 1}{m_L L_i}. \tag{D.6}
$$

I retain the estimated strain coefficient and original centering reference, and recompute offer regions and joint offer–acceptance probabilities.

Table D.9 reports the finite expansions. Increasing effective staffing by 50% and 100% raises expected placements by approximately 9% and 14%, respectively. The raw strain input itself falls as $1/m_L$, so the less-than-proportional response is partly built into the transformation and does not independently establish diminishing returns to matching or congestion relief.

Table D.9: Finite Effective HaH Staffing Expansions

	Staffing +50%	Staffing +100%
Additional expected offers	415	638
Additional expected placements	292	451
Additional potential B&M bed-days	1,363	2,115

Notes: Changes are relative to baseline. The effective staffing multiplier divides the raw strain input in equation (D.6); its estimated coefficient and original centering reference remain fixed. Forecast HaH census, remaining exposure, episode treatment costs, duration predictions, and factual histories are held fixed. Additional staffing acquisition costs and subsequent census changes are excluded. These are strain-channel sensitivities, not net returns to hiring or predictions of a fully scaled program.

E Survey Augmentation of Patient Choice Model

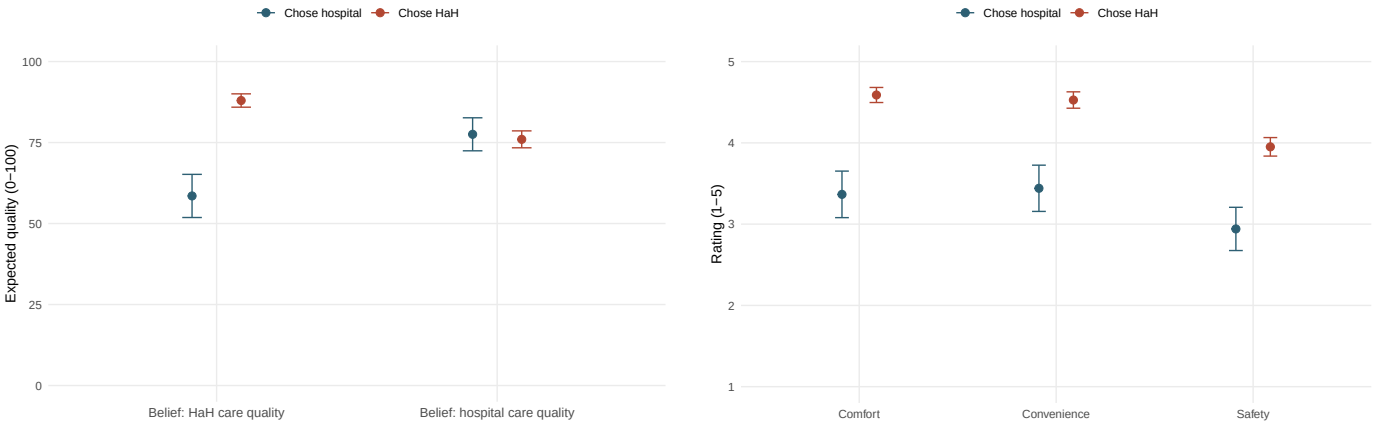
This appendix augments the joint demand–supply model in Section 5 with a novel survey measuring household beliefs over expected relative care quality and relative comfort, safety, and convenience under HaH, as it compares to B&M care. Survey answers enter the model as part of time-invariant patient preferences and are not part of the hospital's information set in estimation. Non-medical spending responses are relatively sparse and therefore enter only as a supplemental specification at the end of this appendix. Table E.1 summarizes survey respondents and response availability. Figure E.1 compares expected care quality and experience by reported treatment choice.

Table E.1: Survey Respondents and Reported Information

	Count (N)
Unique patients linked to a survey	453
Patient respondents	359
Caregiver respondents	94
Complete expected-quality difference	404
Any expected-experience information	453
Complete comfort, safety, and convenience ratings	450
Complete expected non-medical spending responses	297

Notes: Caregiver responses answer questions phrased to address the patient's expected beliefs and therefore are assumed to measure the associated household's care-choice preferences.

Figure E.1: Expected Care Quality and HaH Experience by Reported Choice



(a) Expected Quality

(b) Expected Experience

Notes: The left panel uses expected care-quality ratings on a 0-to-100 scale. The right panel uses a one-to-five Likert scale for expected comfort, convenience, and safety ratings. Points report respondent means and bars report 95 percent confidence intervals.

E.1 Survey-Augmented Demand Model

Expected quality. Let $\Delta Q_j = \mathrm{Q}_j^H - \mathrm{Q}_j^B$ denote patient j's reported expected-quality difference between HaH and B&M, each rated from 0-to-100. I standardize this difference as

$$Q_j = \frac{\Delta Q_j - \bar{\Delta}_Q}{s_{\Delta Q}}, \qquad \bar{\Delta}_Q = 4.61, \quad s_{\Delta Q} = 34.64, \tag{E.1}$$

using the sample mean and standard deviation among the 404 respondents with both ratings. Higher Q_j indicates more favorable expected HaH quality relative to B&M. Conditional on first-assessment covariates X_j^0, I specify

$$Q_j \mid X_j^0 \sim N(\mu_j^Q, s_Q^2), \qquad s_Q = 0.958,$$

where μ_j^Q is a ridge-regression prediction. The prediction rule and residual standard deviation are held fixed during joint estimation. Observed scores are treated as exact measurements of a time-invariant quality-expectation trait, while missing scores are integrated out. Hospital beliefs use this conditional distribution, updated with strictly prior assessment outcomes, without observing the later survey response.

Expected experience. Let $F_j \sim N(0,1)$ denote a persistent expected-experience factor. For component $m \in \{\text{comfort}, \text{safety}, \text{convenience}\}$, let $Y_{jm}^S \in \{1,\ldots,5\}$ denote the ordered rating. The

ordered-probit measurement model is

$$Y^S_{jm} = k \quad \Longleftrightarrow \quad \tau_{m,k-1} < \lambda^S_m F_j + e^S_{jm} \le \tau_{m,k}, \qquad e^S_{jm} \sim N(0,1), \tag{E.2}$$

where $\tau_{m,0} = -\infty$, $\tau_{m,5} = \infty$, and the four finite cutpoints are ordered. Conditional on F_j, component errors are independent, so an observed response contributes

$$h_{jm}(f) = \Phi\Big(\tau_{m,Y^S_{jm}} - \lambda^S_m f\Big) - \Phi\Big(\tau_{m,Y^S_{jm}-1} - \lambda^S_m f\Big). \tag{E.3}$$

The loadings and cutpoints are estimated in a first-stage model and then held fixed. The loadings are 2.389 for comfort, 1.228 for safety, and 1.466 for convenience; higher F_j therefore indicates more favorable expected HaH experience. Missing components contribute a factor of one. Quality and experience are patient-specific traits that apply across the patient's assessments. The interpretation treats later answers as measurements of preferences present at the time of choice.

Augmented utility and hospital beliefs. Using the demand notation in Section 5.1, and letting superscript S denote the survey-augmented demand system, the survey adds two terms to time-invariant patient utility:

$$\Omega_j = \sigma_\xi \xi_j + \theta_Q Q_j + \theta_F F_j, \qquad \xi_j \sim N(0,1), \tag{E.4}$$

$$\Delta U^S_{ij} = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \Omega_j + \varepsilon_i. \tag{E.5}$$

The encounter shocks follow the baseline joint-normal distribution, with r re-estimated. In the survey-augmented model without spending, residual preference variance is estimated freely and the likelihood is maximized at $\sigma_\xi = 0$. The specification conditioning on reported spending fixes this residual variance at zero. The baseline non-medical price difference is zero in the beliefs-only extension. Conditional on first-assessment covariates, the prior distribution of (Q_j, F_j, ξ_j) is

$$\mathsf{G}^0_j = N(\mu^Q_j, s^2_Q) \times N(0,1) \times N(0,1). \tag{E.6}$$

This distinguishes the residual preference SD, σ_ξ, from the total conditional SD of time-invariant private utility,

$$s_\Omega = \Big(\sigma^2_\xi + \theta^2_Q s^2_Q + \theta^2_F\Big)^{1/2}. \tag{E.7}$$

Let G_i^- denote the distribution of these traits updated with the patient's strictly prior observed assessment outcomes. For a trait vector $\mathbf{z} = (\tilde{q}, \tilde{f}, \xi)$, define

$$v^S_i(\mathbf{z}) = X_i'\beta_U + h(\mathcal{H}_i^-)'\theta - \alpha P_i + \theta_Q \tilde{q} + \theta_F \tilde{f} + \sigma_\xi \xi, \qquad b^S_i(\mathbf{z}, \nu) = v^S_i(\mathbf{z}) + r\nu.$$

The type-specific acceptance probability conditional on the hospital disturbance is

$$\rho_i^S(\nu;\mathbf{z}) = \Phi\left(\frac{b_i^S(\mathbf{z},\nu)}{s_\varepsilon}\right), \qquad s_\varepsilon = \sqrt{1-r^2}. \tag{E.8}$$

The hospital integrates over its prior-history trait distribution to form

$$\begin{aligned}
\rho_i^{H,S}(\nu) &= \int \rho_i^S(\nu;\mathbf{z})\, d\mathsf{G}_i^-(\mathbf{z}),\\
Q_i^{H,S}(\nu) &= \int \left[b_i^S(\mathbf{z},\nu)\Phi\left(\frac{b_i^S(\mathbf{z},\nu)}{s_\varepsilon}\right) + s_\varepsilon\phi\left(\frac{b_i^S(\mathbf{z},\nu)}{s_\varepsilon}\right)\right] d\mathsf{G}_i^-(\mathbf{z}).
\end{aligned}$$

Here $Q_i^{H,S}$ is the conditional option surplus, distinct from the patient's surveyed quality trait Q_j. With D_i denoting the non-surplus direct offer terms, the offer probability is

$$\begin{aligned}
\Gamma_i^S(\nu) &= D_i + \kappa_Q Q_i^{H,S}(\nu) + \rho_i^{H,S}(\nu)\Delta W_i,\\
\mathcal{O}_i^S &= \{\nu : \Gamma_i^S(\nu) + \nu \geq 0\}, \qquad p_i^S = \int_{\mathcal{O}_i^S} f_\nu(\nu)\, d\nu.
\end{aligned} \tag{E.9}$$

Here $\tilde{q}$ and $\tilde{f}$ denote draws of the time-invariant quality and experience traits Q_j and F_j, respectively. Survey answers inform the shared preference parameters through the measurement likelihood, but do not update G_i^- or enter the supply index directly. The historical offer–acceptance filter integrates the Gaussian composite Ω_j using 20 Gauss–Hermite nodes. Survey measurement integration is evaluated separately.

Augmented likelihood. Let $\widetilde{\Theta}$ collect the free baseline model parameters together with θ_Q and θ_F. Conditional on a trait vector $\mathbf{z} = (\tilde{q}, \tilde{f}, \xi)$, define the integrated joint placement probability

$$J_i^S(\mathbf{z}) = \int_{\mathcal{O}_i^S} \rho_i^S(\nu;\mathbf{z}) f_\nu(\nu)\, d\nu.$$

The contribution of observed assessment outcome $Y_i \in \{a,d,n\}$ is

$$\ell_i^S(Y_i;\mathbf{z},\widetilde{\Theta}) = \begin{cases} J_i^S(\mathbf{z}), & Y_i = a,\\ q_i\{p_i^S - J_i^S(\mathbf{z})\}, & Y_i = d,\\ 1 - p_i^S + (1-q_i)\{p_i^S - J_i^S(\mathbf{z})\}, & Y_i = n. \end{cases} \tag{E.10}$$

Its predictive probability is $L_i^S = \int \ell_i^S\, d\mathsf{G}_i^-$, and posterior weights update after the current outcome as in Section 6.3. Here q_i retains its baseline meaning as the documentation probability, distinct from the quality draw $\tilde{q}$. The same assessments from the baseline joint demand-

supply contribute to the objective. Let $\mathcal{M}_j$ contain the observed experience components and define $H_j(\tilde{f}) = \prod_{m \in \mathcal{M}_j} h_{jm}(\tilde{f})$. For any trait distribution G, the joint density or probability of patient j's observed survey information is

$$m_j^S(\mathsf{G}) = \begin{cases} g_Q(Q_j^{\text{obs}}; \mathsf{G}) \int H_j(\tilde{f})\, d\mathsf{G}(\tilde{f}, \xi \mid Q_j = Q_j^{\text{obs}}), & Q_j \text{ observed}, \\ \int H_j(\tilde{f})\, d\mathsf{G}(\tilde{q}, \tilde{f}, \xi), & Q_j \text{ missing}, \end{cases} \tag{E.11}$$

where $g_Q(\cdot; \mathsf{G})$ is the marginal density of Q_j under G. An observed quality score enters as a density evaluated at its reported value. Missing quality is integrated out, and a patient with no observed survey components has $m_j^S = 1$.

Let G_j^Y denote the terminal trait distribution after the patient's complete observed assessment history, updated without survey answers, and let $\mathcal{R}$ denote patients with survey responses. The normalized survey-augmented log likelihood is

$$\mathcal{L}^S(\widetilde{\Theta}) = \sum_j \sum_{i \in \mathcal{T}_j} \log L_i^S(\widetilde{\Theta}) + \sum_{j \in \mathcal{R}} \log \left\{ \frac{m_j^S(\mathsf{G}_j^Y)}{m_j^S(\mathsf{G}_j^0)} \right\}. \tag{E.12}$$

Each patient's survey contributes once, irrespective of their number of assessments. The denominator is the marginal measurement density under the fixed first-stage prior and is constant with respect to the jointly estimated parameters. The normalized measurement contribution is zero when $\theta_Q = \theta_F = 0$, without a survey multiplier or population expansion weight. Non-respondents contribute through their outcome histories and the shared parameters. Predictions conditioned on survey answers are used for patient diagnostics separately from the hospital's historical beliefs.

All free demand and supply parameters, including the encounter-shock correlation, are jointly estimated conditional on linkage and the fixed cost and measurement inputs. Residual variance is freely estimated in the beliefs-only model and fixed at zero in the spending specification. Caregiver income, employment, and paid leave remain descriptive survey fields that are explored in the subsection that follows.

E.2 Conditioning on Reported Non-Medical Spending

The survey elicits household spending on transportation, meals, child care, pet care, cleaning, and paid caregiver support under each location. Figure E.2 summarizes expected HaH-minus-B&M spending differences by reported treatment choice. Let $\Delta C_i^{\$}$ denote the signed HaH-minus-B&M difference in dollars per day, and let $D_i^{\$}$ indicate a complete report matched to assessment i. Spending is used in daily units, without multiplication by length of stay, and applies only to the matched assessment.

The 297 matched reports comprise 237 acceptances and 60 documented declinations. This extension retains every assessment and adds a spending slope and reporting indicator to equation (E.5). Denoting this specification by superscript $\{S, \$\}$,

$$\Delta U_{ij}^{S,\$} = \Delta U_{ij}^{S} + \gamma_P D_i^{\$} \frac{\Delta C_i^{\$}}{100} + \delta_{\$} D_i^{\$}. \tag{E.13}$$

For $D_i^{\$} = 0$, the added terms are defined to be zero without imputing missing spending. The unrestricted slope γ_P measures the utility association with an additional \$100 per day of reported HaH spending relative to B&M. The reporting indicator allows the matched group to have a different utility level. This specification jointly re-estimates 175 parameters using equation (E.12), with acceptance probabilities formed from equation (E.13). Spending and report status enter the offer decision through conditional acceptance and patient-surplus beliefs. Neither enters supply directly. Quality and experience answers remain outside the historical filter, and their normalized measurement contribution is unchanged.

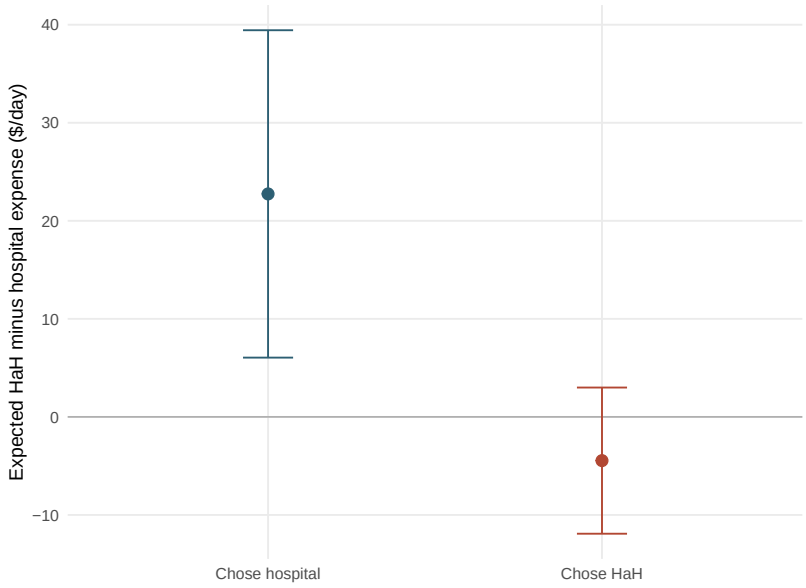
This is a retrospective sensitivity analysis that conditions on report information unavailable in the historical EHR data that informs the offer decision. The reporting indicator absorbs an average group difference without fully correcting for survey selection. Spending answers lack exogenous price variation. Thus, γ_P is considered purely associational and is not used to convert utility into willingness to pay or monetary welfare.

E.3 Results

Table E.2 shows that more favorable expectations about HaH quality and experience are associated with greater willingness to accept. A one-SD increase in the reported quality difference raises acceptance utility by 0.075, while a one-SD increase in the experience factor raises it by 0.265. These estimates connect persistent differences in patients' choices to their stated expectations about care. Previously completed treatment remains positively associated with acceptance. The residual preference term $\sigma_\xi \xi_j$ captures a patient's lasting inclination toward or against HaH after accounting for observed covariates and, when included in the survey-augmented model, quality and experience expectations. It shifts utility across that patient's assessments, unlike the assessment-specific shock ε_i.

Total conditional time-invariant preference variation (s_Ω) includes quality and experience as well as the residual σ_ξ, and its recovered survey-augmented estimated SD is 0.274 compared with 0.232 in the baseline. Supply estimates retain the same substantive pattern as the baseline model. Hospitals favor transfers with greater financial returns and patient option value, while clinical burden and escalation risk discourage offers. The hospital-specific linear capacity terms

Figure E.2: Expected Non-Medical Spending Difference by Reported Choice



Notes: The outcome is expected daily household spending under HaH minus expected daily household spending under hospital care, summed across transportation, meals, child care, pet care, cleaning, and paid caregiver support. Points report means and bars report 95 percent confidence intervals.

are positive and statistically significant at the 10 percent level or better at all four hospitals. HaH strain is negative and statistically significant. Thus, incorporating measured patient expectations leaves the main interpretation of hospital incentives unchanged from the baseline model.

Table E.3 adds reported non-medical spending. Among matched reports, higher HaH spending relative to B&M has a negative coefficient, but it is not statistically distinguishable from zero ($p = 0.133$). The positive reporting-group coefficient indicates that respondents with complete spending reports differ substantially from other assessments, even after the included controls. Quality and experience remain positively associated with acceptance. Changes in the encounter-shock correlation and patient-option-value coefficient also show that conditioning on these reports affects the model's separation of patient choice and hospital targeting.

The joint log-likelihood gain of ~ 322.5 reflects both the spending slope and the reporting indicator. The full-sample accepted-placement Brier score falls from 0.0695 to 0.0677. Because every spending report follows a recorded offer, and the conditional model uses information unavailable to the hospital at the time, this gain does not establish better historical prediction and critically does not identify a population price response.

Table E.2: Baseline and Survey-Augmented Joint Estimates

	Baseline	Survey-augmented	
	Coefficient	Coefficient	SE
Patient acceptance			
Expected-quality difference, θ_Q	–	0.075^{***}	(0.018)
Expected-experience factor, θ_F	–	0.265^{***}	(0.043)
Encounter-shock correlation, r	-0.723^{***}	-0.711^{***}	(0.033)
Home–hospital distance	0.007	0.000	(0.007)
Age	0.050^{***}	0.041^{***}	(0.009)
Male	0.014	0.023^{*}	(0.012)
Household size	−0.001	0.000	(0.008)
College share	0.019^{*}	0.017^{*}	(0.010)
Log per-capita income	−0.015	−0.012	(0.011)
Share below 150% FPL	-0.019^{**}	−0.008	(0.009)
Prior expected rejections	-0.152^{***}	-0.137^{***}	(0.029)
Ever previously treated with HaH	0.151^{***}	0.142^{***}	(0.033)
Count of prior HaH acceptances	-0.050^{*}	-0.064^{**}	(0.027)
Condition acuity weight	-0.034^{**}	-0.035^{**}	(0.015)
Residual preference SD, σ_ξ	0.232^{***}	0.000	–
Total time-invariant preference SD, s_Ω		0.274^{***}	(0.041)
Hospital offers			
Profit, β_Π	0.057^{***}	0.061^{***}	(0.013)
HaH strain, λ_H	-0.004^{***}	-0.004^{***}	(0.001)
Escalation risk, β_H	-0.940^{***}	-0.978^{***}	(0.320)
Care burden	-0.098^{***}	-0.098^{***}	(0.007)
Acute severity	-0.043^{***}	-0.043^{***}	(0.006)
B&M stay-duration exposure	-0.326^{***}	-0.333^{***}	(0.051)
Hospital A occupancy release	0.145^{***}	0.145^{***}	(0.025)
Hospital B occupancy release	0.109^{***}	0.108^{***}	(0.026)
Hospital C occupancy release	0.123^{***}	0.125^{***}	(0.026)
Hospital D occupancy release	0.047^{*}	0.048^{*}	(0.027)
Patient option value, κ_Q	0.876^{***}	0.904^{***}	(0.057)
Likelihood assessments	31,944	31,944	
Unique patients	21,354	21,354	
Linked survey patients	–	453	
Outcome log likelihood	−16,203.7	−16,197.9	
Normalized joint log likelihood	−16,203.7	−16,175.5	
Accepted-placement Brier score	0.0694	0.0695	

Notes: Both models retain the baseline controls, fixed effects, documentation probabilities, and input transformations. Profit is measured in centered $1,000 units. Capacity terms use forecast census, hospital-specific linear relief slopes, and a separate B&M duration exposure. The likelihood is maximized at the boundary $\sigma_\xi = 0$. The approximate one-sided 95% profile upper limit is 0.197 ($\sigma_\xi^2 = 0.039$); no regular Wald standard error is reported. Standard errors for the remaining parameters appear in parentheses, condition on $\sigma_\xi = 0$, and use the observed Hessian, additionally conditioning on patient linkage and first-stage measurement inputs. The normalized survey measurement contribution enters the likelihood once per patient. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table E.3: Survey-Augmented Estimates with Reported Spending

	Survey-augmented		With reported spending	
	Coefficient	SE	Coefficient	SE
Spending per \$100/day, γ_P	–		-0.106	(0.070)
Complete-report indicator, $\delta_{\$}$	–		0.915^{***}	(0.065)
Expected-quality difference, θ_Q	0.075^{***}	(0.018)	0.084^{***}	(0.014)
Expected-experience factor, θ_F	0.265^{***}	(0.043)	0.357^{***}	(0.044)
Encounter-shock correlation, r	-0.711^{***}	(0.033)	-0.506^{***}	(0.037)
Profit, β_Π	0.061^{***}	(0.013)	0.094^{***}	(0.020)
Patient option value, κ_Q	0.904^{***}	(0.057)	1.474^{***}	(0.118)
Likelihood assessments	31,944		31,944	
Matched spending reports used	–		297	
Outcome log likelihood	$-16{,}197.9$		$-15{,}885.1$	
Normalized joint log likelihood	$-16{,}175.5$		$-15{,}853.0$	
Accepted-placement Brier-score: all assessments	0.069		0.068	
Accepted-placement Brier-score: terminal assessments	0.068		0.065	
Accepted-placement Brier-score: first post-treatment	0.165		0.161	

Notes: Both specifications augment the baseline model with quality and experience. The second also conditions on reported non-medical spending and report status in conditional acceptance and surplus beliefs, jointly re-estimating all free demand and supply parameters. The beliefs-only model estimates residual time-invariant variance freely and selects the zero boundary; the spending specification fixes it at zero. Spending enters in units of \$100 per day. Standard errors in parentheses use the observed Hessian, conditional on $\sigma_\xi = 0$, patient linkage, estimated costs, and first-stage measurement inputs. Fit comparisons are in-sample and use different conditioning information. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F Dynamic Offer Model

This appendix develops the dynamic counterpart to Section 5.2. The hospital values the same accounting, capacity, clinical, and offer-stage consequences, but now accounts explicitly for future assessment opportunities and operating states. The dynamic specification replaces the static model's valuation of remaining capacity exposure at the forecasted state with interval payoffs and continuation values. It retains Θ_S and the payoff functions defined in Section 5; the supply coefficients are estimated again under the dynamic specification.

F.1 The Hospital's Dynamic Offer Problem

Assessment timing and state. The risk set and patient information are those of the joint model. Let τ_i denote assessment time in hours. With the daily discount factor $\delta = 0.99$, the continuation discount and interval flow factor are

$$d_i = \delta^{(\tau_{i+1}-\tau_i)/24}, \qquad a_i = \int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s ds = \frac{1-d_i}{-\log \delta}. \tag{F.1}$$

Exactly tied assessments share an event time; their census changes are grouped before evaluating nonlinear capacity flows. Below, binary choice values refer to the focal assessment conditional on the other decisions at that event.

Write z_i for the full information state: the B&M census and staffed capacity at each facility, HaH census and staffing, current patient characteristics, calendar variables, and the prior-history and preference weights used in Section 5. Admitted patients occupying ED beds remain part of B&M census. Including patient histories distinguishes physical capacity changes from information acquired through an offer. The next-event law F_A governs the time to the next assessment, its patient characteristics, and the background operating state. It is estimated from consecutive assessment events, without constructing a separate daily admission model. Future assessment timing, patient mix, and staffing are uncertain; the hospital does not observe their future realizations.

Current payoffs. The direct offer terms $Z_i'\kappa$, remaining accounting return $\Delta\Pi_i$, and escalation risk H_i retain their definitions in equations (6)–(14). Accounting returns are booked once at the placement decision because $\Delta\Pi_i$ already values the expected remaining episode. Escalation remains an ex ante risk term; the baseline does not separately simulate return transfers. No explicit backfill-profit payoff is added.

For capacity flows, let $m^B(z)$ collect the four hospital-specific B&M capacity-pressure functions defined in equation (11). Let $m^H(z)$ be the HaH strain function in equation (8). These retain

the static model's functional forms, evaluated along the evolving operating-state paths. Conditional on an accepted transfer, define the capacity differences over the first interval as

$$M_i^{B,0} = \mathbb{E}\left[\int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s \{m^B(z_i^B(s)) - m^B(z_i^H(s))\}\, ds \mid z_i, A_i = 1\right], \tag{F.2}$$

$$M_i^{H,0} = \mathbb{E}\left[\int_0^{(\tau_{i+1}-\tau_i)/24} \delta^s \{m^H(z_i^H(s)) - m^H(z_i^B(s))\}\, ds \mid z_i, A_i = 1\right]. \tag{F.3}$$

Here B and H denote the focal patient's alternative treatment paths. If the state is constant over the interval, its discounted flow is multiplied by a_i. Survival and capacity reuse determine how long the two paths differ. The current offer-minus-no-offer payoff is

$$g_i(\Theta_S) = Z_i'\kappa + \rho_i\left[\beta_\Pi \Delta\Pi_i + \beta_H H_i + \beta_E E_i + \lambda_B' M_i^{B,0} + \lambda_H M_i^{H,0}\right]. \tag{F.4}$$

The acceptance probability enters once. The interval capacity terms replace, rather than supplement, the full remaining-exposure terms in the static index. Their subsequent flows enter continuation value. [27]

Dynamic problem. Let $\bar{\Gamma}_i(o, z_i; \Theta_S)$ be the expected current payoff from action o. Its capacity-flow component is the interval integral of $-\lambda_B' m^B(z) + \lambda_H m^H(z)$, so that $\bar{\Gamma}_i(1, z_i; \Theta_S) - \bar{\Gamma}_i(0, z_i; \Theta_S) = g_i(\Theta_S)$. These flow levels are retained in the value function; normalizing them away at every state would remove consequences of earlier transfers. The choice-specific value and Bellman equation are

$$v_i(o, z_i; \Theta_S) = \bar{\Gamma}_i(o, z_i; \Theta_S) + \mathbb{E}[d_i V_{i+1}(z_{i+1}; \Theta_S) \mid z_i, O_i = o], \tag{F.5}$$

$$V_i(z_i; \Theta_S) = \mathbb{E}_{\varepsilon_i}\left[\max_{o \in \{0,1\}} \{v_i(o, z_i; \Theta_S) + \varepsilon_i(o)\}\right]. \tag{F.6}$$

The first expectation integrates over future event timing, acceptance, survival, capacity reuse, staffing, and patient information. The discount factor remains inside this expectation because the next event time is uncertain. The second expectation integrates over current private offer shocks. Interval flows are not added again to the continuation term.

[27] Input units follow Appendix C.2: accounting returns are in \$1,000 units and capacity and risk measures retain their units. Reference offsets in centered placement features enter once in the assessment payoff, not again in every subsequent interval of that placement. The physical state-flow functions themselves are evaluated before this assessment-level centering.

Let $z_{i+1}^{a,i}$, $z_{i+1}^{d,i}$, and $z_{i+1}^{0,i}$ denote next states following acceptance, rejection, and no offer. Then

$$\begin{aligned}\Delta v_i(\Theta_S) = g_i(\Theta_S) + \rho_i \mathbb{E}\Big[d_i\{V_{i+1}(z_{i+1}^{a,i};\Theta_S) - V_{i+1}(z_{i+1}^{0,i};\Theta_S)\} \mid z_i, A_i = 1\Big] \\ + (1-\rho_i)\mathbb{E}\Big[d_i\{V_{i+1}(z_{i+1}^{d,i};\Theta_S) - V_{i+1}(z_{i+1}^{0,i};\Theta_S)\} \mid z_i, A_i = 0, O_i = 1\Big]. \end{aligned} \tag{F.7}$$

The counterfactual no-offer state is evaluated using the same underlying draws in each conditional comparison. Rejection and no offer have the same physical census path, but can imply different subsequent histories and beliefs. If these information effects are held fixed, the final line is zero and the continuation difference reduces to the acceptance-weighted HaH-minus-B&M comparison. Future value functions already incorporate acceptance by future patients; they do not multiply the focal ρ_i again.

Counterfactual state paths. An accepted transfer initially removes one patient from B&M census and adds one to HaH census. Let $s_i^B(h)$ and $s_i^H(h)$ be the remaining-stay survival probabilities at elapsed days h. The expected physical footprints are

$$w_i^B(h) = s_i^B(h) e_{b(i)}, \qquad w_i^H(h) = s_i^H(h) e_h, \tag{F.8}$$

where the unit vectors add census to the originating hospital or HaH. At later assessment j, the physical states are constructed as

$$\mathcal{S}_j^{B,i} = \mathcal{S}_j^{-i} + w_i^B(h_{ij}), \qquad \mathcal{S}_j^{H,i} = \mathcal{S}_j^{-i} + w_i^H(h_{ij}), \qquad h_{ij} = \frac{\tau_j - \tau_i}{24}. \tag{F.9}$$

The common background $\mathcal{S}_j^{-i}$ removes the focal patient's realized footprint and any capacity use attributed to its release before either path is overlaid. This avoids adding the same replacement patient twice. Reuse affects the duration of the census difference, not a separate profit payoff. The B&M coefficient continues to value capacity on a composite basis.

Both paths use common draws from $\widehat{F}_A$, with the fitted offer policy reevaluated at their respective states. Patient histories and preference weights update after simulated outcomes using the joint model's filtering rule. The maintained background arrival law does not respond to the focal offer beyond the specified capacity-reuse footprint. Changes in admission behavior or assessment intensity therefore require an extension of that law.

F.2 Conditional Choice Probabilities and Two-Step GMM

Conditional choice probabilities. I use a CCP estimator following Hotz and Miller (1993); Hotz et al. (1994); the equation-matching formulation is related to Pesendorfer and Schmidt-Dengler

(2008). These methods represent continuation values using conditional choice probabilities and state transitions, avoiding repeated solution of the dynamic optimization problem during estimation (Aguirregabiria and Mira, 2010; Arcidiacono and Ellickson, 2011; Aguirregabiria et al., 2021).

The static logit control-function specification, described in Appendix D.2, supplies the fitted ex-ante true-offer policy $\widehat{p}(1 \mid z) = \widehat{p}(z)$ and acceptance belief $\widehat{\rho}(z)$. The former is a logit-based fitted probability, not a documented offer indicator or the posterior classification in equation (18). The documentation probabilities q_i remain empirical measurement inputs.

Conditional on z_i, assume independent standard Type-I extreme-value action shocks, with unit scale, and the maintained acceptance and transition laws. The Hotz–Miller inversion gives

$$\Delta\widehat{cv}_i = \log\left(\frac{\widehat{p}_i}{1-\widehat{p}_i}\right). \tag{F.10}$$

The first-stage policy must capture the state dependence relevant to the dynamic decision. Using the joint model's fitted policy does not equate its static payoff coefficients with the dynamic coefficients.

Fixed-policy returns. I hold the first-stage policy, demand parameters, and transition estimates fixed during supply estimation. At each simulated state, the fitted offer–acceptance fixed point is evaluated using these first-stage parameters. A trial Θ_S changes payoff weights, not the policy or acceptance map used to construct returns. This is a two-step estimator, distinct from joint maximization of the static likelihood.

For fixed policy $\widehat{p}$, the policy-value recursion includes the expected selected offer shock:

$$\begin{aligned} V^{\widehat{p}}(z;\Theta_S) &= \sum_o \widehat{p}(o \mid z)\left\{\bar{\Gamma}(o,z;\Theta_S) + \mathbb{E}[dV^{\widehat{p}}(z';\Theta_S) \mid z,o]\right\} + \eta(z), \\ \eta(z) &= \gamma_E - \sum_o \widehat{p}(o \mid z)\log\widehat{p}(o \mid z), \end{aligned} \tag{F.11}$$

where γ_E is Euler's constant under the standard extreme-value location normalization.

I simulate returns over 90 days, using common transition draws across branches and an approximate stationary tail. A future event j is discounted by $\delta^{(\tau_j-\tau_i)/24}$. Each return component is projected onto a flexible basis of the current state and patient characteristics. Because payoffs are linear in Θ_S under the fixed policy,

$$\Delta v_i(\Theta_S) \approx \Delta\widehat{v}_i^0 + \widehat{\mathcal{D}}_i'\Theta_S, \qquad r_i(\Theta_S) = \Delta\widehat{cv}_i - \Delta\widehat{v}_i^0 - \widehat{\mathcal{D}}_i'\Theta_S. \tag{F.12}$$

The vector $\widehat{\mathcal{D}}_i$ collects current and continuation exposures to every coefficient in equation (F.4);

$\Delta\widehat{v}_i^0$ includes the shock-value contribution. The tail approximation must preserve this affine decomposition.

Two-step GMM. Let $\mathcal{M}_i$ contain projected policy-value derivatives and any additional excluded state instruments used for overidentification. Define

$$g_n(\Theta_S) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{M}_i r_i(\Theta_S), \qquad \widetilde{\Theta}_S \in \arg\min_{\Theta_S} g_n(\Theta_S)' g_n(\Theta_S). \tag{F.13}$$

At $\widetilde{\Theta}_S$, I aggregate assessment-level moments by calendar date and estimate their long-run covariance $\widehat{\Omega}$ using a Bartlett-kernel HAC estimator. The second step is

$$\widehat{\Theta}_S \in \arg\min_{\Theta_S} g_n(\Theta_S)' \widehat{\Omega}^{-1} g_n(\Theta_S). \tag{F.14}$$

Objective coefficients are estimated without sign restrictions.

F.3 Identification and Inference

Identifying variation. The offer-shock normalization fixes the scale of Θ_S. Conditional on the generated demand and transition objects, remaining accounting returns identify β_Π through variation distinct from capacity consequences. The hospital-specific B&M terms use variation in census, capacity, remaining exposure, and their continuation paths. HaH census, staffing, and remaining HaH exposure distinguish λ_H from B&M relief; residual clinical variation identifies β_H and the clinical offer shifters. The coefficient on patient option value is identified separately from acceptance-weighted placement consequences only when their payoff exposures are sufficiently distinct. As in Section 5.2, it can reflect mission-related objectives and directly captures the firm's weight on patient surplus. The common offer term remains a net baseline friction under the stated centering. History and fixed-effect coefficients capture policy heterogeneity without separate welfare labels.

Local identification requires

$$\operatorname{rank}\left\{\mathbb{E}[\mathcal{M}_i \widehat{\mathcal{D}}_i']\right\} = \dim(\Theta_S). \tag{F.15}$$

This condition also requires valid moments, adequate offer-policy support, and a state and transition law that capture the hospital's relevant information. CCP inversion alone does not establish these assumptions. Diagnostics include derivative correlations, the minimum singular value and condition number of the moment Jacobian, and profile objectives. Particular attention is given to accounting versus capacity exposure, and acceptance targeting versus placement benefits.

Inference. For $\mathfrak{m}_i(\Theta_S) = \mathcal{M}_i r_i(\Theta_S)$, define $\mathfrak{m}_t(\Theta_S) = \sum_{i:\text{date}(\tau_i)=t} \mathfrak{m}_i(\Theta_S)$, retaining dates without assessments with zero moments. The HAC matrix allows dependence within dates and serial dependence across nearby dates. Standard errors use calendar-HAC inference with bandwidth seven, conditional on demand parameters, generated inputs, and simulated continuation features. Empirical documentation probabilities remain fixed in this conditional inference; propagating their sampling uncertainty requires reconstructing them as well. The covariance of the replication estimates provides standard errors, including uncertainty in the reestimated generated inputs.

F.4 Dynamic Results and Comparison with the Static Model

Private capacity value. For comparison with the static model, the dynamic private capacity value is the offer-value difference between the full calculation and a no-release calculation that retains focal HaH treatment and HaH strain but removes the B&M capacity change and its continuation consequences. Focal acceptance, history updates, and payoff reference offsets are held common. Its dollar-equivalent value per accepted transfer is

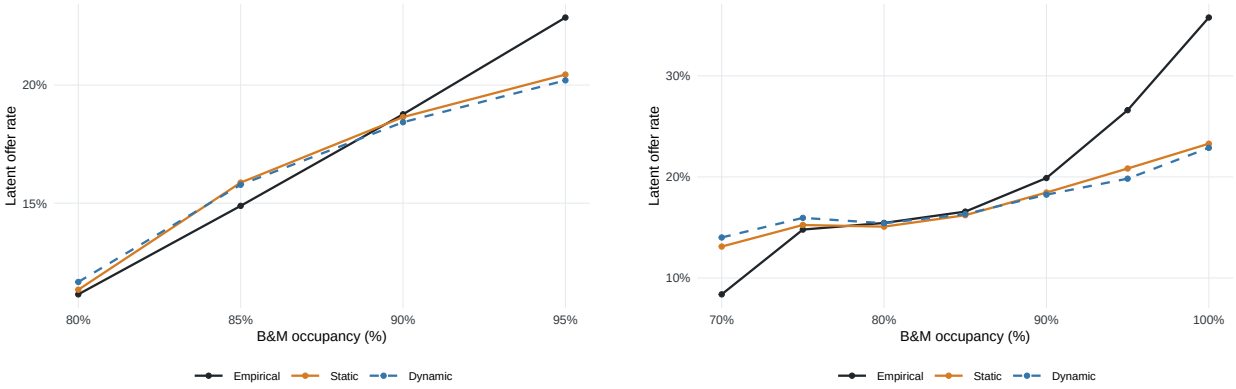
$$\widehat{CV}_i^{\text{Dyn}} = 1{,}000\,\frac{\Delta\widehat{v}_i^{\text{release}} - \Delta\widehat{v}_i^{\text{no-release}}}{\widehat{\rho}_i\widehat{\beta}_\Pi}. \tag{F.16}$$

Table F.1: Dynamic Hospital Offer Estimates

	Estimate	Standard error
Accounting profits (per $1,000)	0.098^{***}	(0.018)
HaH program strain	-0.0143^{***}	(0.0025)
HaH escalation risk	-3.176^{***}	(0.169)
Care-burden	-0.2292^{***}	(0.0020)
Acute-severity	-0.1091^{***}	(0.0016)
B&M stay-duration exposure	-0.753^{***}	(0.066)
Hospital A released-capacity	0.348^{***}	(0.037)
Hospital B released-capacity	0.272^{***}	(0.038)
Hospital C released-capacity	0.303^{***}	(0.040)
Hospital D released-capacity	0.087^{**}	(0.044)
Strictly prior expected offers	0.314^{***}	(0.003)
Patient option value	1.075^{***}	(0.017)
Baseline net offer term	-1.374^{***}	(0.017)
Assessments	31,944	
Unique patients	21,354	
Reestimated supply coefficients	72	
Moments used in GMM	77	
Conditional fit statistics		
Offer-CCP approximation RMSE (logit scale)	0.2977	
Occupancy-bin offer-rate weighted RMSE	0.0193	
Mean predicted offer rate	0.2128	
Mean documentation-adjusted empirical offer rate	0.2213	

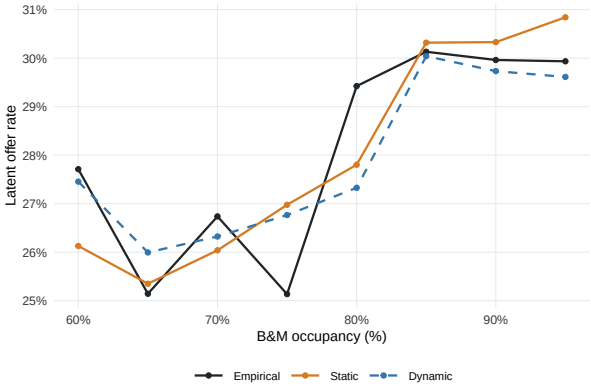
Notes: Estimates recovered using two-step conditional-choice-probability (CCP) GMM. All 72 supply coefficients are reestimated; demand parameters remain fixed at their static estimates described in Appendix D.2. Standard errors use calendar-HAC inference with bandwidth seven and condition upon demand parameters, generated inputs, and simulated continuation features. Estimates retain the static control-function model's logit normalization. Occupancy-bin fit compares predicted offers with the documentation-adjusted empirical reference, weighting cells by assessment counts. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure F.1: Empirical and Predicted Offer Rates by B&M Occupancy and Model Type

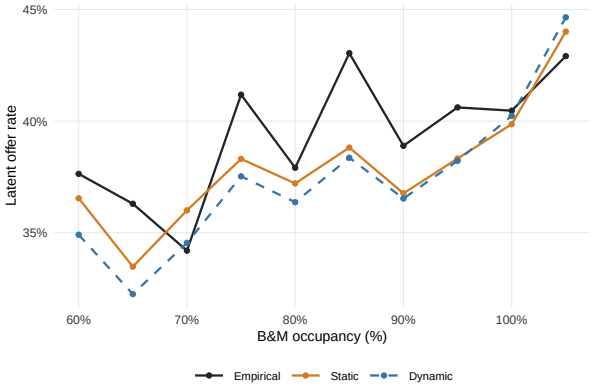


(a) Hospital A

(b) Hospital B



(c) Hospital C



(d) Hospital D

Notes: Points show mean offer rates within hospital-specific five-percentage-point occupancy bins, plotted at each bin's lower bound. Empirical rates are documentation-adjusted; static and dynamic rates average fitted offer probabilities. Bins require at least 50 assessments. All comparisons are in sample.